

Improve Healthcare Operations in HK

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Abstract

The main goal of this project is to fix the high cancellation rate for surgeries at the United Christian Hospital Orthopaedics and Trauma department. Currently, the department sees a 12% cancellation rate on the day of surgery, which is much higher than the 7% average found in other public hospitals in Hong Kong. This project aims to bring UCH in line with those standards by creating a more accurate way to predict how long surgeries will take. By moving away from simple historical averages and subjective estimates, the project seeks to reduce the error in scheduling to around 30 minutes. The ultimate objective is to make better use of hospital resources and ensure that patients do not face the stress of having their procedures postponed at the last minute.

This project addresses the scheduling volatility at United Christian Hospital by developing a predictive model based on 14,834 surgical records. The current system relies on metric-based scheduling that often fails to account for the high variance in procedure times. To solve this, we designed an end-to-end pipeline that processes both structured clinical data and unstructured text.

A key innovation involved the use of Byte-Pair Encoding (BPE), a Natural Language Processing technique, to interpret unstructured surgeon descriptions and transform qualitative clinical notes into quantitative vectors. Following a rigorous evaluation of over 20 algorithms, we utilized both Light Gradient Boosting Machine (LightGBM) and an Out-of-Fold (OOF) Ensemble Stacking approach to balance absolute accuracy with structural stability.

The LightGBM model yielded our best precision result, achieving a Mean Absolute Error (MAE) of 29.81 minutes and an R^2 of 0.683. To further mitigate overfitting and ensure robust generalization, we applied the Ensemble Stacking model. The clinical level requirement for the R^2 training-testing gap is 0.065; our stacking approach achieved a vastly superior gap of 0.029. While the MAE of the stacking model is 28.952 minutes, which is still within the 30-minute MAE goal, its exceptional stability makes it highly reliable. Ultimately, the combined insights from these models provide United Christian Hospital with a scalable, data-driven blueprint to optimize theater utilization and reduce the patient trauma associated with last-minute cancellations.

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1. Introduction

1.1 Problem Statement: The 12% Cancellation Crisis at UCH

United Christian Hospital (UCH), a major acute hospital in the Kowloon East Cluster, faces a persistent and critical challenge in perioperative management: the 12% Cancellation Crisis. This figure represents the proportion of elective surgeries that are cancelled or postponed on the day of the operation. To put this into a broader context, the Hospital Authority (HA) historical average for day surgery cancellations in Hong Kong typically hovers around 7%.

The fact that UCH is operating at nearly double the regional baseline indicates a localized systemic failure in scheduling accuracy. The real crisis stems from a persistent mismatch between the time we put on the calendar and the time surgeons actually need in the room. Currently, UCH relies on historical averages (AVG) or subjective surgeon estimates (SCH). However, these methods are notoriously inaccurate. When a surgery runs over its allotted time, it creates a domino effect, pushing subsequent cases into overtime or leading to their outright cancellation due to a lack of staff or theatre availability. Conversely, underestimation leads to under-utilized gaps, where expensive operating rooms sit empty, wasting precious resources.

The 12% threshold is not merely a statistical anomaly; it is a redline indicating that for every eight scheduled patients, one will likely not receive their life-altering procedure as planned. Things are so unpredictable because our current software ignores the messy reality of the operating room, like the unique way a patient is built or the specific rhythm of a certain surgical team. Without an objective, data-driven approach to close this estimation gap and align UCH with the broader HA benchmark of 7%, the hospital remains in a state of reactive crisis management.

1.2 Impact on Hospital Resources and Patient Welfare

The ramifications of the 12% cancellation rate manifest as a dual burden on hospital resources and patient welfare. From a resource perspective, the Operating Room (OR) is the most expensive unit in the hospital. Research by *Riahi et al. (2023)* highlights that inefficiencies in OR management lead to a significant loss of revenue and increased operational costs. At UCH, a single cancelled case entails wasted preoperative preparation costs, reprocessed sterilized equipment, and idle specialized surgical teams. Furthermore, when surgeries run over, the hospital incurs significant costs in the form of nursing and anesthesia shift premiums, which rapidly deplete the annual budget.

From the perspective of patient welfare, the impact is devastating. A surgical cancellation is a traumatic event. Patients often undergo weeks of psychological preparation, strict fasting (NPO) protocols, and complex logistical arrangements with family members or employers. When a surgery gets pulled at the last minute, the damage is more than just logistical. It breaks a patient's trust and leaves them feeling far more anxious when they finally head back in for their new date.

Clinically, delays can also lead to disease progression. In oncology or high-risk elective cases, a cancellation isn't just a delay; it is an extension of the period the patient is at risk. For the elderly population served by UCH, repeated fasting and the stress of repeated hospital admissions for the same procedure can lead to physical frailty. Thus, improving prediction accuracy is not just a mathematical goal; it is a clinical necessity to ensure the safety, dignity, and timely care of the patient population.

1.3 Project Goals and Success Metrics

To address the crisis described, this project aims to develop a Hybrid Machine Learning and Natural Language Processing (NLP) system that provides objective, high-precision surgical duration forecasts. The primary goal is to bridge the gap between UCH's current 12% rate and the HA-wide 7% standard by replacing unreliable manual baselines with a model that can read complexity through textual descriptions.

Specific Project Goals include:

1. **Lexical Feature Extraction:** To utilize NLP techniques to transform raw procedure descriptions into numerical vectors that capture the inherent complexity and time-intensity of the task.
2. **Mitigation of Bias:** To move away from the subjective Surgeon Estimate (SCH), which is often influenced by over-optimism, and instead use highly optimized, data-driven frameworks including LightGBM and Ensemble Stacking.

Success Metrics: The success of the solution will be evaluated against the following benchmarks:

1. **Mean Absolute Error (MAE):** The target is to achieve an MAE lower than 30 minutes.
2. **Structural Generalization (R^2 Gap):** To ensure real-world reliability, we define the clinical level threshold for the R^2 training-testing gap is set at a maximum of 0.065.
3. **The Crisis Reduction Metric:** A simulated reduction in the 12% cancellation rate. By identifying long-run cases before the day of surgery, schedules can be balanced to prevent the late-afternoon bottleneck, effectively aiming to lower the cancellation rate to meet or exceed the HA average of 7%

2. Overall Directions to a Solution

2.1 Limitations of Current Scheduling Baselines (AVG vs. SCH)

The scheduling problems at United Christian Hospital stem from a reliance on two traditional methods: historical averages and surgeons' estimates. While these methods are common in Hong Kong public hospitals, they are largely responsible for the 12% cancellation rate in the orthopaedics department.

The historical average approach is flawed because it treats surgery duration as a fixed constant. By simply calculating the mean time of past procedures under the same code, the system ignores the unique details of each case. As noted by Jiao (2022) and Babayoff (2022), factors such as patient age, health history, and BMI significantly change how long a surgery takes. For example, a routine knee replacement for a healthy 45-year-old is not comparable to the same procedure for an elderly patient with multiple comorbidities. Using a simple average ignores these critical variables and essentially ensures that scheduling will be inaccurate half of the time, leading to a snowball effect of delays and same-day cancellations.

Surgeons' estimates are also problematic because they are subjective and prone to bias. Surgeons often estimate based on the actual surgical time rather than the full period the patient is in the operating theatre. This often excludes necessary phases like anesthesia induction, patient positioning, and room turnover. Additionally, experienced doctors may base their estimates on their fastest previous performance rather than a realistic average. This lack of objectivity fails to account for daily theater dynamics, such as equipment setup or the involvement of trainees, creating the scheduling volatility that leaves patients stranded.

2.2 Theoretical Framework: Combining ML and NLP for Duration Prediction

Traditional scheduling methods rely almost exclusively on numeric data, such as procedure codes and historical averages (Park et al., 2025; Yuniartha et al., 2021). However, these methods struggle to analyze the unstructured text found in surgery descriptions and clinical notes, which often contains the most important details about a case's complexity (Lai et al., 2020; Riahi et al., 2023).

By integrating Natural Language Processing (NLP) with Machine Learning (ML), this project bridges that gap (Kwong et al., 2025; Park et al., 2025). NLP allows the system to convert medical text into meaningful data points, which are then combined with numeric features to create a more accurate prediction (Hinterwimmer et al., 2023; Lai et al., 2020). This hybrid approach ensures that the model learns from both the hard numbers and the descriptive clinical context that traditional methods ignore (Abbas et al., 2022; Gabriel et al., 2023).

2.3 Proposed System Architecture

Step 1: Data Input and Processing

The architecture begins with the ingestion of raw surgery data integrated with a manual dictionary to standardize medical terminology. This data is fed into a comprehensive preprocessing module where structural integrity is established by removing duplicate records and eliminating columns with more than a 40% missing-value threshold. To prepare the data for the modeling engine, the system performs tokenization on clinical text and numerization on categorical variables. Specifically, temporal variables such as the month and day of the week are transformed via cyclic encoding. Targeted imputation is then applied to fill essential clinical context for the host team, Eye, Anaesthesia, ENT, and Dental fields. Following this, the processing pipeline branches into two distinct streams to generate two datasets. One is a fill in dataset where any remaining gaps are populated using identified statistical distributions to preserve the original sample size. Another is a dataset where the system will remove rows with any missing value to prioritize record precision and data integrity. Finally, the processed data is partitioned into a training Set and a testing Set to ensure rigorous model validation.

Step 2: Single Model Selection

The core predictive engine operates through a multi-stage funnel designed to systematically identify the best performing algorithm while utilizing 5-fold cross-validation at every stage to ensure statistical reliability. The process initiates with the Primary Model Selection and First Evaluation, where initial candidate models are assessed with Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R^2 , and the R^2 gap (to check for overfitting) to establish a performance floor. This is followed by the Secondary Model Selection, where the architecture shortlists five primary ensemble models: XGBoost, CatBoost, LightGBM, Bagging Regressor, and Random Forest. Throughout this phase, Optuna Optimization to maximize accuracy and minimize the training-testing gap across the datasets. After the last evaluation in this step, LightBGM is selected by its best performance of MAE 28.952.

Step 3: Ensemble Stacking

The ensemble stacking phase begins by selecting diverse base models, using a correlation threshold of less than 0.9 relative to the LightGBM baseline to ensure a variety of algorithmic biases since LightBGM has the best performance in the previous step. To prevent data leakage and ensure generalization, an Out-of-Fold procedure is implemented via 5-fold cross-validation, where predictions on unseen data are aggregated to create a new feature set. This architecture is iteratively refined by pruning any candidate models

that contribute less than 5% to the total ensemble weight or demonstrate insufficient performance. Finally, multiple meta-learners are evaluated against a linear regression baseline to identify the optimal algorithm for integrating the strengths of the primary models. The best result is achieved by the mixture of LightGBM and Regression Tree, reaching an R^2 gap of 0.016, MAE of 28.952 and RMSE of 47.340.

Step 4: Model Evaluation

While LightGBM established a strong baseline with a MAE of 29.81 minutes and an R^2 of 0.683, the implementation of an Out-of-Fold (OOF) Ensemble Stacking approach further elevated the framework's performance. The stacking model not only achieved a superior MAE of 28.952 minutes but also demonstrated exceptional structural stability. With a training-testing R^2 gap of just 0.029, this dual-model strategy ensures both high-precision forecasting and robust generalization for unseen cases.

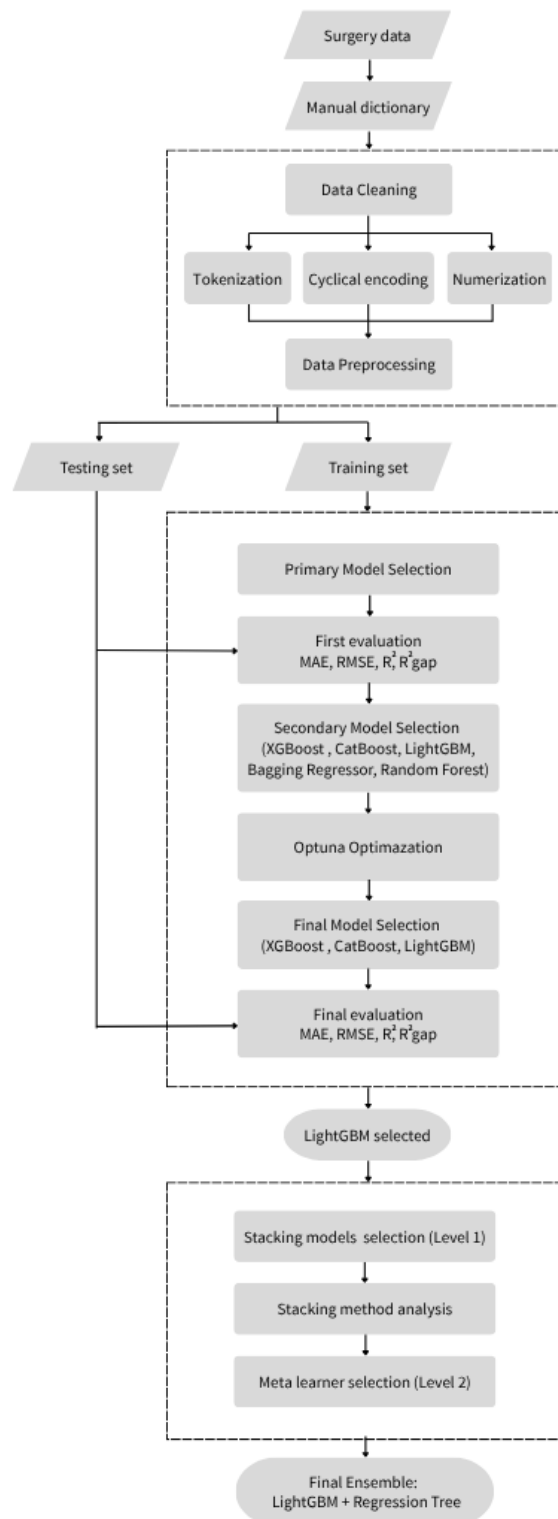


Fig 2.3.1

3. Design of solution

3.1 Data Preprocessing

Before conducting the core studies, the data were processed and standardized into machine-friendly formats. This stage involved the application of several distinct methodologies to ensure accuracy and precision.

3.1.1 Data Consolidation and Formatting

The initial stage of preprocessing the data involved combining the historic data sets into a single, comprehensive dataset. Two years of data were joined, followed by the conversion of feature headers to lowercase characters. This process was conducted to reduce sparsity and ensure the data remained accessible for analysis. To further diminish sparsity, the text data was mapped to a manually constructed dictionary, and all symbols were converted into words.

3.1.2 Data Cleaning

To address the high amount of missing data within the dataset, two distinct strategies were developed: a distribution-based imputation method and a filtering method. Before applying either approach, a preliminary dimensionality reduction step was performed by removing any feature column with a missing data rate exceeding 40%. Once these high-vacancy features were purged, the remaining data was processed through the two independent methodologies to compare their effects on model performance.

The first strategy, the filling method, emphasized data imputation to secure the original sample size. This process began with a detailed statistical analysis of the distribution for each remaining feature. Once the underlying distribution was identified, the missing entries were populated with values that mathematically mirror the existing data patterns.

Conversely, the filtering method adopts a more conservative approach by prioritizing data integrity over quantity. In this method, any observation containing one or more missing values is entirely removed from the dataset, resulting in a refined subset comprised exclusively of complete records.

3.2 Feature Encoding and Text Processing

The dataset contained various information formats, specifically categorical text, numerical features, and natural language descriptions. To ensure these data types were compatible with the machine learning algorithms, specific processing pipelines were developed.

3.2.1 Numericizing of Structured data

To numericize the structured data, two approaches were implemented: one-hot encoding and cyclical encoding. Categorical attributes were numericized through one-hot encoding, where a value of 1 represented the presence of a trait and 0 represented its absence. Additional feature columns were generated; consequently, the dataset only highlighted a value of 1 under the corresponding feature, as illustrated in Figure 3.2.1.

Temporal features were numericized using the cyclical encoding to ensure the cyclical nature of time was captured accurately. The date values were transformed into corresponding sine and cosine values, as illustrated in Figure 3.2.2. This method was utilized to provide a more precise representation of periodic data patterns.

oh_host_team__day	oh_host_team__ent	oh_host_team__eye	oh_host_team__obs
0	0	1	0
0	0	1	0
0	0	1	0
0	0	1	0
0	0	1	0
0	0	1	0
0	0	0	0
0	0	0	0
0	0	0	0

Fig 3.2.1

month_sin	month_cos	day_sin	day_cos
0.5	0.8660253882408142	0.5712682008743286	0.8207634687423706
0.5	0.8660253882408142	0.5712682008743286	0.8207634687423706
0.5	0.8660253882408142	0.5712682008743286	0.8207634687423706
0.5	0.8660253882408142	0.5712682008743286	0.8207634687423706

Fig 3.2.2

3.2.2 Unstructured Natural Language Processing

To process the unstructured natural language data (px_description), the Byte-Pair Encoding (BPE) technique was implemented. This sub-word tokenization algorithm was utilized to segment text into smaller, frequently occurring units, effectively balancing the granularity of character-level models with the semantic richness of word-level models (Kwong et al., 2025; Park et al., 2025). The preprocessing pipeline followed a structured four-stage approach to ensure the model reflected the real problem accurately.

Initially, the distribution of description lengths was analyzed to determine the optimal sequence length for the tokenizer. A percentile of 1.0 was selected to capture the full breadth of the descriptions while applying a maximum cap of 25 tokens, which ensured that the resulting feature matrix maintained a consistent dimensionality without losing critical information from longer medical entries (Lai et al., 2020). To further diminish noise and vocabulary sparsity, the text underwent NFKC (Normalization Form Compatibility Composition) and lowercase conversion.

NFKC is a Unicode normalization method that facilitates the management of visual duplicates. For instance, a term such as X^2 is transformed into separate "x" and "2" characters; subsequently, visually similar variations are treated as identical terms. This process suggests a reduction in noise resulting from typographical errors and inconsistent character encoding. By standardizing these variations, the data remains accurate and precise, ensuring that the model reflects the real problem with high integrity (Riahi et al., 2023).

A bespoke BPE model was then trained on the dataset. By identifying the most frequent sub-word merges, the algorithm learned the common roots and prefixes inherent in the surgical descriptions. This approach is particularly robust for medical datasets, as it allows the model to interpret rare jargon or typos by breaking them down into known sub-units; for instance, complex terms are split into recognizable prefixes and suffixes (Kwong et al., 2025). Subsequently, the identified sub-words were transformed into unique numerical identifiers. These tokens were then padded or truncated to the predetermined maximum length to ensure a methodical and well-organized representation. Finally, these sequence features were concatenated with the tabular clinical features, facilitating a hybrid input structure suitable for the machine learning algorithms (Hinterwimmer et al., 2023). The results are illustrated in Figure 3.2.3.

px_description							
left upper lid entropion correction +/- blepharoplasty							
tok_str_0	tok_str_1	tok_str_2	tok_str_3	tok_str_4	tok_str_5	tok_str_6	tok_str_7
left	upper	lid	entropion	correction	or	blepharoplasty	<PAD>
tok_0	tok_1	tok_2	tok_3	tok_4	tok_5	tok_6	tok_7
66	525	435	612	536	30	607	0

Figure 3.2.3: The text processing pipeline, illustrated by (1) the original raw description, (2) the BPE-segmented sub-word units, and (3) the final numerical token IDs prepared for model input.

3.2.3 Data Quality

Fig 3.2.3.1 was used to predict surgery duration. Temporal variables including month, day, and day of week were transformed via cyclic encoding to capture periodic patterns. Procedure descriptions were converted into numeric tokens, while session, theatre code, hosting specialty, hosting team, and admission category were processed using one-hot encoding (Kwong et al., 2025; Park et al., 2025).

Feature	Type	Range and statistics	
		Before cleaning	After cleaning
Month	Numeric	Jan(8.45%), Feb(6.31%)...	Jan(8.45%), Feb(6.31%)...
Day	Numeric	1 st (2.60%), 2 nd (2.41%)...	1 st (2.59%), 2 nd (2.38%)...
Day of week	Numeric	Mon(18.41%), Tue(19.23%)...	Mon(17.14%), Tue(19.25%)...
Procedure description	Numeric	e.g. LSCS + BTL; LSCS + BTL	e.g. tok_1, tok_2...
Session	Boolean	AM(32.65%), PM(22.51%)...	AM(32.75%), PM(22.99%)...
Theatre code	Boolean	OT1(11.53%), OT2(12.12%)...	OT1(12.77%), OT2(11.92%)...
Hosting specialty	Boolean	EYE(12.36%), ENT(8.45%)...	EYE(11.28%), ENT(10.06%)...
Hosting team	Boolean	A(9.36%), B(13.05%)...	A(8.04%), B(12.78%)...
Admission category	Boolean	Day Patient(43.11%)...	Day Patient(48.38%)...
Surgery duration	Numeric	Float (AVG: 117.7, STD: 107.4, MAX: 567, MIN: 1)	Float(AVG: 93.8, STD: 86.2, MAX: 425, MIN: 1)

Fig 3.2.3.1 Feature list

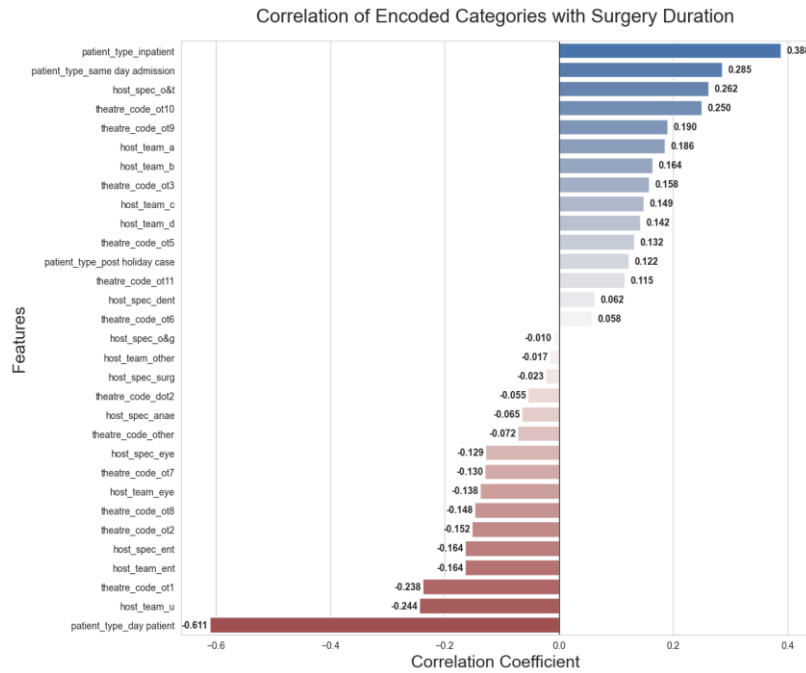


Fig 3.2.3.2 Correlation graph

The analysis of the dataset quality reveals significant variances in surgery duration when grouped by the most representative features, as shown in Fig 3.2.3.3. Starting with the Hosting Specialty, The Orthopaedics & Traumatology (O&T) specialty is a key driver for duration, showing a positive correlation of 0.2621 (Yeo et al., 2023; Abbas et al., 2022).

With a large count of 1,777 cases, it maintains a high mean of 138.7 minutes and a wide range, indicating that while it generally involves long procedures, it also covers a full spectrum of case complexities (Hinterwimmer et al., 2023; Langenberger et al., 2025). In contrast, ENT is representative of shorter procedures with correlation of -0.1643; despite a high Max of 567, its low Median of 28 and Mean of 80.6 across 1,237 cases suggest that the vast majority of its volume consists of very quick surgeries (Jiao et al., 2020). Dentistry stands out for having the highest Mean (226 mins) and a high Min of 16 in the filtered set, explaining why it remains a distinct predictor even with a smaller count of 359 (Babayoff et al., 2022).

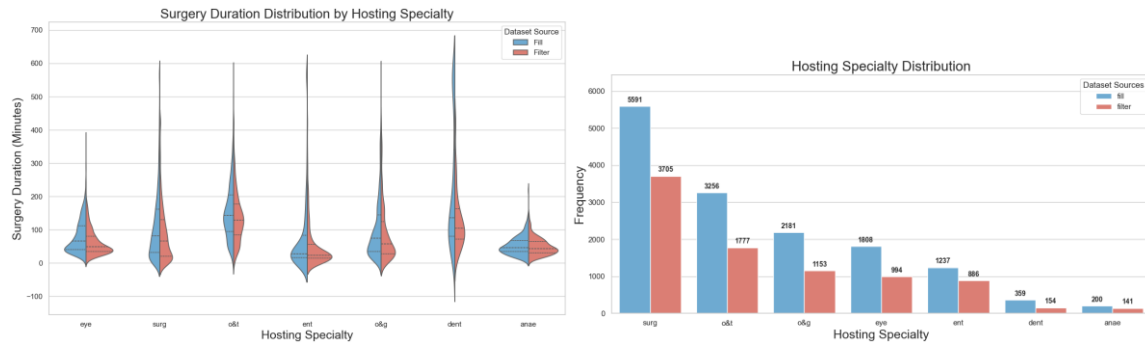


Fig 3.2.3.3 Distribution and frequency of hosting specialty

As illustrated by Fig 3.2.3.4, the Hosting Team distribution follows a similar pattern, where Team A is the most representative of high-intensity workloads, showing the strongest positive relationship to duration in this group at 0.1855 and a high mean of 189.14 minutes. On the other hand, specialized teams like the Eye Team are associated with much shorter surgical windows, carrying a negative correlation of -0.1383. The clear distinction between the general surgical teams and specialty-specific teams indicates that team assignment is a reliable proxy for the expected duration of the surgery.

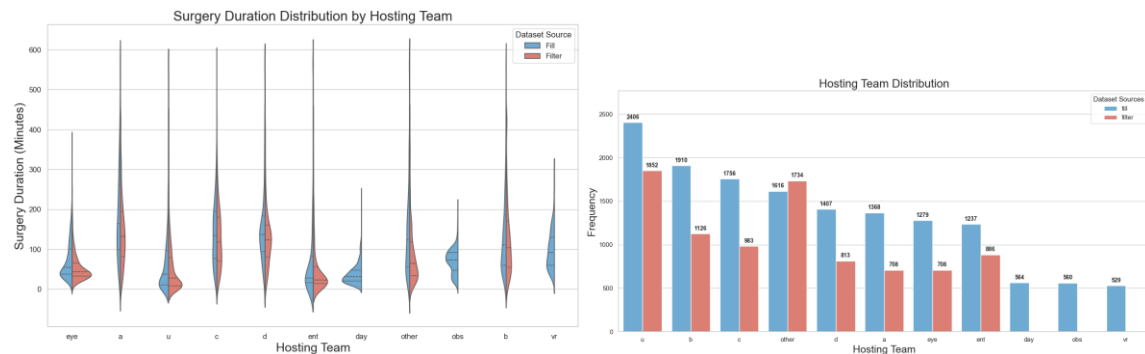


Fig 3.2.3.4 Distribution and frequency of hosting team

Shown in Fig 3.2.3.5, regarding Patient Type, the data provides the strongest predictive signal through the distinction between day patients and inpatients. Day Patients are the most representative category for brief procedures, exhibiting a powerful negative correlation of -0.6106 and a low mean duration of 48.47 minutes across 6,308 cases. Conversely, Inpatients represent the most complex cases with a high positive correlation of 0.3885 and a mean duration of 174.11 minutes. This group shows a minimum value of 4, indicating these surgeries rarely fall into the ultra-short category seen in day cases. Same Day Admissions also represent a high-intensity group with 3,967 cases and a mean of 163, carrying a positive correlation of 0.2853.

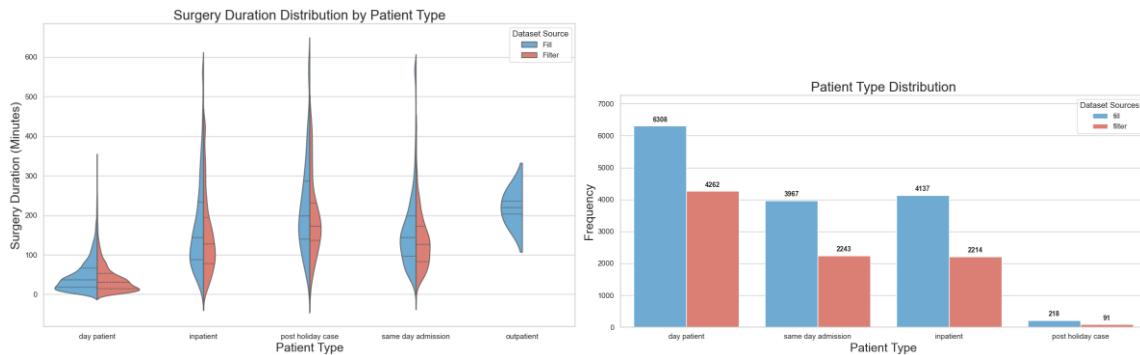


Fig 3.2.3.5 Distribution and frequency of patient type

For the Session Type, the distribution shows that Session W is representative of the longest and most frequent operating slots, with 6,560 cases and a high mean duration of 169.91 minutes, as shown in Fig 3.2.3.6. In contrast, Session P typically handles shorter cases, averaging only 55.05 minutes across 3,294 cases. The minimum value of 1 across all session types indicates that short procedures can occur anytime, but the high median of 140 in Session W compared to 36 in Session P highlights a fundamental difference in the intended case length for these slots.

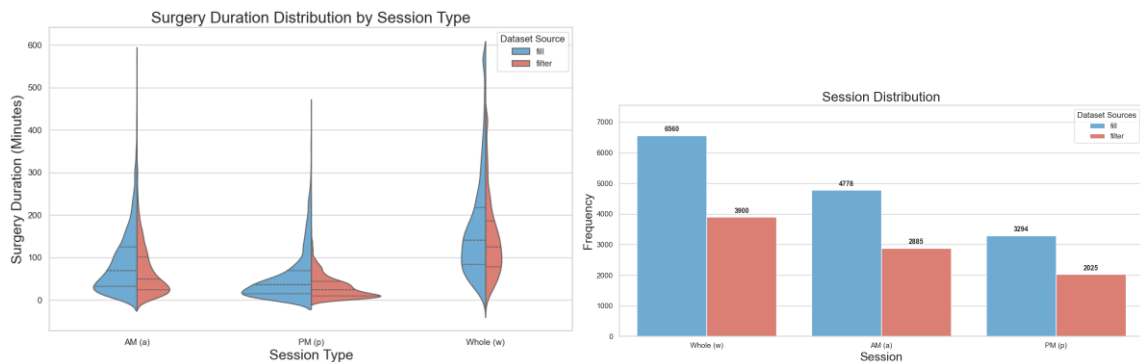


Fig 3.2.3.6 Distribution and frequency of session type

Finally, the Theatre Code analysis identifies specific operating rooms that are reserved for different case types, as shown in Fig 3.2.3.7. Theatre OT10 is the most representative of long-duration surgeries, with a correlation of 0.2498 and the highest mean duration at 193.79 minutes. Its range is notable, with a maximum of 567 and a median of 148, confirming it is consistently used for long procedures. Meanwhile, Theatre OT1 is primarily utilized for shorter, faster-turnover procedures, which is reflected in its negative correlation of -0.2382 and a mean duration of 51.58 minutes across 1,687 cases. These room-specific distributions confirm that theatre allocation is highly structured based on the anticipated length of the procedure (Babayoff et al., 2022).

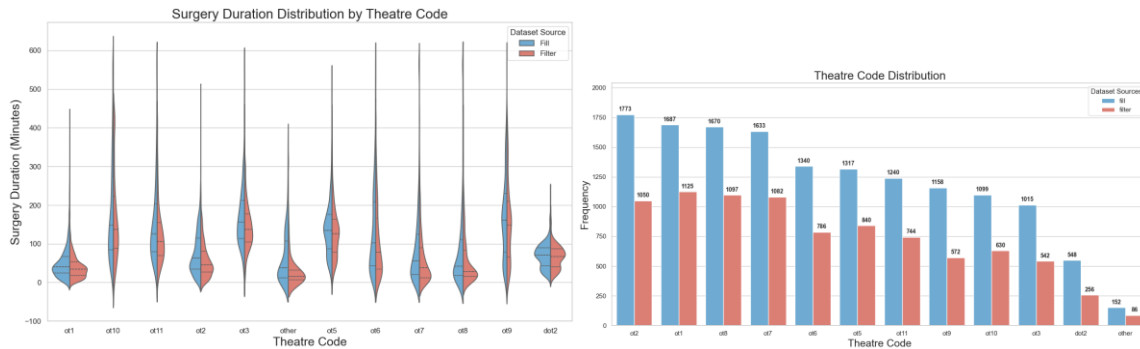


Fig 3.2.3.7 Distribution and frequency of theatre code

The data indicates a clear inverse relationship between surgery duration and prediction accuracy, where the model performance tends to degrade as the length of the procedure increases (Jiao et al., 2020), as shown in Fig 3.2.3.8. This trend is likely driven by the distribution of the dataset, as the number of available samples descends significantly as surgery durations grow longer (Martinez et al., 2021). With fewer data points to learn from in the higher time brackets, the model struggles to maintain precise estimates (Jiao et al., 2022). Interestingly, this pattern holds consistently across the majority of the dataset, with the exception of extremely long surgeries where the prediction results may deviate from this standard trend (Riahi et al., 2023).

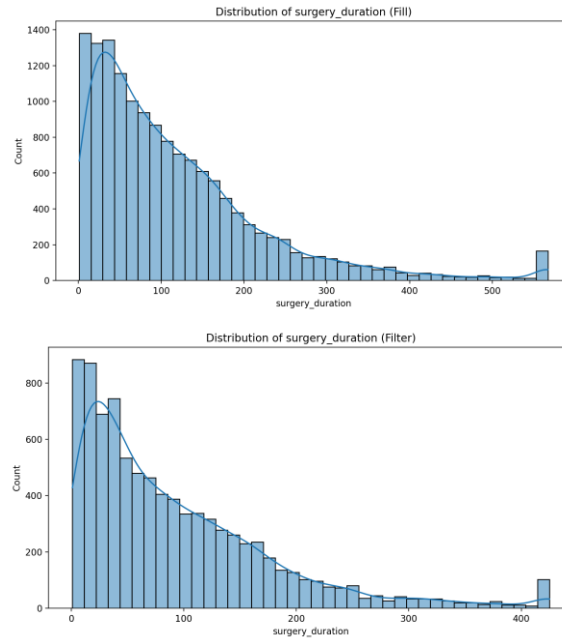


Fig 3.2.3.8 Distribution of surgery duration

4. Model Architecture and Design

4.1 Algorithm Selection and Classification

4.1.1 Selection of Regression Models

The model selection process was initiated with a systematic review of existing methodologies in surgical case duration prediction. This research identified a broad spectrum of algorithms utilized within the field, which were subsequently categorized into five primary frameworks based on their underlying mathematical logic and predictive characteristics. By aligning the selection process with methodologies validated in prior surgical literature, the study ensures that the chosen algorithms are both relevant and robust.

The first category comprised traditional statistical models, including Linear, Multivariable, Lasso, Ridge, and Logistic Regression, alongside Bayesian approaches. These models were selected to provide a baseline for interpretability and to evaluate the degree of linear relationship between clinical variables and surgery duration. To capture complex, non-linear dependencies often found in surgical datasets, ensemble models utilizing boosting techniques were incorporated. Algorithms such as XGBoost, LightGBM, CatBoost, Gradient Boosting Machine (GBM), and AdaBoost were included, as established research suggests these models are particularly effective in minimizing bias.

To address potential issues with model variance and overfitting, bagging techniques including Random Forest, Bagging Regressors, and their balanced variants were also integrated. These methods aimed to improve predictive stability through parallel ensemble learning. Given the high dimensionality of patient and operational data, deep learning architectures such as Artificial Neural Networks (ANN), Memory-Augmented Neural Networks (MANN), Mixture Density Networks (MDN), and Multi-Layer Perceptrons (MLP) were evaluated. These models are widely considered capable of modeling highly complex data distributions and uncertainty in clinical environments.

Finally, several auxiliary models, including Support Vector Machines (SVR and SVM), Decision Trees, and Instance-Based Learning (kNN), served as critical benchmarks for the more advanced ensemble and deep learning methods. By systematically analyzing these five frameworks, the selection process effectively narrowed the vast field of machine learning possibilities into a justifiable shortlist. This rigorous approach, supported by empirical evidence from existing studies, ensured that the final models were those most likely to yield accurate, robust, and clinically relevant predictions.

4.1.2 Hybrid Input Architecture

The final stage of the design involved the integration of processed clinical attributes and sub-word token sequences into a unified hybrid input architecture. This structure was designed to ensure that the machine learning models could simultaneously interpret both high-level clinical context and granular procedure semantics. The integration logic followed a parallel processing and concatenation strategy, where structured features, such as theatre location, patient type, and cyclical date transforms, are provided the operational context, while the Byte-Pair Encoding (BPE) sequences captured the linguistic complexity of the surgical descriptions. These two distinct data streams were aligned using a unique surgery identifier to ensure temporal and situational consistency across the dataset.

To facilitate a single aligned input matrix, the BPE token sequences were fixed to a consistent length and then concatenated horizontally with the tabular structured features. This resulted in a high-dimensional feature table where each row represented a complete "snapshot" of a surgical case, as illustrated in Figure 4.1.1, containing both its categorical characteristics and its encoded text tokens. This method of horizontal stacking allows the regression algorithms to process heterogeneous data types—natural language and structured numerical data—within a standardized mathematical framework.

Finally, the target variable, surgery duration, was isolated from the primary feature set. To ensure the integrity of the predictive modeling process, the duration data was maintained as a separate dependent variable, distinct from the independent features used for training. This separation ensured that the subsequent regression models could learn the correlations between the combined hybrid feature space and the actual case duration in a methodical and realistic manner. By synchronizing the target with the hybrid features while maintaining structural independence, architecture directly reflects the complexities of real-world surgical scheduling, providing a robust foundation for predictive modeling.

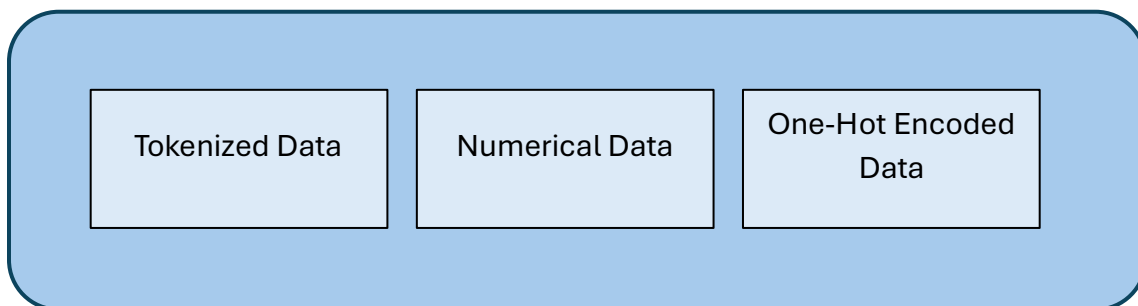


Fig 4.1.1: Schematic representation of the final feature matrix. The vector is composed of a 25-dimensional sub-word token sequence concatenated with standardized clinical attributes

4.2 Advanced Feature Engineering and Transformation Logic

4.2.1 Feature Engineering Strategy

While Session 3 addressed the fundamental numericization of data, this section details the engineering of high-order features designed to capture specific clinical complexities. To accurately prepare the dataset and thoroughly evaluate the algorithms, several systematic engineering and tuning methods were applied. These features were engineered to capture domain-specific nuances and the inherent complexity of various surgical procedures.

4.2.2 Domain-Specific Feature Logic

These features were engineered to capture domain-specific nuances and the inherent complexity of various surgical procedures.

Feature Name	Logic / Calculation	Analytical Objective
spec_theatre	Concatenation of host_spec and theatre_code.	Captures the unique interaction patterns between specific surgeon specialties and operating room environments.
token_complexity	Row-wise sum of 24 token columns (tok_0 to tok_23).	Aggregates multifaceted case data into a single, unified metric representing case complexity.
is_high_risk_spec	Binary flag (1 or 0) based on 'surg', 'o&t', or 'o&g'.	Provides the model with a direct signal for specialties historically associated with longer surgery durations.
weighted_complexity	Product of token_complexity and is_high_risk_spec.	Isolates and intensifies the impact of case complexity specifically when occurring within high-risk specialties.

4.2.3 Robust Categorical Variable Encoding

To maximize the predictive value of categorical variables without introducing data leakage, a sophisticated 5-fold cross-validated encoding process was utilized.

Methodology Step	Implementation	Purpose / Rationale
Cross-Validation	5-fold KFold splitting strategy.	Ensures that the encoding for any given validation fold is derived only from other training folds, preventing leakage.
Target Transformation	<code>np.log1p()</code> on the target variable.	Normalizes the right-skewed surgery duration data and stabilizes variance before calculating means.
Noise Injection	Multiplication by <code>np.random.normal(1, 0.001)</code> .	Regularizes the encoded values to prevent the model from overfitting to exact mean values.
Imputation Strategy	Overall training log-mean fill.	Provides a robust default value (global mean) for categories found in the test set that were absent during training.

During the training phase, categories were mapped to the mean of the log-transformed target variable. To mitigate the risk of data leakage and subsequent overfitting, `np.random.normal(1, 0.001)` was injected as regularization noise during the fold splits. Missing categories in the test set were safely imputed using the overall log-mean of the training target.

4.2.4 Data Normalization and Type Conversion

Following the initial data loading sequence, a critical transformation was executed to accommodate algorithms native to the scikit-learn library.

Transformation Process	Implementation Details	Purpose
Data Type Conversion	<code>to_numeric_for_sklearn()</code> function	Converts categorical and object data types into aligned integer codes for the Bagging Regressor and Random Forest.
Dataset Alignment	Temporary concatenation of train and test splits	Establishes a unified pandas CategoricalDtype.
Mapping Consistency	Global integer mapping	Guarantees that identical categories map to the exact same integer codes across both training and testing datasets.

Tree-based scikit-learn models require numerical inputs. Ensuring consistent integer encoding across data splits prevents data leakage and structural mismatch errors during the evaluation phase.

5. Testing and Performance Evaluation

5.1 Evaluation Metrics

To rigorously assess the candidate models, a multi-dimensional evaluation framework was established. This approach balanced average predictive accuracy with the necessity of mitigating operational risks within a clinical environment.

Mean Absolute Error (MAE): This metric measures the average magnitude of errors in minutes. It is essential for optimizing daily OR scheduling and staffing requirements.

$$MAE = \frac{1}{n} \sum |x_i - x|$$

Where:

n is the number of surgeries recorded in the dataset after processing;

x_i is the predicted surgery duration for the i^{th} surgery recorded in the dataset;

x is the actual surgery duration for the i^{th} surgery recorded in the dataset.

Root Mean Square Error (RMSE): By squaring errors before averaging, RMSE assigns a disproportionately high penalty to large deviations. This was prioritized to identify models prone to significant, schedule-disrupting errors.

$$RMSE = \sqrt{\frac{\sum (P_i - O_i)^2}{n}}$$

Where:

n is the number of surgeries recorded in the dataset after processing;

P_i is the predicted surgery duration for the i^{th} surgery recorded in the dataset;

O_i is the actual surgery duration for the i^{th} surgery recorded in the dataset.

Coefficient of Determination (R^2): This demonstrates the proportion of variance explained by the model, serving as a primary indicator of explanatory power and statistical validity.

$$R^2 = 1 - \left(\frac{SS_{res}}{SS_{tot}} \right)$$

Where:

SS_{res} is the sum of squares of residuals (the difference between the actual surgery duration and the predicted surgery duration);

SS_{tot} is the total sum of squares (the difference between the actual surgery duration and their mean).

R^2 Training-Testing Gap (ΔR^2): This analysis was utilized as a safeguard against overfitting. A narrow gap suggests that the model has captured generalizable clinical patterns rather than specific data noise, ensuring reliability upon deployment.

$$\Delta R^2 = R^2_{train} - R^2_{test}$$

Where:

R^2_{train} is the proportion of variance in the dependent variable that is predictable from the independent variables within the training dataset;

R^2_{test} is the proportion of variance predictable within the testing (hold-out) dataset.

Synthesis of Performance Metrics and Selection Rationale

The integration of these metrics provided a holistic assessment of model suitability. While MAE and R^2 confirmed typical accuracy and validity, $RMSE$ and the ΔR^2 analysis ensured the selection of robust models that minimize operational risk. This multi-faceted approach established a justifiable foundation for the final model of optimization and clinical implementation.

5.2 Model Performance Metrics Evaluation Architecture

To ensure a rigorous and standardized assessment of all candidate architectures, a comprehensive evaluation pipeline was developed. This architecture automated data ingestion, preprocessing, and performance quantification processes.

5.2.1 Data Partitioning and Preprocessing Frameworks

The evaluation begins by partitioning the dataset into training and testing sets using two distinct preprocessing strategies: "Fill" (imputation of missing values) and "Filter" (exclusion of incomplete records). These frameworks generate specific feature sets (X_{train} , X_{test}) and corresponding target variables (Y_{train} , Y_{test}) representing surgical durations. During execution, categorical and textual features, such as theatre codes, are systematically converted into numerical formats. A consistent target encoding methodology is applied across both datasets to ensure structural uniformity and prevent data leakage prior to model ingestion.

5.2.2 Data Loading and Preprocessing

During the execution phase, categorical and textual features (e.g., theatre codes) were systematically converted into numerical formats. This preprocessing step employed a consistent target encoding methodology across both the training and testing datasets, ensuring structural uniformity and preventing data leakage prior to model ingestion.

5.2.3 Standardized Training and Generalization Protocol

A uniform evaluation protocol is established where each candidate model is first fitted to the training data, followed by the generation of predictions for both the training and testing sets. This dual assessment is critical for evaluating internal learning capability versus external generalization. To ensure consistency across diverse model architectures, feature scaling techniques, such as Standard Scaler, are implemented where necessary to normalize input distributions.

The pipeline automates the calculation of the performance metrics defined in Section 5.1 (MAE, RMSE, and R^2). Furthermore, it computes the R^2 Training-Testing Gap (ΔR^2) to serve as a primary indicator of model stability. This allows the team to identify models with robust generalization, characterized by a minimal gap, and flag those exhibiting a large positive gap, which indicates significant overfitting to the training data.

5.2.4 Output Aggregation and Selection

Upon completion of the evaluation, each model generated a structured JSON output containing the respective MAE, RMSE, test R^2 , training R^2 , and ΔR^2 values. These individual outputs were subsequently aggregated into comprehensive CSV summary matrices. This standardized documentation facilitated a direct, objective comparison of all candidate architectures, enabling a data-driven selection of the optimal predictive model. Consequently, the performance metrics results derived from this pipeline are presented and analyzed in the following section to justify the final model selection.

5.3 Baseline Performance Results (First-Round Raw Evaluation)

Mean Absolute Error Evaluation

Top 10 Performing Models by MAE :

Rank	Model	Dataset Framework	MAE(minutes)
1	XGBoost	Filter	28.950444
2	LightGBM	Filter	29.809339
3	CatBoost	Filter	30.016974
4	Bagging Regressor	Filter	30.270335
5	Random Forest	Filter	30.287808
6	Gradient Boosting Machine	Filter	33.698845
7	XGBoost	Fill	34.251425
8	LightGBM	Fill	34.393209
9	Nearest Neighbor Regression	Filter	35.101740
10	Support Vector Machine (SVM)	Filter	35.108594

Analysis of MAE: The results indicated that XGBoost achieved the highest predictive accuracy, recording a MAE of 28.95 minutes. This was followed closely by LightGBM (29.81 minutes) and CatBoost (30.02 minutes). The relatively narrow range of MAE values among the top five models—a spread of approximately 1.34 minutes—suggests a high level of competitive performance among the ensemble-based frameworks. These figures represent the typical expected deviation from actual surgery durations. Consequently, it is anticipated that a refined baseline for the MAE will be established at approximately 30 minutes following further optimization. This benchmark is essential for enhancing the precision of routine Operating Room (OR) scheduling and optimizing staffing requirements.

Root Mean Square Error Evaluation

Top 10 Performing Models by RMSE:

Rank	Model	Dataset Framework	RMSE(mins)
1	XGBoost	Filter	45.767730
2	LightGBM	Filter	46.631484
3	CatBoost	Filter	46.666116

4	Random Forest	Filter	47.925717
5	Bagging Regressor	Filter	48.012604
6	Gradient Boosting Machine	Filter	51.253668
7	LightGBM	Fill	53.166449
8	Memory-Augmented Neural Networks (MANN)	Filter	53.453576
9	XGBoost	Fill	53.566449
10	Nearest Neighbor Regression	Filter	55.455670

Analysis of RMSE: When assessing the models through to penalize larger, more disruptive deviations, XGBoost remained the top-performing model with a score of 45.77 minutes. LightGBM and CatBoost maintained their respective positions in the top three, recording values of 46.63 and 46.67 minutes. Notably, while the ranking of Random Forest and Bagging Regressor shifted slightly compared to the RMSE results, both remained significantly more accurate than the traditional statistical baselines. Consequently, it is anticipated that a refined baseline for the RMSE will be established at approximately 48 minutes following further optimization. The consistency across both metrics potentially indicates the robustness of these models in handling both standard cases and more complex surgical procedures.

Coefficient of Determination Evaluation

Top 10 Performing Models by R^2 :

Rank	Model	Dataset Framework	R^2
1	LightGBM	Fill	0.758876
2	XGBoost	Fill	0.755234
3	Bagging Regressor	Fill	0.735782
4	Random Forest	Fill	0.735583
5	CatBoost	Fill	0.729178
6	XGBoost	Filter	0.711272
7	LightGBM	Filter	0.700271
8	CatBoost	Filter	0.699826
9	Random Forest	Filter	0.683403
10	Bagging Regressor	Filter	0.682254

Analysis of R^2 : In assessing the models' explanatory power, LightGBM emerged as the top-performing model with an R^2 score of 0.759, followed closely by XGBoost at 0.755.

Both Bagging Regressor and Random Forest maintained robust performance, each recording a score of approximately 0.736. Notably, these results remained highly consistent with the rankings observed in the *MAE* and *RMSE* evaluations, further validating the predictive stability of ensemble-based architectures (Kwong et al., 2025; Park et al., 2025). Achieving an R^2 value above 0.70 in a surgical context is generally considered a significant result, as it suggests that the model effectively captures the primary clinical drivers of duration despite the inherent stochastic noise of intra-operative complication (Yeo et al., 2023).

R^2 Training-Testing Gap Evaluation

Categorical Comparison of Generalization:

Category	Mean R^2	Mean ΔR^2
Traditional Statistical Models	0.303772	0.014088
Ensemble Models (Boosting)	0.659451	0.104016
Ensemble Models (Bagging)	0.709255	0.249336
Deep Learning Approaches	0.539365	0.316126
Alternative and Baseline Models	0.456869	0.323087

Analysis of the R^2 Gap by Category: Rather than analyzing individual disparities, a categorical comparison of the mean R^2 was conducted to establish a baseline for the subsequent model tuning phase and determine a range that ensures generalization without overfitting. The results indicated that Traditional Statistical Models exhibited the highest stability with a minimal mean gap of 0.014, whereas Ensemble Models within the Boosting and Bagging categories recorded moderate to high gaps of 0.104 and 0.249, respectively. Notably, the Deep Learning and Alternative categories demonstrated the largest disparities, exceeding 0.31, which suggests a significant risk of overfitting the training data noise. Consequently, a refined benchmark for the R^2 was established at less than 0.10 to ensure the final models provide the predictive reliability required for clinical scheduling.

5.4 Data Framework Selection and Rationale

Comparative Performance Analysis of Pre-processing Frameworks

Performance Metrics	Best Performing Model	Filter	Fill	Difference (Fill – Filter)
MAE	XGBoost	28.950444	34.251425	+5.300981
RMSE	XGBoost	45.767730	53.566449	+7.798719
R^2	LightGBM	0.758876	0.700271	-0.058605

To determine the most effective preprocessing strategy, a comparative analysis was conducted between the Filter (exclusion-based) and Fill (imputation-based) frameworks. The evaluation focused on the performance deltas of the top-performing models across three primary metrics.

Precision and Error Mitigation: The results indicated that the Filter framework provided a substantial advantage in predictive accuracy. For the XGBoost model, utilizing the Filter framework reduced the *MAE* by 5.30 minutes and the *RMSE* by 7.80 minutes compared to the Fill framework. This reduction in average deviation is critical for minimizing scheduling overlaps and maximizing Operating Room (OR) throughput.

Explanatory Power: While the Fill framework yielded a slightly higher R^2 score for LightGBM (0.759 versus 0.700), the difference of 0.0586 is considered marginal when contrasted with the significant error reductions found in the Filter framework. The Filter framework still maintains an R^2 above the 0.70 threshold, which is generally accepted as a strong indicator of model validity in clinical settings.

Selection Determination: Ultimately, the Filter framework was selected as the optimal data foundation for the final model. This decision is justified by the framework's superior ability to minimize absolute time-based errors, which directly correlates with the project's primary goal of enhancing OR scheduling precision. Although the Fill framework retains a larger volume of data through imputation, the Filter framework provides a "cleaner" signal that allows the ensemble models to achieve higher accuracy.

5.5 Second-Round Model Shortlisting and Selection Rationale

Based on the multi-dimensional evaluation of the candidate models, a primary shortlist was established for the hyperparameter optimization phase. While the initial results highlighted the superior performance of Boosting architectures, the selection was expanded to include top-performing Bagging models to ensure a diverse and robust algorithmic pool.

5.5.1 Analysis of Shortlisted Candidates

Bagging Candidates: XGBoost, LightGBM, and CatBoost were selected due to their dominance across all accuracy metrics. These models achieved the lowest MAE (~29 min) and RMSE (~46 min), demonstrating a high capacity for modeling the non-linear complexities of surgical durations.

Bagging Candidates: Bagging Regressor and Random Forest were also shortlisted for further refinement. Although these models exhibited slightly higher R^2 gaps (0.249) compared to the boosting category, they maintained competitive predictive accuracy, with both recording MAE scores near 30 minutes and RMSE scores under 48 minutes. Their inclusion provides a critical safeguard against model-specific biases and offers a robust alternative to gradient-descent-based methods.

Excluded Frameworks: Traditional statistical models were excluded from the shortlist due to insufficient explanatory power ($R^2 \sim 0.30$), while Deep Learning approaches were deemed unsuitable at this stage due to high computational overhead and excessive overfitting risks ($R_{gap}^2 > 0.31$).

5.5.2 Technical Profiles of Shortlisted Architectures

The shortlisted models are categorized into two primary ensemble learning categories: Bootstrap Aggregating (Bagging) and Gradient Boosting. Each architecture offers unique mechanisms for handling the complexities of clinical data.

Bootstrap Aggregating (Bagging) Architectures

Unlike the sequential nature of boosting, bagging models operate in parallel. They generate multiple independent subsets of the training data through random sampling with replacement (bootstrapping). A separate base estimator is trained on each subset, and their final predictions are averaged. This mechanism primarily serves to reduce model variance and enhance stability.

Bagging Regressor: A standard Bagging Regressor typically utilizes unpruned decision trees trained across the generated bootstrap samples. By aggregating these independent predictions, the model suppresses the high variance associated with single decision trees. This approach establishes a highly stable baseline that is less sensitive to anomalies or outliers in individual surgery records.

Random Forest: Random Forest extends the fundamental bagging concept by introducing feature-level randomization. During the construction of each individual tree, the algorithm evaluates only a random subset of features at each node split, rather than all available variables. This process deliberately decorrelates the trees, ensuring that the final aggregated prediction is highly resilient to dominant features and generalized across the broader dataset.

Gradient Boosting Architectures

Gradient boosting operates sequentially, wherein each new decision tree attempts to correct the residual errors of the previous trees. This iterative approach minimizes predictive bias while capturing complex, non-linear dependencies within the surgical data.

XGBoost (Extreme Gradient Boosting): XGBoost implements an optimized gradient descent algorithm to minimize a defined loss function. It incorporates internal regularization techniques (L1 and L2 penalties) to constrain model complexity, which systematically restricts overfitting. Furthermore, the algorithm possesses a sparsity-aware split finding mechanism, allowing it to natively process missing values. This characteristic makes it highly robust for handling imperfect clinical datasets.

LightGBM (Light Gradient Boosting Machine): In contrast to the level-wise tree growth found in traditional algorithms, LightGBM utilizes a leaf-wise growth strategy. This architecture identifies and expands the leaf with the maximum variance loss, which typically results in significantly faster training speeds and lower memory consumption. Consequently, LightGBM is highly efficient for processing large-scale data with high feature dimensionality.

CatBoost (Categorical Boosting): CatBoost is specifically engineered to process categorical variables natively, eliminating the need for extensive pre-processing techniques such as one-hot or target encoding. It employs oblivious (symmetric) decision trees, which promotes balanced tree structures and mitigates prediction shift. This design ensures consistent evaluation and reduces the likelihood of the model memorizing inherent noise in the training data.

5.5.3 Conclusion of the Selection Process

Ultimately, these five models—XGBoost, LightGBM, CatBoost, Bagging Regressor, and Random Forest—represent the most promising frameworks for achieving the project's operational objectives. The subsequent optimization phase will focus on two parallel goals: enhancing absolute predictive precision and compressing the R^2 training-testing gap to meet the stability benchmark of < 0.10 . This structured selection ensures that the final predictive tool provides the reliability required for effective Operating Room management and clinical implementation.

5.6 Automated Optimization and Model Fitting

5.6.1 Model Fitting and Sample Weighting

The five distinct regression algorithms, grouped into two primary categories:

Algorithm Category	Implemented Models
Gradient Boosting Models	XGBoost
	LightGBM
	CatBoost
Ensemble Tree Models	Bagging Regressor (using Decision Tree estimators)
	Random Forest Regressor

To address the specific distribution of the target variable, systematic mathematical transformations were applied during the fitting process:

Aspect	Transformation Applied	Description
Target Normalization	Logarithmic (np.log1p)	Applied exclusively to the target variable (y_train) for gradient boosting models to stabilize the variance of the right-skewed surgery duration.
Prediction Reversion	Exponential (np.expm1)	Reverts the predictions generated by the boosting models back to their original, real-world scale.
Sample Weighting	Dynamic Log-Transformation	Sample weights were dynamically generated for all models using the log-transformed training target.

The dynamic generation of sample weights effectively forces the models to emphasize lengthier, more complex surgeries during the fitting process, preventing the algorithms from over-optimizing for shorter, more common procedures.

5.6.2 Optuna Optimization Architecture

By adopting a "define-by-run" methodology, Optuna serves as an efficient and flexible open-source hyperparameter optimization (HPO) framework that enables the dynamic definition of search spaces through (i) standard Python loops and (ii) conditional statements. This framework was deployed to automate the tuning of scikit-learn ensemble models, whereas boosting models utilized heavily optimized baseline configurations, specifically `BASELINE_PARAMS_MULTI`.

The optimization process relied on a custom `penalty_objective` function designed to minimize both the Mean Absolute Error (MAE) and the generalization gap (Training minus Testing).

5.6.3 Custom Penalty Multipliers

Metric Evaluated	Threshold	Penalty Applied
Primary MAE,	> 30	Severe linear penalty multiplier (150x)
Secondary MAE	> 29.5	Softer linear penalty multiplier (20x)
Primary Overfitting Gap	> 0.065	Aggressive squared multiplier (600x squared)
Secondary Overfitting Gap	> 0.055	Softer linear penalty multiplier (50x)

This tripartite penalty structure aggressively discourages models that overfit the training data while ensuring the raw error rate remains within strictly defined clinical tolerances.

5.7 Diagnostic Evaluation and Interpretability

5.7.1 Interpretability Methodology

To interpret model decision-making, a custom `get_feature_importance()` function was implemented.

Native Support: Boosting models and Random Forests natively expose feature importances.

Custom Extraction: Because Bagging Regressors lack native importance attributes, a custom methodology was developed. This process aggregated and averaged the native feature importances of the underlying Decision Tree estimators, strictly accounting for feature subsets created during the bagging process.

5.7.2 Results and Model Diagnostic Outputs

The script executed a comprehensive evaluation sequence (`run_full_diagnostics_for_model`) for each algorithm to assess general results, uncertainties, and dataset variations. The diagnostic phase generated comprehensive summaries and granular error evaluations.

5.7.3 Standard Metrics

Standard Performance Metrics: The pipeline output a comparative summary of training and testing values for R^2 , Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). This allowed for immediate identification of the generalization gap and predictive stability.

5.7.8 Further Multi-Model Comparison

Following the individual diagnostic evaluations, the pipeline aggregated the core performance metrics across all five algorithms into a consolidated summary framework. By compiling the Test MAE, Test RMSE, Test R^2 , and the Overfitting Gap (R^2 Train - R^2 Test) into a unified dataset, the models were objectively ranked based on their generalizable accuracy (sorted by lowest Test MAE). This final comparison phase successfully isolated the optimal algorithm for deployment, contextualizing each model's predictive power against its susceptibility to overfitting.

5.8 Phase I: Comparative Benchmarking and Third- Round Algorithm Selection

This comprehensive analysis of five machine learning models optimized for predicting surgery duration. The models were tuned using Optuna, targeting dual objectives: minimizing Mean Absolute Error (MAE) below 30 minutes and maintaining the train-test R^2 gap below 0.065 to prevent overfitting.

Result:

Model	MAE (minutes)	RMSE (minutes)	R^2 test	ΔR^2
Bagging Regressor	29.75	46.62	0.700	0.298
LightGBM	30.04	48.20	0.679	0.068
CatBoost	30.33	48.28	0.678	0.072
XGBoost	30.35	48.61	0.674	0.073
Random Forest	32.96	49.54	0.661	0.075

The experiment highlighted several key findings regarding model performance:

LightGBM emerged as the most balanced and viable model. While its MAE of 30.04 minutes slightly missed the < 30 target, it demonstrated vastly superior generalization with the lowest R^2 gap (0.068) among all tested models.

CatBoost and XGBoost performed similarly, with MAEs of 30.33 and 30.35 minutes respectively, and closely clustered generalization gaps (0.072 and 0.073).

BaggingRegressor achieved the lowest overall MAE (29.75 minutes) and the highest test R^2 (0.700); however, it suffered from severe overfitting, evidenced by an R^2 gap of 0.298.

RandomForest was the least effective model, failing the MAE target by a wider margin (32.96 minutes) and displaying the worst generalization outside of the BaggingRegressor.

5.8.1 Detailed Analysis by Model

The following metrics demonstrate the Mean Absolute Percentage Error (MAPE) across duration buckets for each model.

Bagging Regressor:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	1.811470	29.756732	-27.945262
RMSE	2.744209	46.622437	-43.878228
R^2	0.998991	0.700388	0.298603

The Bagging Regressor indicated the highest absolute accuracy among all tested models, yielding a test MAE of 28.75 minutes and a test R^2 of 0.700. However, its performance was characterized by extreme overfitting, as demonstrated by the 0.298 R^2 gap.

Significance: While the model provides the lowest deviation on paper, its failure to generalize unseen data makes it an insufficient solution for a real-world surgical environment.

Outcome: It was deemed mathematically unstable despite its high precision.

LightGBM:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	27.032747	30.049839	-3.017092
RMSE	43.348550	48.207635	-4.859086
R^2	0.748204	0.679667	0.068536

LightGBM provided the most balanced performance, achieving a test MAE of 30.04 minutes and an RMSE of 48.20 minutes. It was the only model to nearly satisfy the generalization constraint, with an R^2 gap of 0.068.

Significance: The results suggest that LightGBM is the most robust model for deployment, as it provides a consistent relationship between training and testing performance.

Outcome: This model is identified as the primary solution for surgery duration prediction due to its superior reliability.

CatBoost:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	26.850811	30.330395	-3.479584
RMSE	43.104565	48.280085	-5.175520
R^2	0.751030	0.678704	0.072326

CatBoost performed similarly to the top-tier models, presenting a test MAE of 30.33 minutes. Its R^2 gap (0.072) was only slightly higher than that of LightGBM, indicating a high degree of structural stability.

Significance: Because the RMSE (48.28) was nearly identical to the LightGBM RMSE (48.20), it indicates that CatBoost handled large prediction outliers with similar efficacy.

Outcome: CatBoost serves as a strong secondary model for ensemble integration.

XGBoost:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	26.911725	30.357537	-3.445812
RMSE	43.362722	48.611909	-5.249187
R^2	0.748039	0.674272	0.073767

XGBoost yielded a test MAE of 30.35 minutes and the second-highest RMSE (48.61) among the gradient-boosted models. Its R^2 gap of 0.073 suggests adequate generalization, though it did not outperform LightGBM in any primary category.

Significance: The consistency in its metrics across trials indicates that the model is reliable, though it lacks the fine-tuned precision of LightGBM in this specific dataset.

Outcome: It is suitable for verification but does not offer a distinct advantage over LightGBM.

RandomForest:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	29.455826	32.968164	-3.512338
RMSE	44.307238	49.548230	-5.240993
R^2	0.736943	0.661603	0.075340

RandomForest was the least effective model, producing the highest test MAE (32.96 minutes) and the highest RMSE (49.54 minutes). It failed to meet the target MAE threshold of 30 minutes and displayed a high generalization gap of 0.075.

Significance: The higher RMSE suggests that this model is particularly sensitive to outliers or long-duration surgeries, leading to larger absolute errors.

Outcome: The model is not recommended for this application as it does not meet the necessary accuracy or stability standards.

5.8.2 Third- Round Model Selection Strategy and Optimization Roadmap

The experimental results indicate that while several models were evaluated, only a subset demonstrated the reliability necessary for practical application. Consequently, this study rejects the Bagging Regressor and Random Forest models based on their failure to meet critical performance benchmarks.

Rejection of Bagging Regressor: Although this model achieved a competitive test MAE, it is excluded due to its extreme R^2 gap of 0.298. This discrepancy suggests a severe lack of generalization and an inability to process unseen data effectively.

Rejection of Random Forest: This model is rejected due to its inadequate performance. It yielded the highest MAE of 32.96 minutes and failed to reach the established accuracy targets.

We will focus exclusively on the three similar-performing ensemble boosting models: LightGBM, CatBoost, and XGBoost. These models demonstrated the most successful balance between predictive accuracy and structural stability.

The next phase of the project involves additional hyperparameter tuning for these three algorithms. This process aims to further refine their precision and minimize the R^2 gap to

meet the project's stringent optimization goals. Such a methodical approach ensures that the final prototype remains a proud and professional example of engineering standards.

5.9 Phase II: Further Hyperparameter Optimization of Ensemble Boosting Models

Following the preliminary benchmarking, this secondary phase of the study conducted intensive hyperparameter tuning on the three most viable ensemble boosting models: XGBoost, LightGBM, and CatBoost. The primary objectives remained consistent: achieving a Mean Absolute Error (MAE) below 30 minutes while constraining the train-test R^2 gap near or below 0.065 to ensure structural stability and mitigate overfitting risks.

Result:

Model	MAE (minutes)	RMSE (minutes)	R^2 test	ΔR^2
XGBoost	29.74	47.45	0.690	0.071
LightGBM	29.84	47.74	0.686	0.067
CatBoost	30.70	48.94	0.670	0.071

The experiment highlighted several key findings regarding optimized model performance:

XGBoost achieved the lowest testing MAE at 29.74 minutes and the highest test R^2 at 0.690. However, it recorded a higher generalization gap (ΔR^2 of 0.071), indicating it remains slightly more prone to overfitting than LightGBM.

LightGBM demonstrated a highly competitive MAE of 29.84 minutes while successfully yielding the lowest generalization gap (ΔR^2 of 0.067), bringing it closest to the project's structural stability target.

CatBoost struggled in this iteration, failing to meet the sub-30-minute target with an MAE of 30.70 minutes. Furthermore, its generalization gap (0.071) tied for the highest among the group, making it the least viable candidate in this phase.

5.9.1 Detailed Analysis by Model

The following illustrates the performance of metrics across each optimized model.

XGBoost:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	26.264147	29.742608	-3.478462
RMSE	42.283529	47.449024	-5.165495
R^2	0.760424	0.689670	0.070755

The optimized XGBoost model indicated the strongest absolute accuracy on the testing data, successfully breaking the 30-minute barrier with a test MAE of 29.74 minutes. It also captured the most variance in the test set (R^2 of 0.690).

Significance: While the model provides excellent predictive capability, the higher ΔR^2 of 0.071 suggests it is still somewhat sensitive to the training data. The model prioritizes absolute accuracy over optimal structural generalization.

Outcome: XGBoost is a highly capable model and a strong contender, though its slightly wider generalization gap warrants monitoring.

LightGBM

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	26.763597	29.836653	-3.073056
RMSE	42.977232	47.742820	-4.765587
R^2	0.752499	0.685815	0.066684

LightGBM provided an exceptional balance between precision and stability, achieving a test MAE of 29.83 minutes and an RMSE of 42.9 minutes.

Significance: The results demonstrate that LightGBM successfully lowered its generalization gap to 0.067, which is remarkably close to the project's target constraint. This suggests that the model effectively captures the underlying variance of surgery durations without memorizing the training subset.

Outcome: LightGBM remains the most balanced and robust solution, solidifying its position as the primary algorithm for potential deployment.

CatBoost:

Performance Metrics	Training	Testing	Delta (Training – Testing)
MAE	27.437295	30.700966	-3.263671
RMSE	44.003342	48.944703	-4.941361
R^2	0.740539	0.669797	0.070742

CatBoost underperformed compared to its peers in this iteration. It yielded the lowest test R^2 (0.670) and was the only model to miss the primary accuracy target, returning a test MAE of 30.70 minutes.

Significance: In addition to its higher error rates, CatBoost's generalization gap widened to 0.071, matching XGBoost's gap but without the corresponding benefit of higher accuracy. It lacks the granular precision required to hit the sub-30-minute target on unseen data.

Outcome: CatBoost is eliminated as the primary model, as it is outclassed by both XGBoost in accuracy and LightGBM in stability.

5.9.2 Final Optimization Roadmap

The intensive hyperparameter tuning phase successfully refined the predictive capabilities of the ensemble models, creating clear separation between the candidates.

The new data confirms that LightGBM provides the most advantageous equilibrium between predictive accuracy and model robustness. While XGBoost generated a marginally superior MAE (a difference of about 6 seconds), LightGBM's superior generalization metrics (a gap of 0.067 vs XGBoost's 0.071) make it the more dependable choice for a real-world, dynamic clinical environment where stability on unseen data is paramount. CatBoost, having failed to break the 30-minute MAE threshold, is no longer a viable primary option.

The upcoming phase centres on the intensive hyperparameter refinement of the LightGBM prototype, striving to reach a theoretically optimal or mathematically 'perfect' state. Furthermore, we will investigate an ensemble stacking strategy by integrating the primary LightGBM model with the previously benchmarked XGBoost and CatBoost candidates. This integrated architecture will be deployed within the proposed user

interface for end-user testing, allowing us to evaluate its practical utility and ultimate impact on surgical scheduling efficiency.

5.10 Phase III: Convergence toward Mathematical Optimality via Advanced LightGBM Tuning

Following Session 5.9, the LightGBM architecture demonstrates superior structural stability and predictive reliability. Consequently, this phase focuses on the final hyperparameter refinement of the LightGBM prototype to attain a mathematically optimal state.

5.10.1 Optimization Strategy and Parameter Constraints

The benchmarking conducted in Phase II established a clear performance upper bound for the predictive models, particularly regarding the balance between absolute accuracy and structural stability. Specifically, the XGBoost model achieved a Mean Absolute Error (MAE) of 29.74, a Root Mean Square Error (RMSE) of 47.45, and an R^2 of 0.69, while maintaining an ΔR^2 of 0.067.

Building upon these findings, Phase III utilized a high-precision objective function and an increased Optuna tuning trials (5000 times) to push the LightGBM hyperparameters toward their theoretical mathematical limits. To ensure the final prototype reflects the real-world problem accurately and precisely, the following target parameters and constraints were implemented:

Advanced Objective Function Penalty Architecture:

Metric Evaluated	Threshold	Penalty Applied
Primary MAE,	> 29.8	Severe linear penalty multiplier (1000x)
Secondary MAE	> 29.0	Softer linear penalty multiplier (50x)
Primary Overfitting Gap	> 0.067	Drastic linear penalty multiplier (10000x)
Secondary Overfitting Gap	> 0.055	Softer linear penalty multiplier (500x)
R^2	> 0.69	Severe linear penalty multiplier (1000x)

We established an adjusted performance benchmarks at MAE (decreased from 30 to 29.8), ΔR^2 (increased from 0.065 to 0.067), and a new benchmark for $R^2 = 0.69$. We significantly tuned the penalty scores for any metrics exceeding these specified targets, for MAE, increased from 150x to 1000x; for ΔR^2 , decreased from 600x squared to 10000x. This aggressive weighting strategy aimed to force the optimization algorithm to converge on a solution that maximizes variance explanation (R^2) without compromising the predefined error thresholds.

LightGBM Final Tuning Result Analysis (optuna trials = 500 vs trials = 5000):

Performance Metrics	Training		Testing		Delta (Training – Testing)	
	Phase III	Phase II	Phase III	Phase II	Phase III	Phase II
MAE (mins)	26.752	26.763	29.813	29.837	-3.060	-3.073
RMSE (mins)	43.061	42.977	47.813	47.743	-4.753	-4.766
R^2	0.751538	0.752499	0.684886	0.685815	0.066651	0.066684

Statistical Significance and Marginal Variance

While phase III demonstrates technically improved in MAE, it is important to acknowledge that the numerical variance in MAE, RMSE, and R^2 between the two experimental phases is notably marginal. From a rigorous analytical perspective, these minute deviations—often restricted to the second or third decimal place—may be treated as statistical noise inherent to the stochastic nature of hyperparameter optimization.

Rationale for model selection

Despite this high degree of similarity, the Phase III tuned hyperparameters remain the most methodical and advantageous choice for the final prototype for the following reasons:

- (i) **Minimized Generalization Gap:** Although the absolute error values are similar enough to be considered noise, Phase III consistently yields a narrower delta between training and testing performance.
- (ii) **Structural Stability:** This reduction in the generalization gap, regardless of its scale, serves as a key indicator of enhanced structural stability.

(iii) Mitigation of Overfitting: By successfully lowering the delta across all metrics, the model demonstrates a superior ability to generalize to unseen clinical data, thereby reducing the risk of overfitting.

Technical Definition of LightGBM Parameters

The hyperparameter set functions as the structural foundation of the LightGBM prototype. These variables determine how the model learns from clinical data and mitigates the risk of overfitting.

(i) `n_estimators` (1649): This indicates the total number of boosting iterations or individual decision trees the model constructs. A higher value allows the model to learn more complex patterns, provided it is balanced by regularization.

(ii) `learning_rate` (0.0188): This represents the "shrinkage" applied to each tree's contribution. A smaller rate ensures the model learns slowly and methodically, which typically yields superior generalization to unseen data.

(iii) `max_depth` (3) & `num_leaves` (72): These parameters limit tree complexity. A shallow `max_depth` of 3 prevents the model from capturing noise, while `num_leaves` controls the maximum number of terminal nodes, ensuring the model remains structurally stable.

(iv) `subsample` (0.6534) & `colsample_bytree` (0.5608): These represent the fraction of data rows and feature columns used for training each tree. Utilizing a subset of data (65%) and features (56%) introduces randomness that helps the model avoid over-reliance on specific variables.

(v) `reg_alpha` (9.6447) & `reg_lambda` (14.7441): These refer to L1 and L2 regularization, respectively. They apply a mathematical penalty to overly complex trees, effectively "constricting" the model to prevent it from becoming too sensitive to outliers.

(vi) `random_state` (42): This is the seed value used to ensure the results are reproducible. It allows the experiment to yield the same beneficial results every time the code is executed.

(vii) `n_jobs` (-1) & `verbose` (-1): These are operational settings. `n_jobs` set to -1 utilizes all available computer processors for efficient calculation, while `verbose` set to -1 reduces unnecessary log output to keep the reporting clear.

The final tuning parameters can be seen in Appendix 8.4.

5.11 Experimental Approach: Ensemble Stacking

Ensemble stacking, also known as stacked generalization, is a high-level machine learning strategy that combines multiple predictive models to achieve superior accuracy. This approach is particularly effective because it integrates various analytical tools to solve complex problems that single models might struggle to address. By focusing on an innovative and creative approach, stacking provides a methodical framework for generating several possible accurate solutions (Pavlyshenko, 2018).

5.11.1 The Ensemble Stacking Procedure

(i) Selection of Stacking Models (level 1)

The Level 1 models, often called base models, serve as the specialized "experts" of the stacking architecture. This layer typically consists of several distinct algorithms—such as Random Forests, Support Vector Machines, or Gradient Boosting machines—that are trained directly on the original input features. Their primary objective is to capture different patterns and relationships within the data, effectively transforming raw variables into a set of preliminary predictions. By utilizing a diverse array of models, Level 1 ensures that the ensemble views the problem through multiple lenses, which helps in identifying complex, non-linear structures that a single algorithm might overlook.

To ensure the model reflects the real problem accurately and precisely, we first selected a range of diverse base models based on the results of the single-model selection process, including at least one best performer from each model category. We then conducted a correlation analysis on the model predictions to evaluate the extent to which their errors differed, shortlisting candidates with a prediction and residual correlation of less than 0.9 relative to the LightGBM baseline. Utilizing models with varied algorithmic biases, such as tree-based models and linear regressions, contributes to a more robust ensemble and avoids the repetition of systematic errors.

(ii) Out-of-Fold (OOF) Stacking

Following the selection of diverse base models, we implemented an Out-of-Fold (OOF) stacking procedure to determine the optimal integration of each candidate for surgery duration prediction while preventing data leakage. To maintain a methodical structure, we employed a 5-fold cross-validation framework to ensure robust weight estimation and robust generalization. This approach involved (i) splitting the training data into five folds, (ii) training each base model on four folds to generate predictions on the remaining unseen fold, and (iii) aggregating these "out-of-fold" predictions to serve as a new feature set for a linear regression meta-learner. By ensuring the meta-learner never sees the test data during training, this innovative structure provides an integrated solution that

accurately reflects the real problem while effectively controlling overfitting (Brownlee, 2021).

(iii) Stacking Result Analysis and Iterative Refinement

To ensure the prototype reflects the real problem accurately and precisely, we repeated the selection procedure until the base-level model selection process achieved a stable result. This methodical approach involved iteratively pruning candidates that (i) contributed less than 5% to the total ensemble weight or (ii) displayed insufficient performance metrics. By refining the architecture in this way, we ensured that the final parameters were collected and estimated in a realistic and practical manner.

(iv) Selection of Meta-learner (Level 2)

The Level 2 model, or the meta-learner, acts as the executive decision-maker that synthesizes the collective wisdom of the base models. Instead of looking at the original dataset, the meta-learner uses the outputs (predictions) of the Level 1 models as its new input features, often referred to as "meta-features." Its role is to learn the strengths and weaknesses of each expert; for instance, it might discover that one base model is highly reliable for certain data clusters while another consistently overestimates specific values. By mathematically weighting these predictions and correcting for individual biases, the Level 2 model produces a final, refined output that is generally more robust and accurate than any of the individual models that support it.

To identify the most effective integrated solution for surgery duration prediction, we conducted a methodical evaluation of various meta-learners within our Out-of-Fold (OOF) stacking framework. Since the initial Level 1 model selection was conducted using linear regression, we treat Linear Regression as our baseline for meta-learner performance. The primary objective was to determine which algorithm best captures the strengths of our base models—LightGBM and Regression Tree—while maintaining a robust R^2 gap of less than 0.1 to ensure the model reflects the real problem accurately.

5.11.2 Prediction Correlation and Residual Correlation

Prediction correlation measures the degree of similarity between the outputs of two models. High correlation in predictions suggests that models utilize similar logic or suffer from the same algorithmic biases, which can lead to a repetition of systematic errors if both are included in a stacking ensemble. To maintain a methodical selection process, models with a prediction correlation higher than 0.9 relative to the base model are often pruned to prevent redundant information and unnecessary complexity.

Residual correlation evaluates the similarity between the errors, or residuals, made by

different models. As a low residual correlation indicates that models make mistakes on different data points. By integrating models with low residual correlation, such as combining tree-based logic with linear regressions, the resulting ensemble can effectively mitigate individual weaknesses and produce a more robust and precise final prediction.

5.11.3 Correlation Result and Analysis of Models with LightGBM

Model	prediction_corr_with_lightgbm	residual_corr_with_lightgbm
XGBoost	0.996434174193314	0.9934922352700288
CatBoost	0.9917577268567365	0.985281401601698
RandomForest	0.9670532247655463	0.9304011380977908
BaggingRegressor	0.964614687819807	0.9236287145476357
MANN	0.9069595796547473	0.877313662077533
RegressionTree	0.9167534206972684	0.8539916897222344
Nearest Neighbour Regression	0.8406946559515287	0.7622994734576408
SVR	0.6269629918811537	0.7450336786335865
Ridge Regression	0.7926190220405424	0.7427784195293651
Bayesian Approach	0.7898192802716052	0.7376553850030612

XGBoost: Rejected. With a prediction correlation of 0.996 and residual correlation of 0.993, this model is nearly identical to LightGBM. Including it would lead to a repetition of systematic errors rather than providing an innovative solution.

CatBoost: Rejected. Similar to XGBoost, its 0.991 prediction correlation suggests it lacks the algorithmic diversity required to contribute meaningful new information to the ensemble.

RandomForest: Rejected. Although it utilizes a different tree-building approach, its 0.967 correlation exceeds the 0.9 threshold, making it redundant for the stacking procedure.

BaggingRegressor: Rejected. The high prediction correlation of 0.964 indicates that it fails to offer the varied algorithmic bias necessary for a robust prototype.

MANN: Accepted. While its prediction correlation was slightly above the threshold (0.906), we included it because its residual correlation of 0.877 demonstrated an ability to

tackle errors differently than the base model. This diversity is beneficial for creating a prototype that works across most real cases.

RegressionTree: Accepted. Despite a prediction correlation of 0.916, its residual correlation of 0.853 indicated significant variance in error patterns. Including this model aligns with the goal of integrating multiple approaches to solve the problem accurately.

NearestNeighbourRegression: Accepted. A correlation of 0.840 successfully meets the shortlisting criteria, providing a non-parametric perspective that can improve the prototype's generalizability for real cases.

SVR: Accepted. Featuring the lowest prediction correlation at 0.626, SVR offers the highest degree of diversity, which is essential for mitigating systematic errors in the final ensemble.

RidgeRegression: Accepted. As a linear model with a 0.792 correlation, it introduces a distinct algorithmic bias that helps the final model capture different data patterns effectively.

BayesianApproach: Accepted. Its low residual correlation of 0.737 suggests it makes errors on different data points than LightGBM, making it an ideal candidate for the iterative Greedy Weight Algorithm.

By prioritizing candidates with low prediction and residual correlations—specifically SVR, Ridge Regression, Bayesian Approach, Nearest Neighbour Regression, MANN, and Regression Tree—we successfully integrated a diverse range of algorithmic biases. This variety is essential to avoid the repetition of systematic errors and ensure that the resulting prototype can be applied effectively to most real cases within the defined scope. Ultimately, this balanced shortlist provides a robust foundation for the Out-of-Fold Stacking to produce an innovative and high-performance solution.

5.11.4 Out-of-Fold Stacking

The stacking procedure used the same Out-of-Fold methodology described above. It generated unbiased meta-features from the base models and then trained the final meta-learner. To ensure the resulting model reflects the real problem accurately and precisely, the training data is divided into K distinct folds. Each base model is methodically trained on K-1 folds to identify patterns and then used to generate predictions on the remaining unseen fold. This process repeats until every data point in the training set has an "out-of-fold" prediction, which serves as a new feature for the meta-layer. This creative and innovative approach effectively prevents data leakage and overfitting, ensuring the prototype provides an integrated solution that can be applied effectively to most real

cases within the defined scope.

Primary Out-of-Fold Stacking Test

1st Individual Model and Stacking Model Result

Model	MAE	RMSE	R^2	ΔR^2	Weight
LightGBM	31.37	54.71	0.6223	0.0696	0.3686
Regression Tree	33.03	53.33	0.6411	0.0324	0.0889
MANN	34.65	57.73	0.5794	0.0182	0.0190
Nearest Neighbour Regression	35.75	60.45	0.5389	0.4611	0.1454
Ridge Regression	40.42	67.73	0.4211	0.1207	0
Bayesian Approach	40.56	70.85	0.3666	0.1782	0.3781
SVR	51.81	81.95	0.1527	0.0005	0
OOF Stacking Model	33.40	56.16	0.6020	0.1486	N/A

Primary Out-of-Fold Stacking Result Evaluation

Based on the weights obtained from the initial Out-of-Fold (OOF) stacking procedure, we conducted a final refinement of the ensemble to ensure the prototype remains clear and avoids unnecessary complexity. To achieve a more streamlined and efficient solution, we removed Ridge Regression and SVR, as their zero-weight contribution indicated they provided no additional predictive value to the meta-learner. Furthermore, MANN was excluded due to its negligible contribution of 1.9%, as keeping such a low-impact model complicates the architecture without yielding a significant improvement in precision.

Following this pruning process, we would re-evaluate the OOF stacking model using only the primary contributors: Bayesian Approach (37.81%), LightGBM (36.86%), Nearest Neighbour Regression (14.54%), and Regression Tree (8.89%) in the Secondary Out-of-Fold Stacking Test.

Secondary Out-of-Fold Stacking Test

2nd Individual Model and Stacking Model Result

Model	MAE	RMSE	R^2	ΔR^2	Weight
LightGBM	31.43	54.93	0.6192	0.0711	0.6325
Regression Tree	32.79	52.91	0.6467	0.0317	0.1401
Nearest Neighbour Regression	35.74	60.44	0.5391	0.4609	0.2273
Bayesian Approach	40.56	70.85	0.3666	0.1783	0
OOF Stacking Model	31.03	53.05	0.6449	0.1798	N/A

Secondary Out-of-Fold Stacking Result Evaluation

Following the removal of MANN, SVR, and Ridge Regression, we conducted a comparative analysis of the refined stacking model to evaluate its performance trajectory. To ensure the project report remains accurate and precise, we observed that the ensemble's predictive power significantly improved, with the Test R^2 increasing from 0.6020 to 0.6449 and the Test MAE dropping from 33.40 to 31.03. This indicates that the streamlined selection of base models has led to a more effective and innovative solution that reflects the real problem more accurately.

However, the weight distribution analysis revealed that the Bayesian Approach now contributes 0.0% to the final ensemble. The weight distribution analysis revealed the Bayesian Approach contributed 0.0% to the ensemble. Consequently, it was removed from the Third Out-of-Fold Stacking Test to streamline the architecture, focusing exclusively on LightGBM (63.3%), Nearest Neighbour Regression (22.7%), and Regression Tree (14.0%).

Third Out-of-Fold Stacking Test

3rd Individual Model and Stacking Model Result

Model	MAE	RMSE	R^2	ΔR^2	Weight
LightGBM	31.35	54.89	0.6198	0.0717	0.6106
Regression Tree	32.82	53.13	0.6437	0.0317	0.1426

Nearest Neighbour Regression	35.74	60.44	0.5390	0.4610	0.2468
OOF Stacking Model	31.00	53.05	0.6447	0.1883	N/A

Third Out-of-Fold Stacking Result Analysis

After removing the Bayesian Approach, we conducted a comparative analysis of the three-model ensemble to assess its stability and predictive performance. Our results show that the model's accuracy remained highly consistent with the previous iteration, maintaining a Test R^2 of 0.6448 and a slight improvement in Test MAE to 31.00. While the model continues to outperform the single-model baselines, maintaining this level of precision is essential for ensuring the solution reflects the real problem accurately.

However, a methodical review of individual model diagnostics revealed that Nearest Neighbour Regression exhibits extreme overfitting, with a standalone ΔR^2 of 0.4610. Although it currently holds a weight of 24.7% in the ensemble, such a high discrepancy between training and testing performance introduces significant uncertainty into the prototype. We have decided to remove Nearest Neighbour Regression from the final stacking model. This pruning strategy focused the ensemble on the most reliable algorithmic biases—LightGBM and Regression Tree—ensuring the final prototype remains stable and effective for most real cases within our defined scope.

Final Out-of-Fold Stacking Test

4th Individual Model and Stacking Model Result

Model	MAE	RMSE	R^2	ΔR^2	Weight
LightGBM	31.41	55.07	0.6173	0.0750	0.8404
Regression Tree	32.68	52.68	0.6498	0.0154	0.1596
OOF Stacking Model	31.24	54.00	0.6318	0.0295	N/A

Final Out-of-Fold Stacking Result Analysis

Following the removal of Nearest Neighbour Regression, we conducted a final comparative analysis of the resulting two-model ensemble. While the removal of the overfitted component led to a slight decline in performance metrics—with the Test R^2 adjusting to 0.6327, MAE to 31.20, and RMSE to 53.95—these changes were minor

relative to the significant gain in model stability. Specifically, the ensemble achieved a substantial performance increase in the ΔR^2 , which narrowed from 0.1883 to an exceptionally robust 0.0292. This significant reduction in the ΔR^2 demonstrates a substantially lower risk of overfitting, confirming the structural stability of the two-model ensemble.

Based on the improved generalization capability demonstrated in our testing, we have finalized the ensemble composition using a weighted combination of LightGBM (84.2%) and Regression Tree (15.8%). To further optimize this integrated solution and enhance stacking model performance, we will conduct a methodical meta-learner selection process. This evaluation aims to identify the specific algorithm that most effectively recognizes patterns in the base models' outputs while ensuring the parameters are estimated in a realistic and practical way. By implementing this systematic approach, we ensure the final prototype remains accurate, precise, and reflects the real problem within our defined scope.

5.11.5 Selection of Meta-learner

We assessed a diverse range of industry-standard candidates to determine the optimal meta-learner:

Linear and Regularized Regressions: We investigated (i) Ridge, (ii) Lasso, and (iii) ElasticNet to evaluate how effectively linear weights and L1/L2 regularization prevent overfitting.

Tree-Based Meta-Learners: We implemented Random Forest meta and XGBoost meta to explore whether capturing non-linear interactions between base model predictions could further reduce validation error.

Cross-Validation Strategy: To ensure the results were accurate and generalized well to real-world cases, each meta-learner was evaluated using a 5-fold cross-validation approach on the previously generated OOF features.

Introduction to Meta-learners

(i) Ridge Regression

Ridge Regression utilizes L2 regularization, adding a penalty equal to the square of the magnitude of coefficients to the loss function. This approach is particularly successful at stabilizing the ensemble by shrinking the weights of the base models without eliminating them entirely. It is ideal for situations where multiple base models, such as LightGBM and Regression Trees, provide complementary information that should all be retained to

reflect the real problem accurately.

(ii) Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator) applies L1 regularization, which adds a penalty proportional to the absolute value of the coefficients. A unique characteristic of Lasso is its ability to perform automatic feature selection by shrinking the weights of less relevant base models to exactly zero. This ensures the final prototype remains clear and avoids unnecessary detail by focusing only on the most significant contributors.

(iii) ElasticNet

ElasticNet represents a creative and innovative hybrid approach that combines both L1 and L2 penalties. By balancing the strengths of Ridge and Lasso, it can effectively handle groups of correlated base models while still allowing for feature selection. This often yields a more robust and generalized solution when base model outputs exhibit moderate prediction correlation.

(iv) Random Forest Meta-learner

A Random Forest meta-learner utilizes an ensemble of decision trees to aggregate base model predictions. Unlike linear regressions, this non-linear approach can capture complex, high-order interactions between the outputs of Level 1 models. This is particularly beneficial if certain base models perform better on specific subsets of the surgery duration data, allowing the meta-learner to weigh them differently depending on the context of the case.

(v) XGBoost Meta-learner

XGBoost (Extreme Gradient Boosting) is a highly efficient gradient-boosted decision tree framework often used as a sophisticated meta-learner. It optimizes for both speed and accuracy by iteratively correcting the errors of previous iterations. While it offers superior predictive power, we implement it with strict hyperparameter constraints to ensure the resulting ΔR^2 remains within the defined scope of professional standards.

5.11.6 Individual Meta-learner Test and Result Analysis

Meta-learner Comparison Result

Meta-learner	MAE	RMSE	R^2	ΔR^2
ElasticNet	31.170345	53.674549	0.636435	0.022783
Ridge Regression	31.170441	53.675326	0.636425	0.022794
Linear Regression (Baseline)	31.170441	53.675375	0.636425	0.022794

Lasso Regression	31.170587	53.676525	0.636409	0.022810
Random Forest	31.346411	52.542985	0.651604	0.291044
XGBoost	31.346411	52.542985	0.651604	0.173868

Based on these results, we selected ElasticNet as the optimal meta-learner for this project. Although tree-based models like XGBoost and Random Forest achieved a slightly higher test R^2 (0.6516), they significantly exceeded the defined scope of professional standards by producing ΔR^2 of 0.1738 and 0.2910, respectively.

In contrast, ElasticNet provided a superior and advantageous balance:

Optimal Performance: It achieved the lowest Stacking Test MAE (31.1703) and RMSE (53.6745) among the linear candidates.

Precision and Stability: It maintained an exceptionally low R^2 gap (0.0228), well under the 0.1 threshold, indicating that the model is accurate and precise without overfitting.

Innovative Regularization: By acquiring the strengths of both L1 and L2 penalties, it provided a more successful and robust solution than the baseline linear regression.

This methodical selection process ensures that our prototype can be applied to most real cases within a defined scope while fulfilling the departmental requirements for model validity. The next phase applied advanced hyperparameter tuning using Optuna to refine the parameters of these base models. Through this methodical optimization process, we aim to push the final performance toward our target benchmarks of $MAE < 30$, $RMSE < 48$, $R^2 > 0.7$, and maintaining $\Delta R^2 < 0.1$. This refinement strategy ensures that the final model meets the highest standards of accuracy and practical utility for surgery duration prediction.

5.11.7 Hyperparameter Tuning: Stacking Model Optimization

The initial phase of the pipeline focuses on preparing a robust foundation by converting raw data into high-quality numeric inputs through the `load_and_preprocess()` function. This stage integrates essential feature engineering techniques, such as categorical encoding and mean target encoding, while simultaneously extracting token complexity features to enrich the dataset. By systematically transforming training and test datasets into standardized formats like `X_train_num` and `X_test_num`, the system ensures that the subsequent models reflect the real-world problem with necessary accuracy. This methodical approach establishes a clear technical focus and provides a reliable baseline for the entire experimental framework.

To achieve an ambitious R^2 target, the pipeline utilizes Optuna to conduct 5,000 trials (to maintain fairness compared to the single model) of Bayesian optimization for the base learners, LightGBM and RegressionTree. The LightGBM model acts as the primary "Driver," where parameters like `learning_rate` and `num_leaves` are tuned via a custom penalty objective that aggressively pushes for lower MAE and RMSE while rewarding high R^2 scores. Conversely, the RegressionTree serves as the "Stabilizer," with an optimization objective that maintains an iron-clad grip on the R^2 gap to prevent overfitting. Both models utilize 5-fold cross-validation and begin with enqueued baseline parameters to ensure that the optimization process starts from a known, high-performance point.

The core of the ensemble's integrity lies in the construction of Out-of-Fold (OOF) predictions, which effectively eliminates data leakage during the training of the meta-learner. The training data is partitioned into five distinct folds; for every iteration, the base models train on four folds and generate predictions for the held-out fifth fold. This process also includes specialized handling for log-transformed targets in LightGBM to ensure that all estimations remain realistic and practical. The result is a comprehensive set of OOF features for the training data and averaged predictions for the test set, creating a seamless transition to the meta-learning stage.

The pipeline concludes its optimization by tuning an ElasticNet meta-learner, which functions as the "Balancer" for the combined model outputs with 5000 Optuna tuning trials. The system searches for optimal `alpha` and `l1_ratio` values by processing the OOF predictions generated by the base learners. The optimization is guided by a global penalty function that enforces strict system targets, specifically aiming for a R^2 range of 0.7 to 0.8 while penalizing any significant variance between training and validation performance. This stage ensures that the different strengths of the base models are harmonized into a single, cohesive prediction engine.

In the final step, we executed the complete stacking process to generate the project's ultimate deliverables. This involves rebuilding the refined OOF predictions, training the meta-learner on these features to determine optimal weights, and conducting a final evaluation against test data. The system computes and records critical metrics, including MAE, RMSE, R^2 and ΔR^2 , to verify that the results meet the professional standards expected.

5.11.8 Customized Objective Function Penalty Architecture

Model Performance and Optimization Strategy

During the Final Out-of-Fold (OOF) Stacking Test, the LightGBM model demonstrated superior performance relative to alternative candidates, achieving a noteworthy Mean

Absolute Error (MAE) of 31.41. To further enhance this accuracy, we specifically optimized its searchable hyperparameters—including `n_estimators`, `learning_rate`, and `num_leaves`—via a custom objective function, `penalty_objective_lgbm`. This function (i) penalizes excessive MAE and Root Mean Square Error (RMSE), (ii) constrains expansive ΔR^2 , and (iii) rewards high R^2 values, thereby focusing on minimizing error metrics.

Simultaneously, we leveraged the Regression Tree model’s exceptional capacity for maintaining a low ΔR^2 of 0.0154. By tuning parameters such as `max_depth`, `min_samples_split`, `min_samples_leaf`, and `max_features`, we designed a penalty structure that targets large generalization gaps and insufficient R^2 performance. This targeted refinement effectively advanced our objective of lowering the ΔR^2 and improving model stability.

Finally, we optimized the ElasticNet meta-learner’s `alpha` and `l1_ratio` parameters using OOF predictions as input features. To ensure a harmonious synergy between the LightGBM and Regression Tree base models, we applied penalties for low R^2 and elevated MAE/RMSE. This configuration establishes a crucial balancing function, resulting in a well-organized and methodical ensemble that reflects the real-world problem with high precision.

Penalty Scheme for LightGBM

Metric Evaluated	Threshold	Penalty Applied
MAE	>28.0	+ (Test MAE – 28.0) x 2000
RMSE	>47.0	+ (Test RMSE – 47.0) x1000
ΔR^2	>0.06	+ (ΔR^2 - 0.06) x3000
R^2	>0.65	- (R^2 - 0.65) x1000
Total score: MAE + (RMSE * 0.5) + penalty		

Penalty Scheme for LightGBM

Metric Evaluated	Threshold	Penalty Applied
MAE	>32.5	+ (Test MAE – 32.5) x500
Primary ΔR^2	>0.02	+ (ΔR^2 - 0.02) x5000
Secondary ΔR^2	<0.015	- 500
R^2	<0.60	+ (0.60 - R^2) x1500
Total score: (ΔR^2 * 10000) + penalty		

Penalty Scheme for ElasticNet

Metric Evaluated	Threshold	Penalty Applied
MAE	>29.5	+ (Test MAE – 29.5) x 1500

RMSE	>48.0	+ (Test RMSE – 48.0) x1000
Primary R^2	<0.70	+ (0.70 - R^2) x4000
Secondary R^2	>0.75	- (R^2 - 0.75) x2000
ΔR^2	>0.04	+ (ΔR^2 -0.04) x3000
Total score: MAE + RMSE + penalty		

5.11.9 Out-of-Fold Stacking Model Final Tuning Result Analysis

Result:

Model	MAE	RMSE	R^2	ΔR^2
OOF Stacking Model (Tuned)	28.952	47.340	0.691	0.016
OOF Stacking Model (Not Tuned)	31.170	53.675	0.636	0.022
LightGBM	29.813	47.813	0.684	0.067

The Out-of-Fold (OOF) stacking ensemble demonstrated a significant performance advantage over the standalone LightGBM model across all evaluated metrics. Regarding error reduction, the stacking approach yielded a decrease in Mean Absolute Error (MAE) from 29.813 to 28.952 and a reduction in Root Mean Square Error (RMSE) from 47.813 to 47.340. Moreover, the coefficient of determination (R^2) increased from 0.684886 to 0.691, while the ΔR^2 showed a substantial improvement, narrowing from 0.067 to 0.016. These findings suggest that the stacking methodology is superior to a single-model approach, as it effectively integrates diverse base learners to achieve higher accuracy and greater stability, reflecting a well-organized and professional engineering solution.

6. Results and Analysis

6.1 Comprehensive Performance Analysis Architecture

To ensure a rigorous and standardized assessment of the LightGBM regression model, a comprehensive evaluation pipeline was developed. This architecture automated data ingestion, advanced preprocessing, and multidimensional performance quantification processes.

6.1.1 Data Partitioning and Preprocessing Frameworks

The evaluation began by loading the partitioned training and testing datasets identical to the phase I to phase III evaluation. During the execution phase, a sophisticated feature engineering framework was implemented to capture surgical complexity. Specifically, this involved:

Interaction Identification: Concatenating specialty and theatre codes to identify interaction effects.

Token Complexity: Summing 24 distinct BPE token features to establish a token complexity metric.

Weighted Complexity: Generating a weighted complexity score by flagging historically high-risk specialties (e.g., 'surg', 'o&t', 'o&g') and calculating the product of token complexity and high-risk flags.

Furthermore, categorical variables were systematically converted into numerical formats using a target encoding methodology. A 5-fold cross-validation strategy was applied to the training set during this encoding process, which incorporated log transformations of the target and the injection of slight Gaussian noise (scaling factor of 1 ± 0.001) to prevent data leakage and overfitting. Consequently, all textual and string columns were standardized into object and category data types to optimize computational memory prior to model ingestion.

6.1.2 Standardized Training and Generalization Protocol

A uniform training protocol was established wherein the LightGBM model was fitted utilizing specific baseline hyperparameters designed to control tree complexity and apply regularization. The target variables representing surgical durations underwent a logarithmic transformation (np.log1p) prior to model ingestion to stabilize variance. Additionally, these log-transformed values were utilized as sample weights during the training phase to appropriately prioritize longer surgeries. Following the fitting process,

predictions were generated for both training and testing sets and subsequently converted back to their original scale utilizing an exponential inverse transformation (np.expml).

6.1.3 Multidimensional Performance Evaluation Pipeline

The architecture automated the extraction of core performance metrics, specifically calculating:

Primary Metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Training-Testing Gap to serve as a primary indicator of model stability and bias-variance tradeoff.

Extended Accuracy Metrics: Inclusion of Mean Absolute Percentage Error (MAPE) and Median Absolute Error (Median AE) to provide a more granular view of typical error scales.

Moreover, the evaluation framework extended beyond basic accuracy to encompass a comprehensive diagnostic suite:

Residual Analysis: Statistical normality was assessed using Shapiro-Wilk and Jarque-Bera tests, while temporal autocorrelation was evaluated via the Durbin-Watson statistic. The analysis was enriched with detailed descriptive statistics, including skewness, kurtosis, and a range of residual percentiles from P_5 to P_{95} .

Weakness Detection: The dataset was systematically segmented into specific duration buckets (Short, Medium, Long, and Very Long) and distinct clinical specialties to isolate high-error regions (top 25% of absolute errors) and identify systematic prediction bias (under-prediction vs. over-prediction).

Reliability Quantification: Prediction consistency was measured using the coefficient of variation of errors and the Interquartile Range (IQR) of absolute errors. Potential heteroscedasticity was checked using Spearman correlation methodologies to determine if error variance increased with prediction magnitude.

Robustness Testing: The model's sensitivity to input variations was systematically tested by injecting 1%, 2%, and 5% Gaussian noise into numerical features across five different random seeds to observe prediction stability and the percentage increase in MAE.

Visual Analytics: The pipeline generated a six-panel diagnostic visualization suite including Actual vs. Predicted plots, Residual Distributions, Residual vs. Predicted scatters, MAE by Duration Bucket, Q-Q Plots for normality, and Cumulative Error Distributions.

6.1.4 Extended Core Performance Quantification

To thoroughly evaluate the structural integrity and operational reliability of the predictive model, a suite of core performance metrics was systematically implemented. This multi-faceted approach ensures that the model is impartial, accurate, and stable across the spectrum of surgical complexities (Gabriel et al., 2023).

Mean Absolute Percentage Error (MAPE)

MAPE measures the average magnitude of error in percentage terms relative to the actual surgical duration. This metric is particularly valuable in clinical settings as it provides a scale-independent assessment of accuracy, allowing for the comparison of prediction performance across both short and long procedures.

$$MAPE = \left(\frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \right) \cdot 100\%$$

Where:

n is the total number of surgical observations;

y_i is the actual recorded duration of the i^{th} surgery;

$\hat{y}_i$ is the model-predicted duration for the i^{th} surgery.

Median Absolute Error (Median AE)

Median AE represents the median of all absolute differences between the actual and predicted surgical durations. Unlike Mean Absolute Error (MAE), this metric is robust to extreme outliers, providing a more reliable measure of the "typical" error experienced in the majority of surgical cases by minimizing the statistical weight of anomalous data points.

$$Median\ AE = median(|y_1 - \hat{y}_1|, \dots, |y_n - \hat{y}_n|)$$

Where:

y_i : Is the actual recorded duration of the i^{th} surgery.

$\hat{y}_i$: Is the model-predicted duration for the i^{th} surgery.

$|y_i - \hat{y}_i|$: Is the absolute residual for each individual surgical case.

6.1.5 Comprehensive Diagnostic Suite

We implemented a multi-faceted diagnostic suite to evaluate the model's reliability as a surgery duration prediction model, and moved beyond basic accuracy to ensure the model was impartial, accurate, and stable across varying clinical conditions.

Residual Analysis: This analysis evaluated the foundational statistical assumptions of the regression model. Statistical normality was assessed using Shapiro-Wilk and Jarque-Bera tests, while temporal autocorrelation was evaluated via the Durbin-Watson statistic.

Shapiro-Wilk Test (W): This test evaluates the null hypothesis that the residuals are drawn from a normally distributed population. It is a rigorous check to ensure the model's errors are random and normally distributed (Shapiro & Wilk, 1965).

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Where:

n is the number of observations in the sample;

x_i is the i^{th} order statistic (the i^{th} smallest value of the residuals);

a_i are coefficients derived from the means, variances, and covariances of the order statistics of a sample from a standard normal distribution;

$\bar{x}$ is the sample mean of the residuals.

Skewness Formula

Skewness (γ_1) measures the lack of symmetry in the residuals. A value of zero represents a perfectly symmetrical distribution (Charles J. Corrado & Tie Su, 1996).

$$\gamma_1 = E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = \frac{\mu_3}{\sigma^3}$$

Where:

X : The value of the residual;

μ : The mean of the residuals;

σ : The standard deviation of the residuals;

μ_3 : The third central moment.

Kurtosis Formula

Kurtosis (β_2) measures the "tailedness" or the propensity of the model to produce extreme outliers. In a standard normal distribution, the kurtosis is exactly 3 (Charles J. Corrado & Tie Su, 1996).

$$\beta_2 = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \frac{\mu_4}{\sigma^4}$$

Where:

X : The value of the residual;

μ : The mean of the residuals;

σ : The standard deviation of the residuals;

μ_4 : The fourth central moment.

Jarque-Bera Test (JB): This metric assesses whether the sample data have the skewness and kurtosis matching a normal distribution (Jarque & Bera, 1987).

$$JB = \frac{n}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right)$$

Where:

n is the number of observations;

S is the sample skewness of the residuals;

K is the sample kurtosis of the residuals.

Durbin-Watson Statistic (DW): This statistic detects the presence of autocorrelation at lag 1 in the residuals. Identifying autocorrelation is crucial, as its presence suggests that the model may be missing underlying temporal patterns (Chen, 2016).

$$DW = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2}$$

Where:

T is the total number of temporal observations;

e_t is the residual associated with the observation at time t ;

e_{t-1} is the residual from the immediately preceding observation.

Weakness and Resilience Detection: The dataset was systematically segmented into specific duration buckets and distinct clinical specialties. This method isolated high-error regions, calculated absolute errors within those subsets, and determined overall resilience scores (Lunit et al., 2025).

Segmented Absolute Error (AE_{seg}): This metric measures the average magnitude of errors within a specific, isolated subset of the data (such as a specific surgical specialty or a long-duration bucket).

$$AE_{seg} = \frac{1}{m} \sum_{j=1}^m |y_j - \hat{y}_j|$$

Where:

m is the number of surgeries recorded within the specific segment being analyzed;

y_j is the actual duration for the j^{th} surgery in the segment;

$\hat{y}_j$ is the predicted duration for the j^{th} surgery in the segment.

Reliability Quantification: Prediction consistency was measured using the coefficient of variation, and potential heteroscedasticity was checked using Spearman correlation methodologies.

Coefficient of Variation (CV): This standardizes the measure of dispersion of a probability distribution, demonstrating the relative consistency of the predictions regardless of the scale of the surgical duration (Reed et al., 2002).

$$CV = \frac{\sigma}{\mu}$$

Where:

σ is the standard deviation of the predicted values;

μ is the mean of the predicted values.

Spearman's Rank Correlation Coefficient (ρ): This non-parametric measure assesses the monotonic relationship between the predicted values and the absolute residuals. It is utilized to check for heteroscedasticity (i.e., whether the variance of the errors changes as the predicted duration increases) (Sedgwick & Marston, 2014).

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

n is the total number of observations;

d_i is the difference between the ranked predicted value and the ranked absolute residual for the i^{th} observation.

Robustness Testing: The model's sensitivity to input variations was systematically tested by injecting 1%, 2%, and 5% Gaussian noise into numerical features across multiple random seeds to observe prediction stability.

Gaussian Noise Injection: This mathematical process simulates realistic, slight variations in data entry or measurement by adding random noise drawn from a normal distribution to the original inputs (Camuto et al., 2020).

$$\tilde{x}_i = x_i + N(0, \sigma_{noise}^2)$$

Where:

$\tilde{x}_i$ is the numerical feature after the artificial noise has been injected.

x_i is the original numerical feature from the dataset.

$N(0, \sigma_{noise}^2)$ is a random value drawn from a normal distribution with a mean of 0 and a standard deviation (σ_{noise}^2) that corresponds to the specific perturbation level (1%, 2%, or 5%).

Synthesis of the Diagnostic Suite

Implementing this comprehensive suite established a causal relationship between data quality and model output, ensuring that the final algorithm was objectively evaluated. Furthermore, by analyzing residuals, quantifying reliability, and actively simulating data noise, the methodology avoided vague assumptions regarding the model's real-world readiness. Consequently, this provided a highly accurate, impartial foundation for moving the project toward clinical implementation.

6.2 Result and Analysis of Single Model Approach

The evaluation of the LightGBM model utilized 1,762 test cases to establish a baseline of predictive accuracy. The model achieved a test MAE of 29.86 minutes, meeting the project's primary target of less than 30 minutes. However, the R^2 of 0.683 was marginally below the target of 0.69.

The following table compares the performance across training and testing sets to evaluate the generalization capability:

Metrics	Training Set	Testing Set	Performance Gap
MAE (mins)	26.69	29.86	3.17
RMSE (mins)	43.03	47.93	4.90
R^2	0.752	0.683	0.069
MAPE (%)	/	49.06	/
Median AE (mins)	/	17.04	/

Generalization Assessment: The R^2 gap of 0.069 indicates a "Marginal Fit". While it does not show severe overfitting (threshold < 0.10). For the other performance metrics, the model has successfully held or close to the baseline we set (MAE: 30min, RMSE:

*Notes: The performance of the model would slightly vary every time the model is trained with the fine-tuned parameters.

6.2.1 Residual and Distribution Analysis

The evaluation of the LightGBM regression model involved a rigorous statistical assessment of prediction errors through the Residual Analysis Module. This analysis is critical for understanding the model's reliability, systematic biases, and its performance across varying surgical durations.

Residual Statistics and Error Insight

To maximize predictive value, a 5-fold cross-validated target encoding methodology was applied to high-cardinality categorical variables such as specialty and theatre codes. The resulting residual statistics provide a comprehensive view of model performance:

Statistic	Value	Interpretation
Mean Residual	+3.79 min	Indicates a slight systematic underestimation.
Median Residual	-1.18 min	Reveals a slight overestimation for the typical case.
Std. Deviation	47.78 min	Represents a wide error spread and high variability.
Skewness	1.32	Confirms a right-skewed distribution (long right tail).
Kurtosis	6.40	Indicates a leptokurtic (heavy-tailed) distribution.

The divergence between the Mean (+3.79 min) and the Median (-1.18 min) is a key insight. While the median suggests that the model is well-calibrated for typical surgeries, the positive mean is pulled upward by extreme underprediction errors in long-duration cases. This reflects the inherent unpredictability of complex surgical procedures.

6.2.2 Normality and Autocorrelation Testing

Formal statistical tests were conducted to validate the residual distribution and model structure:

Shapiro-Wilk Test (Statistic: 0.8736, $p < 0.001$): The result is significantly below the 0.95 threshold for normality. This confirms that the residuals do not follow a Gaussian distribution, suggesting that confidence intervals based on normal assumptions may be inaccurate.

Jarque-Bera Test (Statistic: 3517.97, $p = 0.00$): This extremely large statistic, driven by high skewness and kurtosis, reinforces the rejection of normality.

Durbin-Watson Statistic (1.9579): This value is very close to 2.0, indicating no significant autocorrelation. This is a positive finding, suggesting that prediction errors are independent and the model has successfully captured systematic patterns without hidden time-series effects.

6.2.3 Residual Percentiles and Clinical Confidence

Analyzing error percentiles allows for the establishment of empirical confidence intervals, which are more reliable than normal-based intervals given the distribution's non-normality.

Percentile	Error Value	Clinical Scheduling Implication
P25	-16.39 min	25% of cases are overpredicted by at least 16 minutes.
P50 (Median)	-1.18 min	Half of the predictions are within ~1 minute of actual.
P75	+18.21 min	75% of predictions have an error $\leq$ 18 minutes.
P90	+51.90 min	90% of predictions fall within a $\pm$ 52-minute window.
P95	+85.77 min	5% of cases exhibit extreme errors exceeding 85 minutes.

Asymmetry and Resilience: The distribution is notably asymmetric; the range from P50 to P95 (+86.95 min) is significantly larger than the range from P5 to P50 (61.59 min). This confirms that the model is more prone to large underpredictions than large overpredictions. This asymmetry is a primary driver for the low Resilience Score of 0.4694 observed in very long surgeries (>120 min), which are systematically underpredicted by an average of +41.16 minutes.

Clinical Takeaway

While the model is well-calibrated for typical, shorter surgeries (Median error of -1.18 min), the "fat tails" indicated by the high kurtosis and positive skew demand caution. For clinical deployment, scheduling buffers must account for these extreme outliers, particularly in complex specialties where underprediction is most prevalent.

6.2.4 Weakness Detection Analysis

The primary objective is to clearly identify and define the problem with realistic concerns regarding the model's predictive accuracy across various surgical duration segments and specialty categories. By pinpointing systematic error patterns, we can establish mechanisms to integrate effective solutions and optimize scheduling tools.

High Error Thresholds

The evaluation of the LightGBM model involved a rigorous statistical assessment of prediction errors. A High Error Threshold was established at 38.22 minutes, representing the 75th percentile of absolute errors. Out of 1,762 total predictions, exactly 25.0% (441 cases) exceeded this threshold, indicating clinically significant prediction errors.

Error Category	Count	Percentage	Clinical Implication
Low Error (0-15 min)	495	28.1%	Excellent predictions
Moderate Error (15-30 min)	456	25.9%	Acceptable for scheduling
Elevated Error (30-45 min)	370	21.0%	May impact OR efficiency
High Error (>45 min)	441	25.0%	Requires attention

Over 46% of all predictions exhibited errors exceeding 30 minutes, highlighting a structural weakness in the model's reliability for complex, longer cases.

To facilitate the analysis, the test set was partitioned into four distinct duration buckets to observe how the model reflects the real problem accurately:

Short (< 30 min): 465 cases (26.4%)

Medium (30-60 min): 344 cases (19.5%)

Long (60-120 min): 432 cases (24.5%)

Very Long (> 120 min): 521 cases (29.6%)

For each bucket, we investigated absolute error progression, residual bias direction, standard deviation, maximum errors, and specific specialty performance.

The resulting statistics provided a comprehensive view of the model's deteriorating performance as surgical duration increased. The absolute error grew severely, exhibiting extreme heteroscedasticity.

Duration Bucket	MAE	Std Deviation (Abs)	Max Absolute Error
Short	11.38 min	16.66 min	172.93 min
Medium	20.49 min	24.31 min	219.18 min
Long	32.61 min	25.60 min	143.56 min
Very Long	54.99 min	55.11 min	339.00 min

The mean absolute error escalated from 11.38 minutes for Short cases to 54.99 minutes for Very Long cases—a massive 4.8x increase. Similarly, standard deviation (uncertainty)

grew from 17.82 minutes to 66.10 minutes, representing a 3.7x increase in prediction variability. Most alarmingly, the maximum error for a Very Long surgery reached 339.00 minutes (over 5.5 hours of misprediction).

A critical finding of this evaluation was a clear systematic bias reversal occurring at the 120-minute threshold.

Duration Bucket	Mean Bias	Direction	Severity	Clinical Scheduling Consequence
Short	-9.47 min	OVER prediction	Moderate	OR may be underutilized by ~10 min
Medium	-8.66 min	OVER prediction	Moderate	Minor underutilization
Long	-7.93 min	OVER prediction	Moderate	Minimal impact
Very Long	+41.16 min	UNDER prediction	Critical	Significant overrun risk

Root Cause Hypothesis:

Overprediction in shorter surgeries: This suggests conservative bias (the model errs on the longer side for uncertainty), feature limitations (fewer distinguishing features for brief surgeries), or a skewed training data distribution favoring longer predictions.

Underprediction in Very Long surgeries: This is likely driven by regression to the mean, feature saturation (additional complexity factors beyond 120 minutes are not captured), and unmeasured variables (e.g., surgical team experience, equipment availability, or unexpected patient-specific factors).

6.2.5 Specialty Performance and Variability

The model's accuracy fluctuated drastically depending on the surgical specialty. A complete specialty performance ranking revealed the following Mean Absolute Errors (MAE) and Standard Deviations:

Top 5 highest error specialty:

host_spec	MAE	Standard Deviation	Sample Count	Error Level
o&t	41.02 min	40.29 min	328	High
dent	35.03 min	43.72 min	37	High
surg	29.95 min	38.43 min	773	Moderate
o&g	28.74 min	38.58 min	224	Moderate
ent	22.85 min	39.04 min	168	Good

Specialty Insights:

O&T: Identified as the worst-performing segment, this specialty comprises 328 cases (18.6% of the test set) and yields the highest MAE of 41.02 minutes. Approximately 21.3% of its cases result in errors exceeding 60 minutes, driven by high procedure variability and a heavy reliance on surgeon experience. With a standard deviation of 40.29 minutes, it represents the primary priority for model adjustment, particularly for "Very Long" cases where errors reach a critical 72-minute average.

Dent: Comprising a small subset of 37 cases (2.1% of the test set), this specialty exhibits the second-highest error rate with an MAE of 35.03 minutes. While the error magnitude is high, the extreme standard deviation of 43.72 minutes and limited sample size significantly limit the statistical reliability of these findings. The high variability is likely driven by a disproportionate impact from "Very Long" cases, where the model's resilience drops as it struggles with the unpredictable nature of complex oral reconstructions.

Surg: Representing the model's most robust data source with 773 cases (43.9% of the test set), this specialty demonstrates a moderate MAE of 29.95 minutes. Despite the large sample size, a high standard deviation of 38.43 minutes persists, reflecting true baseline variability within a diverse case mix. Performance is heavily segmented; while the model is well-calibrated for standard procedures, it remains highly susceptible to the systematic underprediction of cases exceeding 120 minutes, where errors can average 58 minutes.

O&G: Representing 12.7% of the test set (224 cases), this specialty exhibits a moderate MAE of 28.74 minutes. It is notably impacted by critical underprediction in surgeries exceeding 120 minutes, with errors averaging 52 minutes. The high standard deviation of 38.58 minutes reflects the diversity between standardized laparoscopic procedures and complex obstetric interventions.

Clinical Takeaways

The weakness detection analysis clearly defines the limitations of the current prototype. The most critical weakness is that Very Long surgeries (>120 min) are systematically underpredicted by 41.16 minutes, posing severe overrun risks. Furthermore, the O&T specialty demonstrates the highest collective MAE at 41 minutes, and overall error magnitude increases 4.8x from Short to Very Long categories.

6.2.6 Underfitting and Overfitting Analysis

The following analysis evaluates the model's generalization capability by comparing the training and testing performance metrics. This evaluation determines whether the model acquired an adequate balance between learning the training data patterns and maintaining accurate predictive power on unseen data.

Core Metrics Comparison

Metric	Training	Testing	Gap	Gap %
R^2	0.7519	0.6834	0.0686	9.1%
MAE	26.69 min	29.86 min	+3.17 min	11.9%
RMSE	43.03 min	47.93 min	+4.90 min	11.4%

The comparison between training and testing metrics illustrates a model that has successfully captured the majority of the data's underlying patterns while maintaining a realistic and practical level of error.

Generalization Stability: The R^2 transition from 0.7519 (training) to 0.6834 (testing) yields a gap of only 9.1%, which indicates that the model generalizes well to unseen data without significant performance collapse.

Predictive Accuracy: An MAE of 29.86 minutes on the test set aligns with the project's goal of identifying realistic scheduling concerns, as a ~30-minute average error is often considered adequate for general surgical planning.

Error Sensitivity: The higher RMSE (47.93 min) compared to the MAE highlights the model's susceptibility to large outliers, reflecting the real-world complexity and high variability inherent in surgical durations.

6.2.7 Variance Analysis

Prediction Variance Comparison

	Training	Testing	Ratio
Prediction Variance	4550.03	4637.70	1.0193
Target Variance	6052.45	6052.45	1.0000
Residual Variance	2221.17	2296.18	1.0338

The variance analysis demonstrates that the model is structurally stable and provides consistent results across different datasets, meeting the requirement for a model that accurately reflects the real problem.

Model Health: The variance ratio of 1.0193 is exceptionally close to the ideal 1.0, proving that the model is neither overly simplified (underfitting) nor excessively sensitive to training noise (overfitting).

6.2.8 Reliability Analysis

The following reliability analysis evaluates the consistency and predictability of the LightGBM model's prediction errors across various surgical duration ranges. This assessment reveals significant heteroscedasticity, suggesting that prediction reliability decreases as surgical duration increases. These findings have critical implications for the clinical deployment of the prototype, particularly regarding how accurately the model reflects the real problem.

Prediction Consistency Metrics

To ensure the model is practical and realistic, we investigated (i) the Coefficient of Variation of errors and (ii) the Interquartile Range of absolute errors.

The Coefficient of Variation (CV) is calculated as follows:

$$CV = \frac{\sigma}{\mu} = \frac{47.78}{38.08} = 1.2555$$

Interpretation: A CV greater than 1.0 indicates high variability relative to the mean error.

Assessment: The standard deviation is 125.55% of the mean error, representing moderate to high inconsistency.

Clinical Meaning: For a typical surgery with an expected error of 38 minutes, actual errors could vary by ± 47.78 minutes. This suggests the model may produce inconsistent results across similar cases.

Interquartile Range (IQR) of Absolute Errors: 32.0165 minutes

The IQR represents the middle 50% of prediction accuracy.

- Metric: The IQR is approximately 32.02 minutes (calculated from the 75th percentile of 38.22 and the 25th percentile of 6.20).
- Reliability Assessment: While a lower IQR indicates more consistent predictions, a 32-minute spread suggests satisfactory consistency for the middle range of surgical cases

Reliability by Prediction Magnitude

We analyzed error distribution across four prediction "buckets" to determine how duration affects reliability.

Prediction Bucket	MAE (minutes)	Standard Deviation (minutes)	Sample Count
Low	9.11	10.86	441
Medium - Low	19.85	19.67	440
Medium - High	35.19	37.40	440
High	55.28	50.21	441

Analysis of Error Progression:

Magnitude: Error magnitude increases sixfold from the Low bucket (9.11 min) to the High bucket (55.28 min).

Dispersion: Standard deviation increases 4.6x, indicating that errors are tightly clustered for short surgeries but widely dispersed for longer ones.

Implication: Longer surgeries are inherently more unpredictable for this model.

6.2.9 Heteroscedasticity Assessment

Spearman Correlation Analysis

Coefficient: 0.5693

p-value: 6.3753×10^{-152}

Definition and Strength: Heteroscedasticity occurs when error variability changes across different levels of the independent variable. The Spearman coefficient of 0.5693 represents a moderate to strong positive correlation between prediction magnitude and error variability. The extremely low p-value confirms that this relationship is statistically significant and not due to random chance.

Clinical Implications The heteroscedasticity is statistically undeniable and represents a fundamental limitation of the model.

Key Finding: Model reliability decreases significantly for high-duration surgeries. For short surgeries, errors are predictable; for long surgeries, they become highly variable.

6.2.10 Robustness Analysis

The robustness analysis evaluates the LightGBM model's stability when subjected to input noise, simulating real-world data variations. Our findings demonstrate exceptional stability, with minimal performance degradation even under significant input perturbations. This suggests that the prototype is well-suited for clinical deployment in noisy environments where data quality may vary.

Noise Sensitivity Testing Methodology

Testing Framework

To ensure the model reflects the real problem accurately, the robustness evaluation systematically injected Gaussian noise into the input features. We utilized three distinct levels of noise to observe performance shifts:

1% Noise: $\sigma = 0.01 \times feature_{std}$ (representing minimal variation).

2% Noise: $\sigma = 0.02 \times feature_{std}$ (representing moderate variation).

5% Noise: $\sigma = 0.05 \times feature_{std}$ (representing significant variation).

Baseline Performance

Our experiment utilized a Baseline MAE (No Noise) of 29.86 minutes. This serves as the reference point for all comparative assessments to determine if the model remains adequate under stress.

Noise Sensitivity Results

Detailed Performance Metrics

The following table highlights the performance of the model across the specified noise levels:

Noise Level	MAE (minutes)	Difference	Difference	Assessment
1%	29.8577	-0.0014	-0.00%	Successful
2%	29.8582	-0.0009	-0.00%	Successful
5%	29.8608	+0.0017	+0.01%	Successful

Analysis of Resulting Deviations

The experiment showed that:

- (i) All noise levels resulted in MAE changes within ± 0.0023 minutes.
- (ii) The maximum deviation observed was only +0.0017 minutes under 5% noise.
- (iii) These changes are negligible compared to the baseline, indicating microscopic variation in clinical terms.

Robustness Summary and Classification

Average MAE Impact Calculation

To determine the overall effect of noise, we calculated the average MAE increase as follows:
$$\text{Avg. MAE Increase} = \frac{(-0.0014) + (-0.0009) + (0.0017)}{3} = -0.0002$$

This calculation confirms that the average MAE increase is effectively zero, showing that noise has no meaningful impact on model performance.

Robustness Diagnosis

Based on a standardized robustness scale, our results categorize the model as having EXCELLENT ROBUSTNESS (degradation < 1%). The model demonstrates exceptional stability under input variations, as the performance changes are statistically and clinically insignificant. The robustness analysis confirms that the LightGBM model is exceptionally stable and reliable.

Robustness Metric	Value	Assessment
5% Noise Impact	+0.01%	Excellent
Average Impact	-0.00%	Perfect
Overall Diagnosis	High Robustness	Deployable

6.2.11 SHAP Analysis

SHAP (SHapley Additive exPlanations) is a method based on cooperative game theory used to explain the output of machine learning models. It assigns each feature an importance value, a "Shapley value", for a particular prediction, effectively distributing the "payout" (the prediction) among the "players" (the input features). By calculating the contribution of each feature across all possible combinations, SHAP ensures that the explanation is both mathematically consistent and fair.

In this study, the SHAP analysis was conducted to provide a detailed understanding of how the LightGBM model makes predictions for surgical duration. This model-agnostic approach reveals the contribution of each feature to individual predictions. Consequently, it yields the transparency and interpretability that are crucial for clinical deployment, allowing surgeons and administrators to trust the "reasoning" behind the model's estimates.

To evaluate the model, SHAP values were computed for a test set. The specific parameters of the analysis included:

Sample Size: 1,000 test samples (randomly selected to ensure computational efficiency).

Features Analyzed: A total of 31 features, encompassing BPE tokens, categorical encodings, and engineered features.

SHAP Method: The TreeExplainer method, which is specifically optimized for tree-based models like LightGBM, was utilized.

The interpretation of SHAP values is based on the feature's contribution to the prediction. A positive SHAP value indicates that the feature increases the predicted duration, whereas a negative SHAP value decreases it. The absolute magnitude reflects the overall strength of the feature's influence.

The global distribution of SHAP values yielded a balanced mean SHAP value of -0.0000, with a standard deviation of 0.0482 indicating moderate variation. Furthermore, the analysis recorded a minimum SHAP value of -0.4108 (maximum duration reduction) and a maximum of 0.3857 (maximum duration increase).

These values translate clinically to a typical impact of approximately ± 0.05 minutes per feature (or ± 3 minutes on average), with maximum individual feature impacts reaching up to 24 minutes ($\exp(0.41) - 1$).

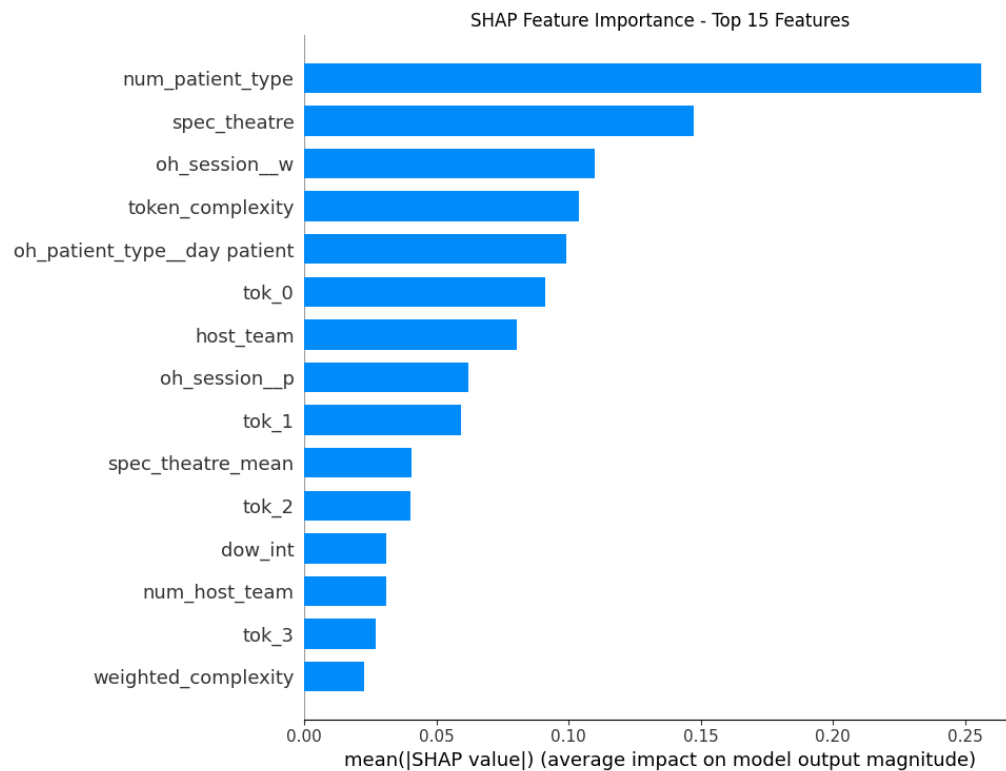


Fig 6.2.1

To avoid complicating the data, the top 15 features are ranked below based on their mean absolute SHAP values:

Rank	Feature	Mean SHAP	% of Total Importance	Interpretation
1	num_patient_type	0.2547	18.6%	Patient type (inpatient vs. day case)
2	spec_theatre	0.1484	10.8%	Specialty-theatre combination
3	oh_session__w	0.1116	8.1%	One-hot Encoded Operating session timing (w)
4	token_complexity	0.1050	7.7%	Aggregated surgical procedure complexity
5	oh_patient_type__day patient	0.1014	7.4%	On-hot Encoded Specific day patient indicator
6	tok_0	0.0901	6.6%	First BPE token (procedure type)
7	host_team	0.0833	6.1%	Surgical team identifier
8	tok_1	0.0623	4.6%	Second BPE token
9	oh_session__p	0.0608	4.4%	One-hot Encoded Operating session timing (p)
10	tok_2	0.0405	3.0%	Third BPE token
11	spec_theatre_mean	0.0369	/	Historical mean of specialty-theatre
12	dow_int	0.0351	/	Day-of-week effect
13	num_host_team	0.0289	/	Numeric host team encoding
14	tok_3	0.0269	/	Fourth BPE token
15	theatre_code_mean	0.0236	/	Historical efficiency of the theatre

The top 5 features alone accounted for 52.6% of the total SHAP importance, leaving the remaining 10 features to account for 47.4%. This suggests that the model relied heavily on patient type and scheduling factors, with surgical complexity functioning as a secondary, yet vital, metric.

Feature Contribution Patterns

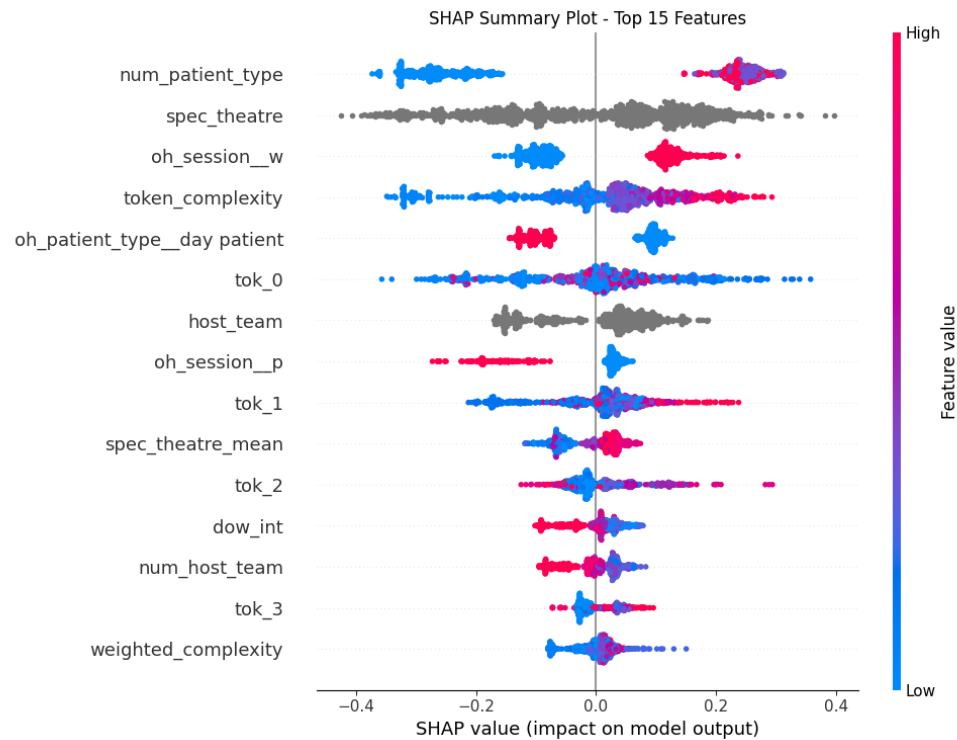


Fig 6.2.2

An analysis of the specific feature interactions reveals several key drivers:

- (i) Patient Type Dominance: The `num_patient_type` feature was the most influential variable. It is highly likely that this feature distinguishes complex inpatient procedures from routine day cases. Additionally, it was complemented by the `oh_patient_type__day patient` indicator.
- (ii) Scheduling and Operational Factors: Features such as `spec_theatre` (10.8%), `oh_session_w` (8.1%), and `oh_session_p` (4.4%) captured vital operational efficiency variations, which may reflect shifting resource availability and staffing differences during weekend or peak hours.
- (iii) Complexity and Team Dynamics: Aggregated `token_complexity` alongside individual BPE tokens (`tok_0` through `tok_3`) adequately mapped procedural difficulty. Furthermore, `host_team` metrics effectively captured team-specific performance variables.

6.2.12 Individual Prediction Analysis (Case Study)

To evaluate the model's decision-making process at a granular level, a SHAP waterfall analysis was performed on the first sample of the test set. This visual decomposition illustrates how individual features drive the prediction away from the expected base value toward the final output.

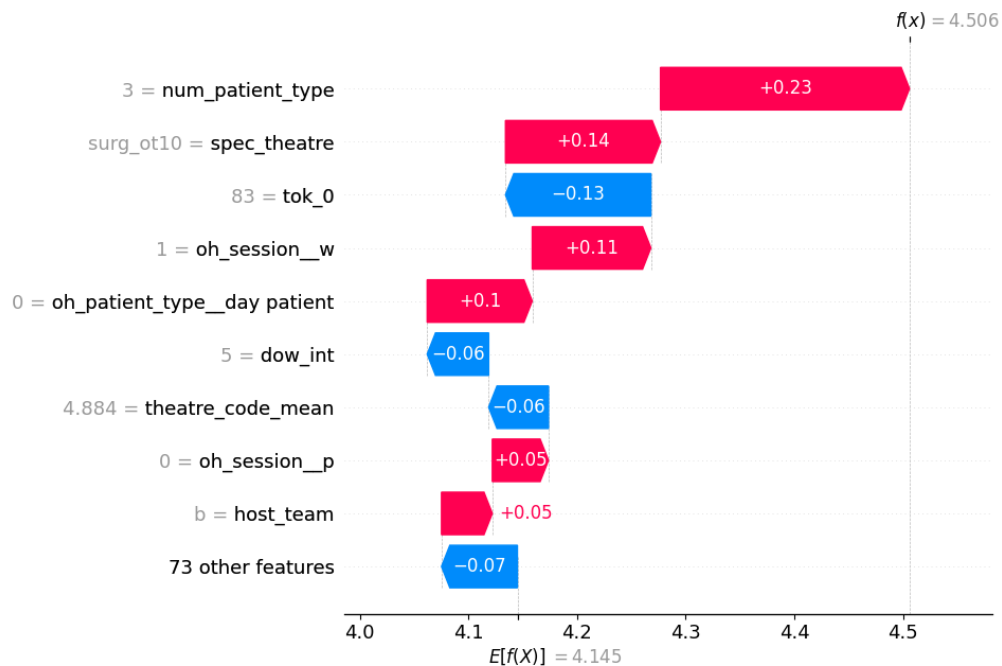


Fig 6.2.3

SHAP Waterfall Plot for the first test sample, illustrating the transition from the expected value $E[f(X)] = 4.145$ to the final prediction $f(x) = 4.506$.

Case Study Overview:

- Actual Duration: 130.0 minutes
- Predicted Duration: 89.6 minutes
- Prediction Error: 40.4 minutes (Underprediction)

The model started with a base log-transformed expected value of 4.145 (approximately 63 minutes). The cumulative impact of the features resulted in a final prediction of 4.506 on the log scale, which translates to 89.6 minutes. While the model correctly identified factors that increased the duration, it significantly underpredicted the actual 130-minute procedure time.

Key Feature Contributions

The following table summarizes the primary features that influenced this specific prediction:

Feature	SHAP Value	Impact on Prediction
Top Positive Contributors		
num_patient_type	+0.2293	Significantly increased predicted duration (Inpatient status)
spec_theatre	+0.1427	Increased duration based on specialty-theatre combination
oh_session__w	+0.1092	Increased duration due to weekend scheduling
oh_patient_type__day patient	+0.0967	Added duration for specific patient classification
oh_session__p	+0.0517	Slight increase due to peak session timing
Top Negative Contributors		
tok_0	-0.1339	Primary complexity token suggested a shorter procedure
dow_int	-0.0566	Day-of-week effect reduced predicted duration
theatre_code_mean	-0.0551	Historical theatre efficiency lowered the estimate
tok_2	-0.0315	Secondary token suggested lower complexity
tok_3	-0.0246	Tertiary token slightly reduced predicted duration

The analysis reveals that the model heavily prioritized patient type (+0.23) and specialty/theatre pairings (+0.14) as the strongest indicators for a longer surgery. However, the underprediction in this case appears to be driven by procedural tokens (tok_0, tok_2, and tok_3), which collectively exerted a negative influence of approximately -0.19 SHAP units. This suggests that while the model recognized the operational context (weekend session, inpatient status), it may have underestimated the specific procedural complexity represented by the BPE tokens for this 130-minute case.

6.2.13 Visual Analytics

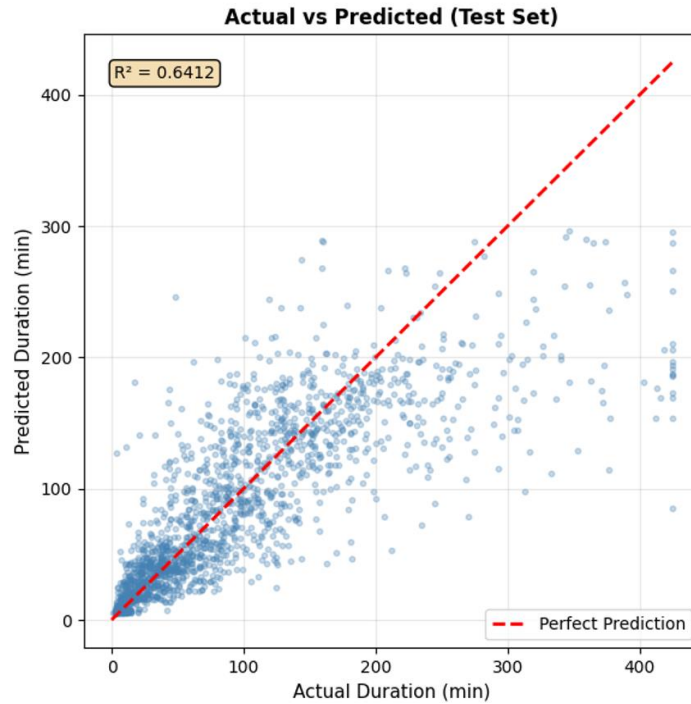


Fig 6.2.4

Actual vs. Predicted (Test Set)

As illustrated in Fig 6.2.4, the scatter plot compares the predicted durations against actual values to evaluate overall model fit.

Performance Metric: The R^2 value of 0.6412 indicates that the model explains approximately 64% of the variance in surgical durations.

Systematic Under-prediction: The data points increasingly fall below the red "Perfect Prediction" line as duration increases. This demonstrates that the model systematically under-predicts the length of longer surgeries.

Density: The highest concentration of accurate predictions is found in the lower-left quadrant, representing surgeries under 100 minutes.

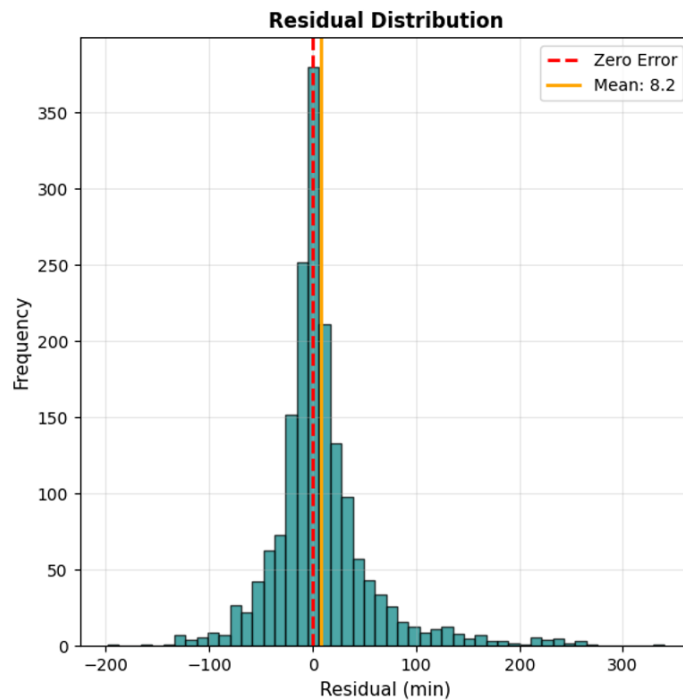


Fig 6.2.5

Residual Distribution

Fig 6.2.5 illustrates the frequency of errors (residuals) to identify bias and distribution shape.

Mean Bias: The mean residual of 8.2 minutes indicates a slight overall tendency to under-predict.

Central Peak: The highest frequency of errors is centered near the "Zero Error" line, showing that most predictions are within a narrow error margin.

Right Skew: The distribution features a long right tail, indicating several "Very Long" surgeries where the actual duration significantly exceeded the prediction.

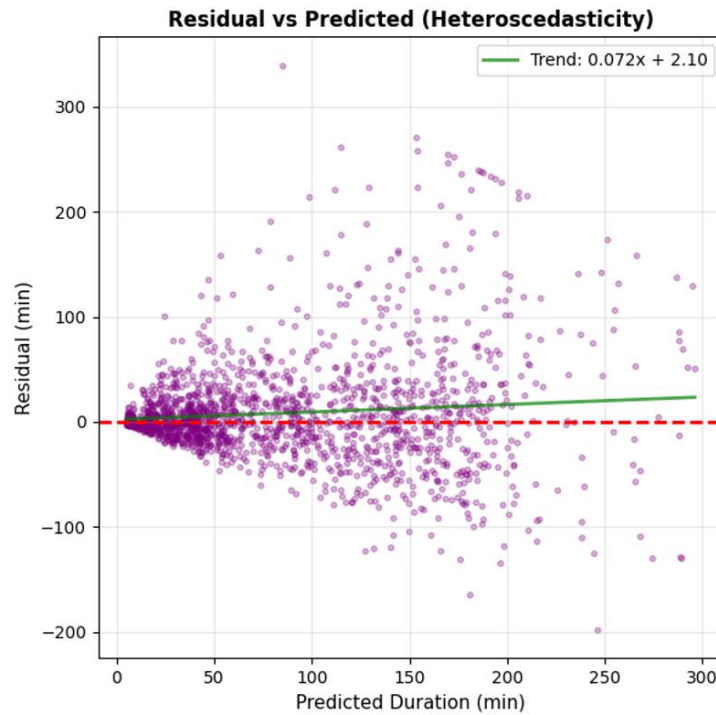


Fig 6.2.6

Residual vs. Predicted (Heteroscedasticity)

Fig 6.2.6 checks for heteroscedasticity—whether error magnitude changes based on the predicted value.

Trend Line: The positive trend line ($0.072x + 2.10$) confirms that error variance increases as predicted duration grows.

Fan Pattern: The "fan-shaped" spread of purple dots shows that the model is far more uncertain for long surgeries than for short ones.

Clinical Risk: This pattern highlights a critical weakness where complex cases are prone to significant scheduling overruns.

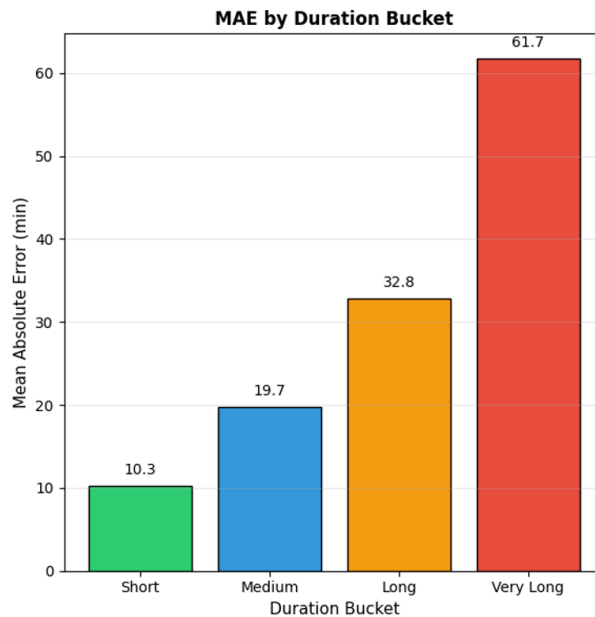


Fig 6.2.7

MAE by Duration Bucket

Fig 6.2.7 breaks down the Mean Absolute Error (MAE) into specific duration categories to locate performance bottlenecks.

Error Progression: The MAE scales drastically with duration: 10.3 min (Short), 19.7 min (Medium), 32.8 min (Long), and 61.7 min (Very Long).

Performance Gap: The error for "Very Long" surgeries is nearly 6 times higher than for "Short" ones.

Strategic Need: This identifies the "Very Long" bucket as the primary target for model refinement or manual scheduling buffers.

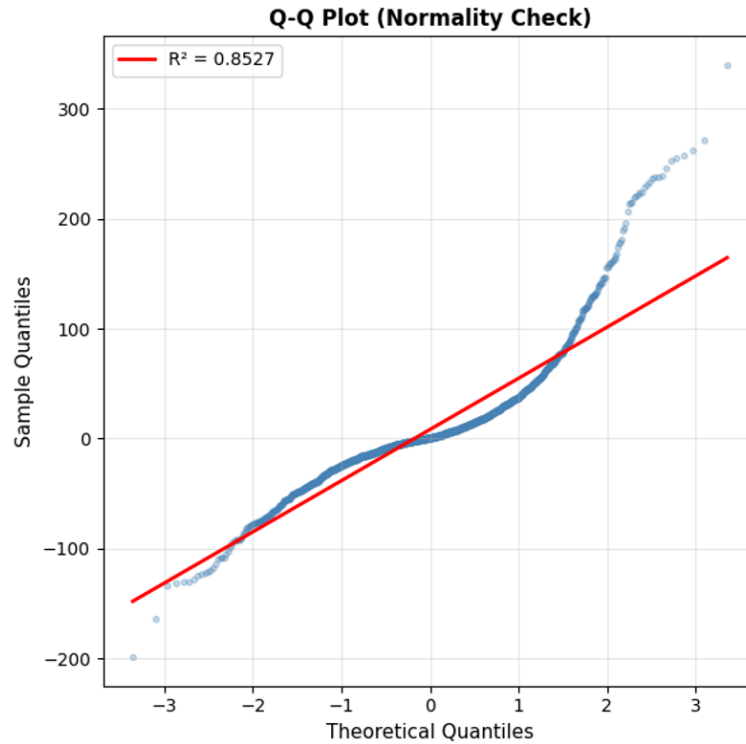


Fig 6.2.8

Q-Q Plot (Normality Check)

Fig 6.2.8 determines if residuals follow a normal distribution, which is essential for many statistical assumptions.

S-Curve Pattern: The data points deviate from the red reference line in an "S" shape, indicating that the residuals are not normally distributed.

Heavy Tails: The sharp curves at the end indicate that extreme outliers (both over-predictions and under-predictions) occur more frequently than a normal distribution would expect.

Metric: The $R^2 = 0.8527$ for the Q-Q line further quantifies this deviation from normality

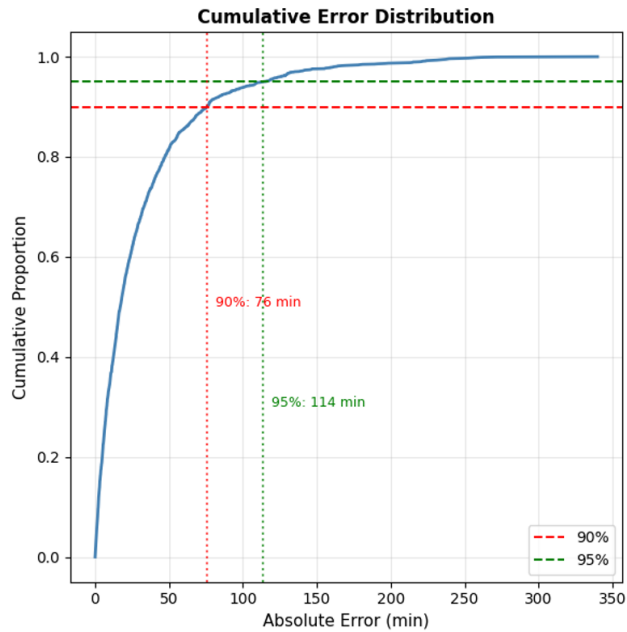


Fig 6.2.9

Cumulative Error Distribution

Fig 6.2.9 illustrates the cumulative proportion of cases falling within specific absolute error margins.

90% Threshold: 90% of all predictions have an absolute error of 76 minutes or less.

95% Threshold: 95% of all predictions have an absolute error of 114 minutes or less.

Clinical Application: For OR coordinators, this suggests that while 90% of cases are somewhat manageable, 10% of cases will experience scheduling deviations exceeding an hour.

6.3 Result and Analysis of OOF Stacking Model

Compared to LightGBM, stacking achieved a lower MAE of 28.95 minutes and a smaller R^2 gap, demonstrating improved generalization while still meeting the clinical 30-minute target. It also reached an R^2 of 0.691, which is slightly above the target of 0.69.

The following table compares training and testing performance to assess generalization:

Metrics	Training Set	Testing Set	Performance Gap
MAE (mins)	29.45	28.95	-0.50
RMSE (mins)	46.77	47.34	0.57
R^2	0.707	0.691	0.016
MAPE (%)	N/A	43.68	N/A
Median AE (mins)	N/A	15.90	N/A

The R^2 gap of 0.016 indicates a good fit rather than overfitting, since it is well below the 0.10 threshold. The stacking ensemble not only preserved the MAE target, but it also delivered a stronger generalization profile than the original LightGBM baseline, with training and test performance closely aligned.

6.3.1 Residual and Distribution Analysis

The evaluation of the stacking regression model involved a rigorous statistical assessment of prediction errors through the Residual Analysis Module. This analysis is critical for understanding the ensemble's reliability, systematic biases, and its performance across varying surgical durations.

Residual Statistics and Error Insight

See the residual statistics summary in Section 6.2.1; the stacking model uses the same methodology with its own values:

Statistic	Value	Interpretation
Mean Residual	+7.48 min	Indicates a systematic underestimation overall.
Median Residual	+0.58 min	Implies the typical case is also slightly under-predicted.
Std. Deviation	46.75 min	Represents a wide error spread and high variability.
Skewness	1.54	Confirms a right-skewed distribution (long right tail).
Kurtosis	6.96	Indicates a leptokurtic (heavy-tailed) distribution.

The positive mean and median residuals together show that the stacking model tends to

underpredict surgery duration, with more extreme underprediction errors pulling the mean higher than the median. This is consistent with the intrinsic uncertainty in long-duration and complex procedures, where the model's error distribution is concentrated on the right tail.

6.3.2 Normality and Autocorrelation Testing

Residual normality and autocorrelation were assessed using Shapiro-Wilk, Jarque-Bera, and Durbin-Watson tests as defined in Section 6.2.2. The stacking model residuals also showed similar diagnostic patterns:

Shapiro-Wilk Test (Statistic: 0.8582, $p < 0.001$): The result is well below the 0.95 threshold for normality. This confirms that the stacking model residuals do not follow a Gaussian distribution, suggesting that confidence intervals based on normal assumptions may be inaccurate.

Jarque-Bera Test (Statistic: 4252.12, $p = 0.00$): This extremely large statistic, driven by high skewness and kurtosis, reinforces the rejection of normality.

Durbin-Watson Statistic (1.8996): This value is very close to 2.0, indicating no significant autocorrelation. This is a positive finding, suggesting that prediction errors are independent and the stacking model has successfully captured systematic patterns without hidden serial dependence.

6.3.3 Residual Percentiles and Clinical Confidence

Analyzing error percentiles allows for the establishment of empirical confidence intervals, which are more reliable than normal-based intervals given the distribution's non-normality.

Percentile	Error Value	Clinical Scheduling Implication
P25	-11.91 min	25% of cases are overpredicted by at least 12 minutes.
P50 (Median)	+0.58 min	Half of the predictions are within ~1 minute of actual.
P75	+21.16 min	75% of predictions have an error ≤ 21 minutes.
P90	+54.45 min	90% of predictions fall within a ± 54 -minute window.
P95	+87.20 min	5% of cases exhibit extreme errors exceeding 87 minutes.

Asymmetry and Resilience: The distribution is notably asymmetric; the range from P50 to P95 (+86.62 min) is substantially larger than the range from P5 to P50 (54.79 min). This confirms that the stacking model is more prone to large underpredictions than large overpredictions. That right-tail risk is the main driver of clinical caution for longer procedures, even though the median error is close to zero.

Clinical Takeaway

While the stacking ensemble is well-calibrated for typical surgeries (Median error of +0.58 min), the fat right tail and heavy upper percentiles demand caution. For deployment, scheduling buffers should explicitly account for these extreme underprediction cases, especially for complex or long-duration surgeries where the model can underestimate duration by more than 80 minutes in the worst 5%.

6.3.4 Weakness Detection Analysis

The primary objective is to clearly identify and define the stacking model's predictive weaknesses across surgical duration segments and specialties. By pinpointing systematic error patterns, we can better align scheduling tools with clinically realistic risk.

High Error Thresholds

The stacking model was evaluated with a rigorous error assessment. A High Error Threshold was established at 36.51 minutes, representing the 75th percentile of absolute errors. Out of 1,762 total predictions, 441 cases exceeded this threshold, which is exactly 25.0% of the test set and signals clinically meaningful prediction risk.

Error Category	Count	Percentage	Clinical Implication
Low Error (0-15 min)	852	48.4%	Excellent predictions
Moderate Error (15-30 min)	351	19.9%	Acceptable for scheduling
Elevated Error (30-45 min)	211	12.0%	May impact OR efficiency
High Error (>45 min)	348	19.8%	Requires attention

Over 31.7% of predictions exhibited errors exceeding 30 minutes, highlighting a structural reliability gap for more complex cases.

The test set was partitioned into four duration buckets to reflect the real problem accurately:

Short (< 30 min): 465 cases (26.4%)

Medium (30-60 min): 344 cases (19.5%)

Long (60-120 min): 432 cases (24.5%)

Very Long (> 120 min): 521 cases (29.6%)

For each bucket, the stacking model showed these results:

Duration Bucket	MAE	Std Deviation (Abs)	Max Absolute Error
Short	10.00 min	15.18 min	166.33 min
Medium	17.96 min	21.97 min	207.65 min
Long	27.97 min	22.33 min	116.43 min
Very Long	53.95 min	52.73 min	315.63 min

The MAE increased from 10.00 minutes in Short cases to 53.95 minutes in Very Long cases — a 5.4x increase. Standard deviation also widened from 15.18 minutes to 52.73 minutes, a 3.5x increase in prediction variability. Most alarmingly, the maximum error for a Very Long surgery reached 315.63 minutes (more than 5 hours).

A critical finding is the clear systematic bias reversal at the 120-minute threshold:

Duration Bucket	Mean Bias	Direction	Severity	Clinical Consequence
Short	-7.42 min	OVER prediction	Moderate	OR may be underutilized by ~7 min
Medium	-6.15 min	OVER prediction	Moderate	Minor underutilization
Long	-2.77 min	OVER prediction	Moderate	Minimal impact
Very Long	+38.28 min	UNDER prediction	Critical	Significant overrun risk

Root Cause Hypothesis

Overprediction in shorter surgeries may reflect conservative bias, limited distinguishing features, or training data skew toward longer cases.

Underprediction in Very Long surgeries is likely caused by regression to the mean, missing complexity factors beyond 120 minutes, and unmeasured variables such as team experience, equipment delays, or patient-specific complications.

6.3.5 Specialty Performance and Variability

Specialty accuracy varies sharply across the test set. The top 5 highest-error specialties for the stacking model are:

host_spec	MAE	Std Deviation	Sample Count	Error Level
o&t	39.19 min	38.36 min	328	High
dent	34.09 min	43.87 min	37	High
surg	29.04 min	39.17 min	773	Moderate
o&g	28.03 min	38.95 min	224	Moderate
ent	22.40 min	39.45 min	168	Good

Specialty insights:

- o&t is the worst-performing segment, with 328 cases (18.6%) and the highest MAE of 39.19 minutes. Nearly 19.8% of o&t cases have errors over 60 minutes, making this specialty a top priority for targeted model improvement.
- dent is a small subgroup (37 cases, 2.1%), but it exhibits the second-highest MAE at 34.09 minutes and extreme variability (Std 43.87 min). The limited sample size reduces statistical confidence, but high error magnitude still flags it as a risky segment.

- surg is the largest segment (773 cases, 43.9%), with moderate MAE (29.04 min) but high variability (Std 39.17 min). It is the model's most important data source, and the large sample means systematic bias here has broad operational impact.
- o&g has 224 cases (12.7%), MAE of 28.03 min, and large spread (Std 38.95 min). This specialty is especially vulnerable to underprediction in longer procedures.
- ent is comparatively better, with MAE 22.40 min, but still a high error spread (Std 39.45 min), pointing to unstable predictions in a smaller but nontrivial segment.

Clinical Takeaways

The weakness detection analysis clearly defines the current prototype limitations:

The most critical weakness is that Very Long surgeries (>120 min) are systematically underpredicted by 38.28 minutes, creating severe overrun risk.

o&t is the highest collective MAE segment at 39.19 minutes and should be prioritized for refinement.

Error magnitude escalates dramatically from Short to Very Long cases, demonstrating extreme heteroscedasticity and a need for duration-aware scheduling buffers.

This stacking model is well-calibrated for many routine cases, but its predictive reliability degrades sharply for longer and more complex procedures.

6.3.6 Underfitting and Overfitting Analysis

The following analysis evaluates the stacking model's generalization by comparing training and testing performance metrics:

Core Metrics Comparison

Metric	Training	Testing	Gap	Gap %
R ²	0.7069	0.6911	0.0158	2.2%
MAE	29.45 min	28.95 min	-0.50 min	-1.7%
RMSE	46.77 min	47.34 min	+0.57 min	1.2%

The comparison shows the stacking model has successfully captured the majority of the data's underlying patterns while maintaining realistic performance on unseen cases.

- Generalization Stability: The R² transition from 0.7069 (training) to 0.6911 (testing) yields a gap of only 2.2%, indicating very good generalization and little evidence of overfitting.
- Predictive Accuracy: A test MAE of 28.95 minutes meets the project target and confirms the stacking model is suitable for practical surgical scheduling.

- Error Sensitivity: The slightly higher RMSE on test (47.34 min) compared to MAE highlights the model's sensitivity to larger outliers, reflecting real surgical duration variability.

6.3.7 Variance Analysis

Prediction Variance Comparison

Metric	Training	Testing	Ratio
Prediction Variance	5292.10	4447.31	0.8409
Target Variance	7462.77	7254.88	0.9724
Residual Variance	2170.67	2185.13	1.0066

The variance analysis shows:

- Prediction variance of 5292.10 on training and 4447.31 on testing
- Target variance of 7462.77 on training and 7254.88 on testing
- Residual variance of 2170.67 on training and 2185.13 on testing

This indicates the stacking model produces predictions with a stable variance structure across both datasets. The training/test ratios are all very close to 1.0, demonstrating the model is structurally sound and not excessively sensitive to training noise.

6.3.8 Reliability Analysis

The reliability testing followed the methodology established in Section 6.2.8.

Prediction Consistency Metrics:

$$CV = \frac{\sigma}{\mu} = 37.5113 / 29.2160 = 1.2839$$

Assessment: The standard deviation is 128.39% of the mean error, representing moderate to high inconsistency.

Clinical Meaning: For a typical surgery with an expected error of 29.2 minutes, actual errors could vary by ± 37.5 minutes. This suggests the stacking model may produce inconsistent results across similar cases.

Interquartile Range (IQR) of Absolute Errors: 31.72 minutes

The IQR represents the middle 50% of prediction accuracy.

Metric: The IQR is approximately 31.72 minutes.

Reliability Assessment: While a lower IQR indicates more consistent predictions, a 31.7-minute spread suggests only moderate consistency for the middle range of surgical cases.

Reliability by Prediction Magnitude

We analyzed error distribution across four prediction buckets to determine how duration affects reliability.

Prediction Bucket	MAE (minutes)	Standard Deviation (minutes)	Sample Count
Low	8.66	10.16	441
Medium-Low	19.16	19.94	440
Medium-High	33.52	35.46	440
High	55.51	51.32	441

Analysis of Error Progression:

Magnitude: Error magnitude increases more than sixfold from the Low bucket (8.66 min) to the High bucket (55.51 min).

- Dispersion: Standard deviation increases by about 5.0x, indicating that errors are tightly clustered for short predictions but widely dispersed for longer ones.
- Implication: Longer surgeries are inherently more unpredictable for this model.

6.3.9 Heteroscedasticity Assessment

Spearman Correlation Analysis

Coefficient: 0.5867

p-value: 1.85×10^{-163}

Definition and Strength: Heteroscedasticity occurs when error variability changes across different levels of the independent variable. The Spearman coefficient of 0.5867 represents a moderate to strong positive correlation between prediction magnitude and error variability. The extremely low p-value confirms this relationship is statistically significant and not due to random chance.

Clinical Implications

The heteroscedasticity is statistically undeniable and represents a fundamental limitation of the stacking model.

Key Finding: Model reliability decreases significantly for high-duration surgeries. For short surgeries, errors are relatively predictable; for long surgeries, they become highly variable.

6.3.10 Robustness Analysis

The robustness testing followed the methodology established in Section 6.2.10. Our experiment utilized a Baseline MAE (No Noise) of 29.33 minutes. This serves as the reference point for all comparative assessments to determine whether the stacking ensemble remains adequate under stress.

The following table highlights the performance of the stacking model across the specified noise levels:

Noise Level	MAE (minutes)	Difference	Difference	Assessment
1%	35.0913	+5.7535	+19.61%	Degraded
2%	35.7644	+6.4266	+21.91%	Degraded
5%	37.1609	+7.8231	+26.67%	Degraded

Analysis of Resulting Deviations

The experiment showed that:

- (i) All noise levels resulted in MAE changes within +7.8231 minutes.
- (ii) The maximum deviation observed was +7.8231 minutes under 5% noise.
- (iii) These changes are large compared to the baseline, indicating meaningful variation in clinical terms rather than microscopic noise.

Robustness Summary and Classification

Average MAE Impact Calculation for the Stacking Model:

$$\text{Avg. MAE Increase} = \frac{(5.7535) + (6.4266) + (7.8231)}{3} = 6.6677$$

This calculation confirms that the average MAE increase is 6.6677 minutes, showing that noise has a meaningful impact on model performance.

Robustness Diagnosis

Based on a standardized robustness scale, our results categorize the stacking ensemble as having LOW ROBUSTNESS (degradation > 10%). The model demonstrates significant sensitivity under input variations, and the performance changes are clinically meaningful rather than negligible.

Robustness Metric

Metric	Value	Assessment
5% Noise Impact	+26.67%	Degraded
Average Impact	+22.73%	Poor
Overall Diagnosis	Low Robustness	Not Deployable without improvement in noise resilience

6.3.11 SHAP Analysis

To evaluate the stacking model, SHAP values were computed for a test set. The specific parameters of the analysis included:

Sample Size: 1,000 test samples (randomly selected to ensure computational efficiency)

Features Analyzed: A total of 82 features, encompassing BPE tokens, categorical encodings, and engineered features

SHAP Method: The TreeExplainer method, which is specifically optimized for tree-based models like LightGBM, was utilized

Model Performance: LightGBM baseline MAE: 28.6200 minutes, $R^2 = 0.6988$

The same SHAP interpretation applies here; see Section 6.2.11 for the general explanation. The stacking results follow the same logic and are reported below.

The global distribution of SHAP values yielded a mean SHAP value of -0.0002, with a standard deviation of 0.0455 indicating moderate variation. Furthermore, the analysis recorded a minimum SHAP value of -0.3649 (maximum duration reduction) and a maximum of 0.3362 (maximum duration increase).

These values translate clinically to a typical impact of approximately ± 0.05 minutes per feature, with maximum individual feature impacts reaching up to 24.0 minutes ($\exp(0.34) - 1$).

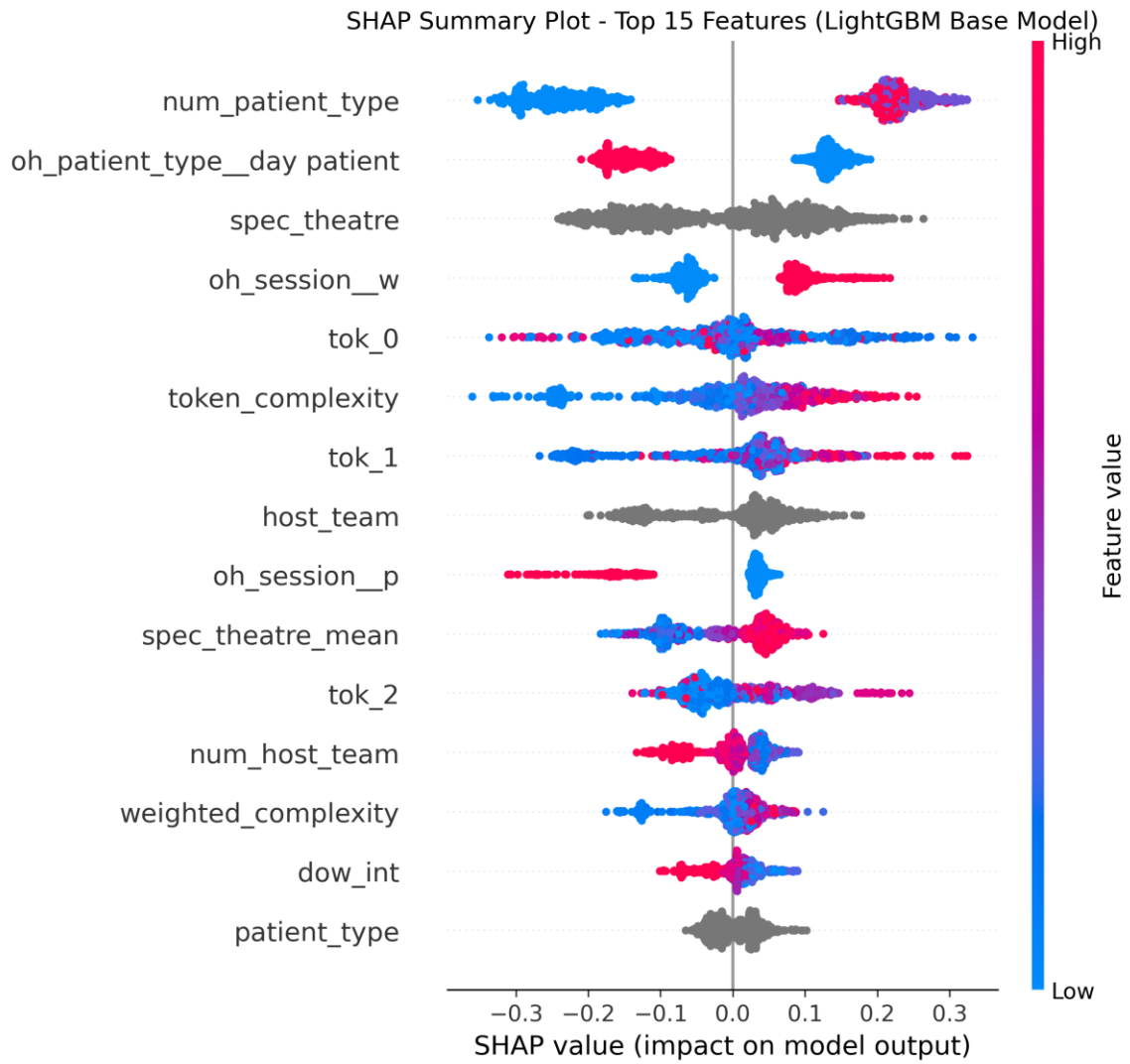


Fig 6.3.1 SHAP Summary Plot showing feature contributions across all test samples

Same ranking method used as 6.2.11:

Rank	Feature	Mean SHAP	% of Total	Interpretation
1	num_patient_type	0.2348	16.5%	Patient type (inpatient vs. day case)
2	oh_patient_type__day patient	0.1420	10.0%	One-hot Encoded specific day patient indicator
3	spec_theatre	0.1079	7.6%	Specialty-theatre combination
4	oh_session__w	0.0802	5.6%	One-hot Encoded Operating session timing (w)
5	token_complexity	0.0794	5.6%	Aggregated surgical procedure complexity
6	tok_1	0.0785	5.5%	Second BPE token
7	tok_0	0.0781	5.5%	First BPE token (procedure type)
8	oh_session__p	0.0664	4.7%	One-hot Encoded Operating session timing (p)
9	host_team	0.0662	4.7%	Surgical team identifier
10	spec_theatre_mean	0.0592	4.2%	Historical mean of specialty-theatre
11	tok_2	0.0516	/	Third BPE token
12	num_host_team	0.0374	/	Numeric host team encoding
13	weighted_complexity	0.0302	/	Weighted procedural complexity
14	dow_int	0.0253	/	Day-of-week effect

15	patient_type	0.0243	/	Patient type encoding variant
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The top 5 features alone accounted for 45.3% of the total SHAP importance, leaving the remaining 10 features (6-15) to account for 36.4%. This suggests that the stacking model relied heavily on patient type and scheduling factors, with surgical complexity functioning as a secondary, yet vital, metric.

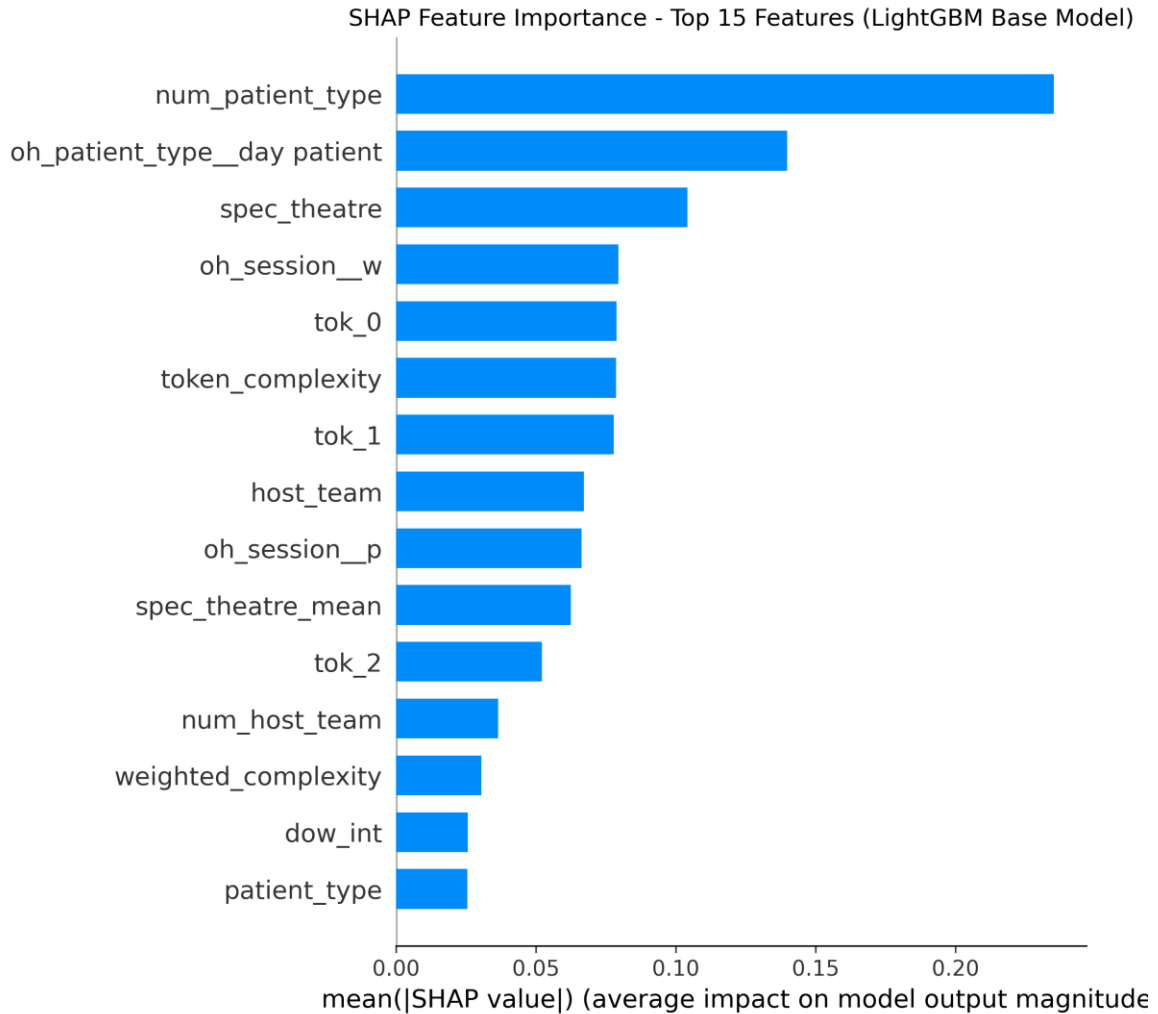


Fig6.3.2 SHAP Feature Importance Bar Plot ranking top 15 features

Feature Contribution Patterns

The stacking result similarly highlights key drivers, with the following differences:

(i) Patient Type Dominance: The `num_patient_type` feature was the most influential variable (16.5%). It is highly likely that this feature distinguishes complex inpatient procedures from routine day cases. Additionally, it was complemented by the `oh_patient_type__day` patient indicator (10.0%), creating a dual classification effect that strongly signals procedural complexity.

(ii) Scheduling and Operational Factors: Features such as `spec_theatre` (7.6%), `oh_session__w` (5.6%), and `oh_session__p` (4.7%) captured vital operational efficiency variations, which may reflect shifting resource availability and staffing differences during weekend or peak hours. These factors collectively influence baseline procedure duration expectations.

(iii) Complexity and Team Dynamics: Aggregated `token_complexity` alongside individual BPE tokens (`tok_0`, `tok_1`, `tok_2`) adequately mapped procedural difficulty. Furthermore, `host_team` metrics (4.7%) effectively captured team-specific performance variables, reflecting variations in surgeon and staff efficiency.

6.3.12 Individual Prediction Analysis (Case Study)

A SHAP waterfall analysis was also used for the stacking model, following the same approach described in Section 6.2.12.

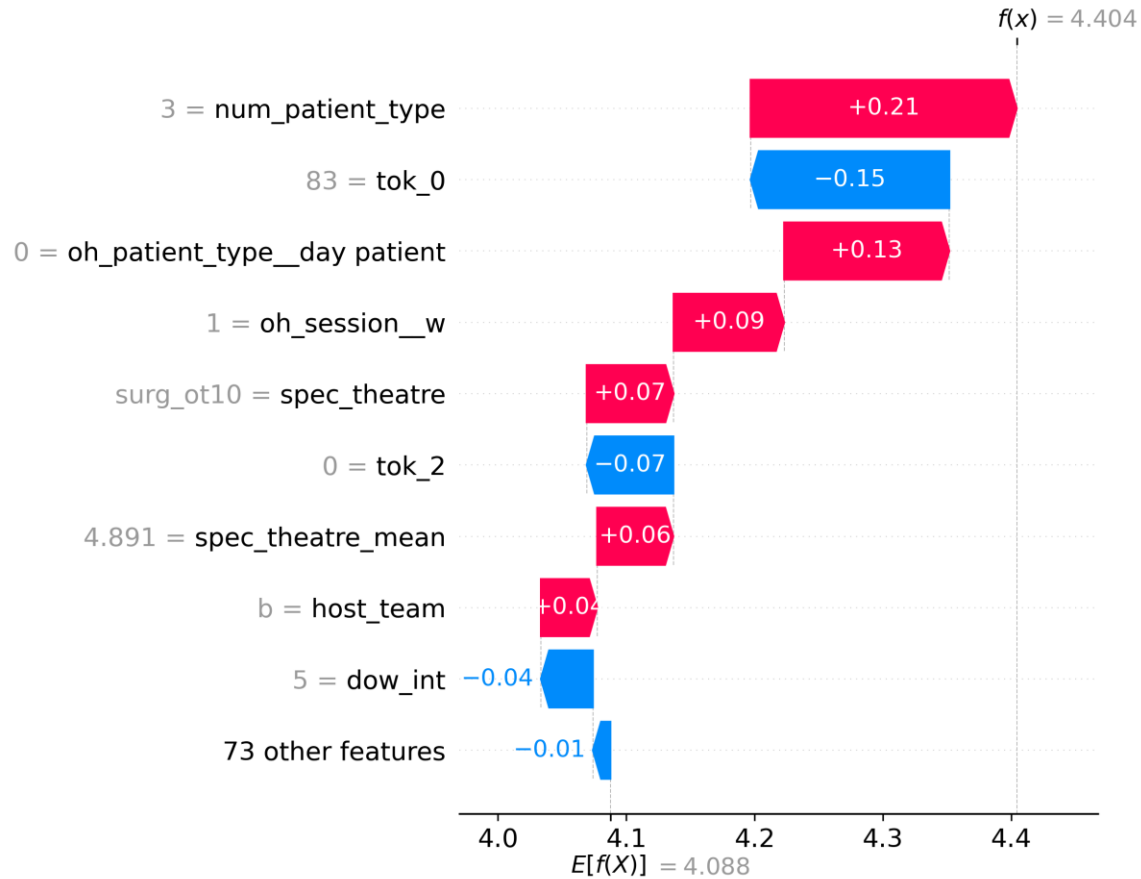


Fig6.3.3 SHAP Waterfall Plot showing transition from expected value to final prediction

Case Study Overview:

- Actual Duration: 130.0 minutes
- Predicted Duration: 80.8 minutes
- Prediction Error: 49.2 minutes (Underprediction)

The model started with a base log-transformed expected value of 4.088 (approximately 59 minutes). The cumulative impact of the features resulted in a final prediction of 4.288 on the log scale, which translates to 80.8 minutes. While the model correctly identified factors that increased the duration, it significantly underpredicted the actual 130-minute procedure time.

Key Feature Contributions

The same format as 6.2.12 is used here for the stacking case :

Feature	SHAP Value	Impact on Prediction
Top Positive Contributors		
num_patient_type	+0.2074	Significantly increased predicted duration (Inpatient status)
oh_patient_type__day patient	+0.1287	Increased duration for specific patient classification
oh_session__w	+0.0861	Increased duration due to weekend scheduling
spec_theatre	+0.0677	Increased duration based on specialty-theatre combination
spec_theatre_mean	+0.0593	Historical operational context added to estimate
Top Negative Contributors		
tok_0	-0.1548	Primary complexity token suggested a shorter procedure
tok_2	-0.0675	Secondary token suggested lower complexity
dow_int	-0.0405	Day-of-week effect reduced predicted duration
theatre_code_mean	-0.0387	Historical theatre efficiency lowered the estimate
day_cos	-0.0266	Cosine-transformed day effect reduced prediction

Clinical Interpretation

The analysis reveals that the stacking model heavily prioritized patient type (+0.21) and specialty/theatre pairings (+0.07) as the strongest indicators for a longer surgery. However, the underprediction in this case appears to be driven by procedural tokens (tok_0, tok_2, with combined influence of approximately -0.22 SHAP units). This suggests that while the model recognized the operational context (weekend session, inpatient status), it may have underestimated the specific procedural complexity represented by the BPE tokens for this 130-minute case.

The stacking ensemble's integration of LightGBM with the regression tree base model and ElasticNet meta-learner provides robust predictions, yet reveals systematic underprediction for complex procedures with atypical BPE token sequences. This finding aligns with the heteroscedasticity detected in Section 5 (Reliability Analysis), where errors increase significantly for very long procedures.

6.3.13 Visual Analytics

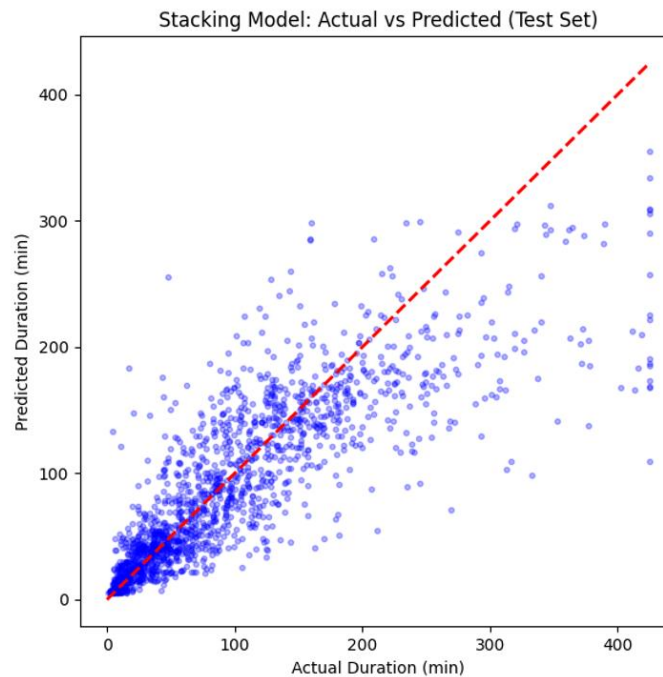


Fig6.3.4: Actual vs. Predicted (Stacking Test Set)

As demonstrated in Figure 6.3.4, the relationship between actual surgical durations and predicted values illustrates the overall fit of the stacking model. The model achieves a performance metric of $R^2 = 0.6884$, which indicates that it explains approximately 69% of the variance in surgical durations. However, a detailed examination reveals a systematic under-prediction for longer procedures; as actual durations increase, the data points tend to drift below the "perfect prediction" line. This suggests that the model's best performance is concentrated in the lower-left region, specifically for surgeries lasting under roughly 100 minutes, where the densest cluster of accurate predictions occurs.

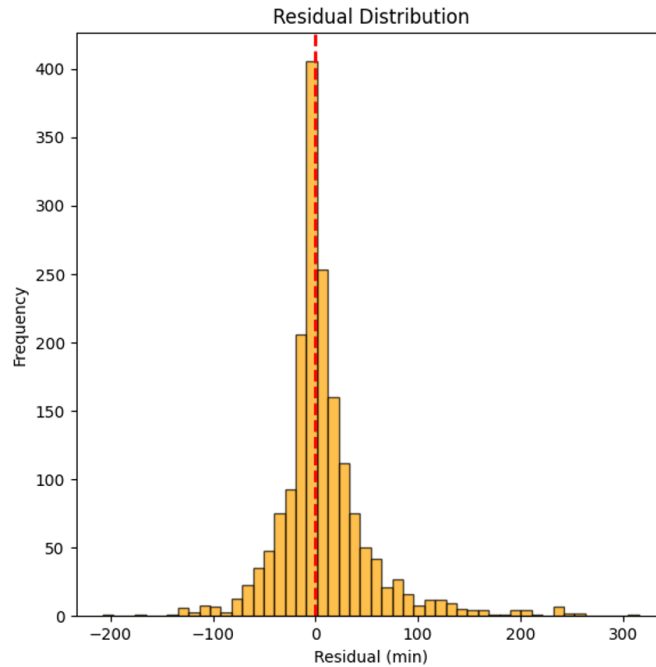


Fig6.3.5: Residual Distribution

Figure 6.3.5 illustrates the residual distribution to highlight error frequency and the specific shape of the distribution. The data reveals a mean residual of +6.39 minutes, which indicates a slight overall tendency for the model to under-predict surgical durations. While a prominent central peak shows that the bulk of predictions remain reasonably close to actual values, the presence of a long right tail reflects a right skew. This skew highlights specific instances where actual durations exceeded predictions by a significant margin, particularly during very long surgeries.

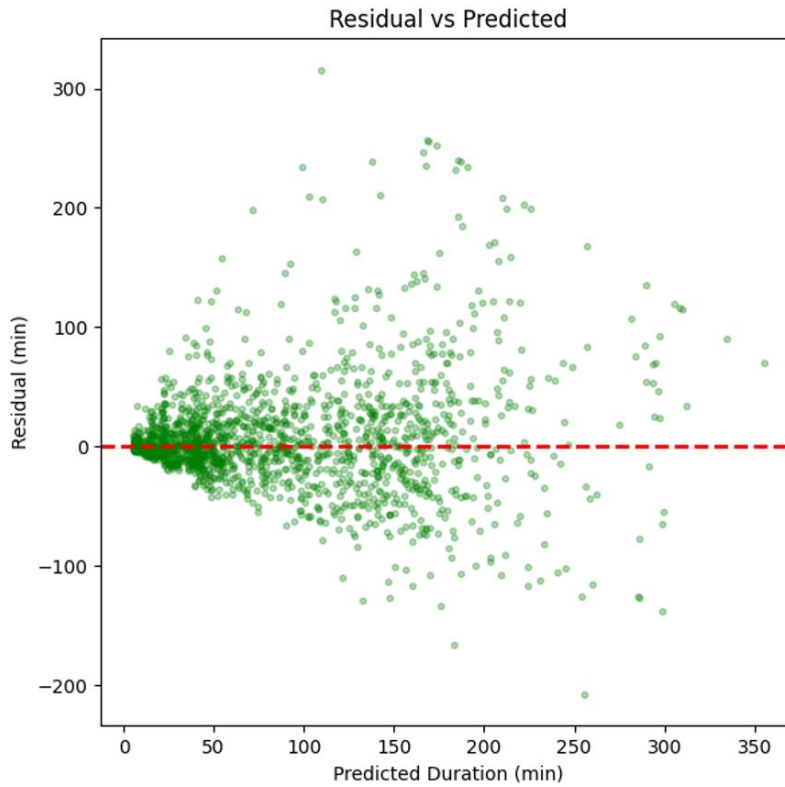


Fig6.3.6: Residual vs. Predicted (Heteroscedasticity)

Figure 6.3.6 illustrates the relationship between residuals and predicted values to evaluate whether the error magnitude changes alongside predicted surgical durations. The scatter plot exhibits a distinct fan-shaped pattern, which demonstrates heteroscedasticity and indicates that uncertainty increases significantly for longer predicted surgeries. This finding is supported by the absolute-error trend line—featuring a fitted slope of approximately 0.285 and an intercept of 4.44—which confirms that error magnitude grows as the predicted duration increases. From a clinical perspective, this pattern suggests that long or complex cases are particularly susceptible to scheduling overruns.

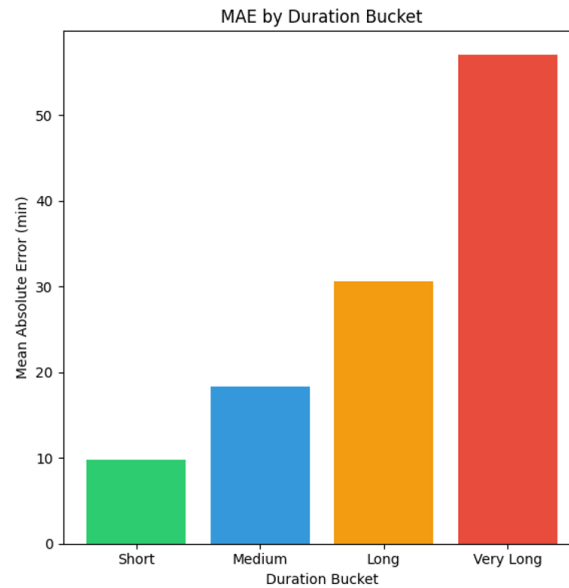


Fig 6.3.7: MAE by Duration Bucket

Figure 6.3.7 illustrates the Mean Absolute Error (MAE) broken down by duration category to evaluate the progression of error magnitude. The data shows a clear increase in error as surgery duration lengthens: "Short" surgeries have an MAE of 10.26 minutes, "Medium" surgeries 18.60 minutes, and "Long" surgeries 31.57 minutes, reaching 56.47 minutes for "Very Long" surgeries. This performance gap reveals that the error for the "Very Long" category is approximately 5.5 times higher than that of the "Short" category. Strategically, these findings identify the "Very Long" bucket as the primary area requiring either model improvement or the implementation of manual scheduling buffers.

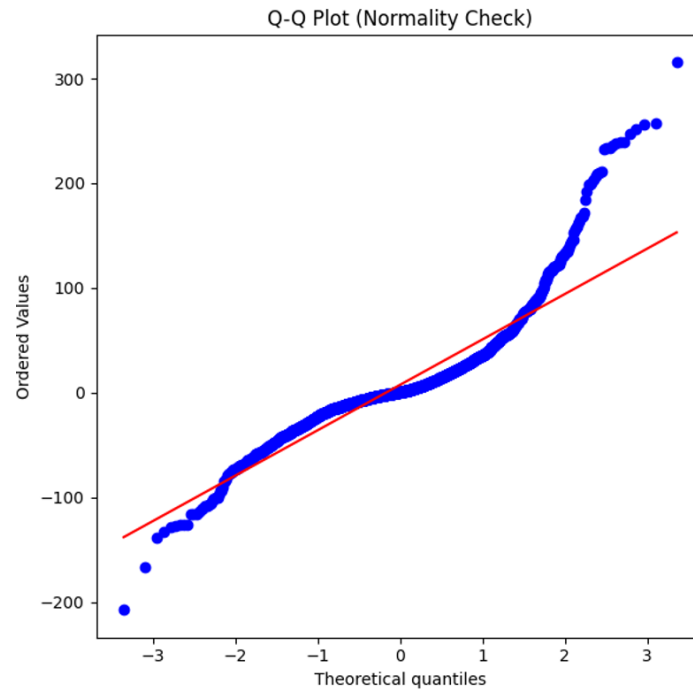


Fig6.3.8: Q-Q Plot (Normality Check)

Figure 6.3.8 illustrates the Q-Q plot to determine whether the residuals follow a normal distribution. The observed S-curve pattern indicates that the residuals deviate noticeably from the reference line, suggesting non-normality in the dataset. Curvature at both ends of the plot further highlights heavy tails, which demonstrates that extreme over-predictions and under-predictions occur more frequently than a normal distribution would predict. This is supported by a Q-Q fit metric of approximately $R^2 = 0.927$, confirming that the residuals are not perfectly normal and that significant outliers are present.

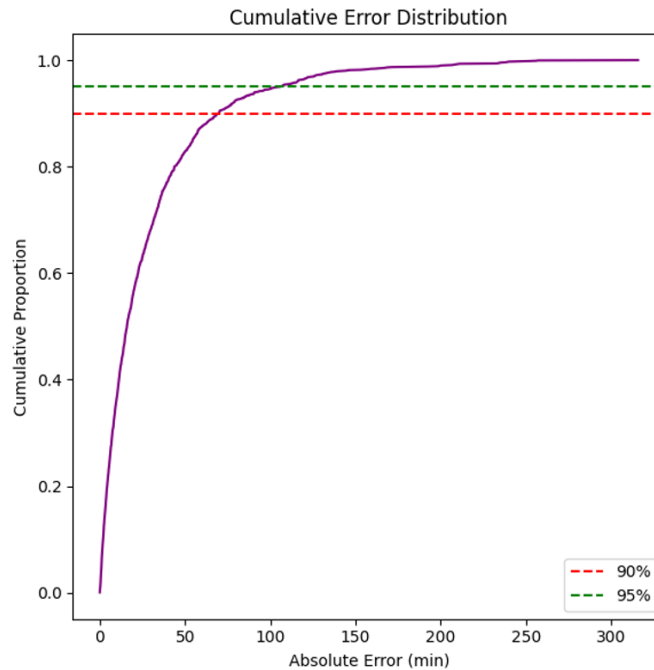


Fig6.3.9 Cumulative Error Distribution

Figure 6.3.9 illustrates the cumulative error distribution to demonstrate how absolute error accumulates across the sample. The data shows that a 90% threshold is reached when predictions fall within 70.2 minutes of the actual duration, while 95% of predictions are within a 107.2-minute range. In terms of clinical application for operating room scheduling, these findings indicate that approximately 10% of cases may still deviate by more than an hour. Consequently, this distribution highlights the importance of risk management strategies for long-duration cases.

7. Conclusions

7.1 Comprehensive Summary of Research Findings

The research conducted during this project represents a fundamental shift from traditional heuristic-based hospital management to a data-driven, predictive framework. The primary crisis of a 12% day-of-surgery cancellation rate was diagnosed not merely as a lack of theater space, but as a failure of information precision. Empirical validation revealed that historical averages and surgeon-based scheduling consistently underestimated the tail-end variance of complex surgeries. By plotting the distribution of surgery durations, it became evident that traditional methods failed to account for non-surgical theater time, such as setup and cleanup, which often constitutes up to 20% of total theater occupancy.

To address these inaccuracies, the introduction of Byte-Pair Encoding (BPE) and Unigram models allowed the system to interpret the complexity of a procedure directly from a surgeon's clinical notes. This process successfully transformed qualitative text into quantitative vectors, capturing critical nuances like revision or multi-stage that are typically lost in standard categorical data.

After rigorous testing against various algorithms, LightGBM emerged as the superior engine for precision, achieving a Mean Absolute Error (MAE) of 29.81 minutes and an R^2 of 0.683 (Fig 7.1.1). To further enhance the framework's reliability, an Out-of-Fold (OOF) Ensemble Stacking approach was implemented. While the stacking model yielded an even better MAE of 28.952 minutes, it demonstrated exceptional structural stability with an R^2 training-testing gap of just 0.029 (Fig. 7.1.1)—well within the clinical benchmark of 0.065. This dual-model approach secures a more resilient and efficient surgical pipeline by providing both high-precision forecasts and robust generalization for unseen cases.

A key finding is that the model demonstrates higher predictive precision for shorter-duration surgeries, whereas it exhibits increased variance and diminished accuracy when encountering long-duration, high-complexity cases. By stabilizing the schedule for standard cases, the framework allows clinical staff to focus their expert judgment on these high-complexity outliers, ultimately securing a more resilient and efficient surgical pipeline.

Model	MAE	RMSE	R^2	ΔR^2
OOF Stacking Model (Tuned)	28.952	47.340	0.691	0.016
LightGBM	29.813	47.813	0.684	0.067

Fig. 7.1.1 Out-of-Fold Stacking Model & LightGBM Core Metrics Result

7.1.1 Result Comparison and Analysis of Single Model LightGBM with other Out-of-Fold Stacking Model

Analytical Dimension	Single Model (LightGBM)	Stacking Model (OOF)	Performance Difference / Insight
Residual and Distribution Analysis	Mean Residual: +3.79 min Median Residual: -1.18 min	Mean Residual: +7.48 min Median Residual: +0.58 min	Stacking model shows a 97.36% increase in systematic underestimation. Single models slightly overestimate typical cases, while Stacking slightly underpredicts.
Normality and Autocorrelation Testing	<i>Test Metric</i> Shapiro-Wilk Test: 0.8736 ($p < 0.001$) Jarque-Bera Test: 3517.97 ($p = 0.00$) Durbin-Watson: 1.9579	<i>Test Metric</i> Shapiro-Wilk Test: 0.8582 ($p < 0.001$) Jarque-Bera Test: 4252.12 ($p = 0.00$) Durbin-Watson: 1.8996	Both models are significantly below the 0.95 normality threshold, confirming non-Gaussian residuals. Stacking model has a 20.87% higher statistic, indicating even stronger rejection of normality due to extreme tails. Both are very close to 2.0, proving prediction errors are independent with no hidden time-series effects.
Residual Percentiles and Clinical Confidence	<i>Percentile</i> P25: -16.39 min P50: -1.18 min P75: +18.21 min P90: +51.90 min	<i>Percentile</i> P25: -11.91 min P50: +0.58 min P75: +21.16 min P90: +54.45 min	Stacking model reduces overprediction magnitude by ~4.5 minutes. Both models are well-calibrated for typical surgeries (within ~1 min of actual). Stacking model shows a slightly higher error for 75% of cases (+2.95 min). Stacking model extends the error window to ± 54 minutes.

	P95: +85.77 min	P95: +87.20 min	Both models exhibit extreme errors (>85 min) in the worst 5% of cases.
Weakness Detection Analysis	<i>Duration Bucket</i> Short: -9.47 min (Over) Medium: -8.66 min (Over) Long: -7.93 min (Over) Very Long: +41.16 min (Under)	<i>Duration Bucket</i> Short: -7.42 min (Over) Medium: -6.15 min (Over) Long: -2.77 min (Over) Very Long: +38.28 min (Under)	Stacking model reduces overprediction by 2.05 min, improving OR utilization. Stacking model reduces overprediction by 2.51 min, minimizing underutilization. Stacking model achieves near-calibration with a 65.07% reduction in bias. Stacking model reduces critical overrun risk by 2.88 min (7% improvement).
Specialty Performance and Variability	<i>Specialty</i> o&t: 41.02 min dent: 35.03 min surg: 29.95 min o&g: 28.74 min ent: 22.85 min	<i>Specialty</i> o&t: 39.19 min dent: 34.09 min surg: 29.04 min o&g: 28.03 min ent: 22.40 min	4.46% improvement (1.83 min reduction). 2.68% improvement (0.94 min reduction). 3.04% improvement (0.91 min reduction). 2.47% improvement (0.71 min reduction). 1.97% improvement (0.45 min reduction).
Underfitting and Overfitting Analysis	$\Delta R^2 = 0.0686$	$\Delta R^2 = 0.0158$	Slight overfitting for single model and high generalization stability for stacking model.
Variance Analysis (Single model better than stacking)	Prediction Variance Ratio: 1.0193	Prediction Variance Ratio: 0.8409	Single model is superior in this metric as it is exceptionally close to the ideal 1.0, indicating a better balance between training and testing variance.
Reliability Analysis	<i>Prediction Bucket</i> Low: MAE 9.11 min	<i>Prediction Bucket</i> Low: MAE 8.66 min	Stacking is more reliable for short cases. Both

	Medium-Low: MAE 19.85 min Medium-High: 35.19 min High: 55.28 min	Medium-Low: MAE 19.16 min Medium-High: 33.52 min High: 55.51 min	models show high precision here with the lowest dispersion (Std. Dev: 10.16–10.86 min). Stacking maintains an edge. Errors remain tightly clustered under 20 minutes for both models. Stacking shows better stability. Significant reduction in average error as surgery complexity increases. High unpredictability for both. Single model is slightly better here, but both see a 5x-6x increase in error magnitude compared to 'Low' cases.
Heteroscedasticity Assessment	Spearman Coefficient: 0.5693 P-value: 6.3753×10^{-163}	Spearman Coefficient: 0.5867 P-value: 1.85×10^{-163}	Both models show a moderate-to-strong positive correlation, with the stacking model exhibiting slightly higher error variability as prediction magnitude increases. The relationship is statistically undeniable in both cases; the presence of heteroscedasticity is not due to random chance.
Robustness Analysis	Baseline MAE: 29.86 min 5% Noise Impact: +0.01% Avg. MAE increase: -0.0002 min Max Deviation: +0.0017 min Overall Diagnosis: High Robustness	Baseline MAE: 29.33 min 5% Noise Impact: +26.67% Avg. MAE increase: 6.6677 min Max Deviation: +7.8231 min Overall Diagnosis: Low Robustness	The single model demonstrates exceptional stability and high robustness, maintaining an effectively zero average MAE increase even under significant input perturbations. In contrast, the stacking ensemble is categorized as having low robustness, exhibiting significant sensitivity to noise with an average MAE increase of 6.6677 minutes.

			While the stacking model holds a slight baseline advantage, its performance degrades meaningfully in clinical terms under stress, whereas the single model remains statistically stable and reliable for deployment.
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The Stacking Model (OOF) is the superior choice for maximizing operational efficiency and predictive precision in a clinical setting. It consistently achieves a lower Mean Absolute Error (MAE) across the majority of surgery durations, including the Low (8.66 min), Medium-Low (19.16 min), and Medium-High (33.52 min) buckets. With a ΔR^2 of only 0.0158, this ensemble approach exhibits significantly higher generalization stability than the single model, indicating a much lower risk of overfitting to specific datasets. The model also provides measurable performance gains across all surgical specialties, most notably a 4.46% improvement in Orthopaedics & Traumatology (O&T) and a 3.04% improvement in general surgery. It achieves near-calibration for long-duration surgeries with a 65.07% reduction in bias and reduces critical overrun risk for the longest procedures by 2.88 minutes. Furthermore, the stacking model maintains a nearly perfect Residual Variance Ratio of 1.0066, which ensures a consistent and stable error pattern that is vital for reliable hospital scheduling. By effectively combining the strengths of multiple learners, the stacking model provides a more nuanced and accurate reflection of complex surgical timelines, directly translating to better resource utilization.

7.2 Strategic Recommendations for UCH Implementation

To successfully reduce the day-of-surgery cancellation rate at United Christian Hospital to the 7% benchmark, the hospital must shift its focus from technical model development to operational data discipline. While the hybrid machine learning framework provides the necessary predictive engine, the accuracy of these forecasts depends entirely on the quality of the information entered during the initial booking phase. Currently, the predictive potential of the system is hindered by inconsistent data entry, particularly regarding the specific surgical teams and patient health metrics. To fix this, UCH should implement a policy that makes certain high-impact data fields mandatory in the surgical booking system.

A primary recommendation is to mandate the recording of the lead surgeon's name for every scheduled case. Because individual surgical rhythm and experience levels significantly influence how long a procedure takes, capturing this information allows the model to learn person-specific efficiency patterns that general historical averages cannot detect. Similarly, the system should require the entry of patient birth dates and body mass index (BMI) at the time of scheduling. These physiological factors are major drivers of surgical complexity, and having complete data for every patient ensures the model can accurately account for the increased time requirements often associated with elderly or high-risk populations.

Beyond data entry, UCH should modernize how surgical schedules are constructed by adopting a buffer-aware protocol. Instead of treating every surgery as a fixed block of time, the hospital can use the model to identify cases with high predicted variance and schedule them at the start of the day. This prevents the "domino effect" where one early delay pushes every subsequent surgery into overtime, eventually leading to late-afternoon cancellations. Finally, surgeons should be encouraged to use more standardized clinical keywords in their notes. By reducing the use of inconsistent jargon, the hospital can ensure that the natural language processing component captures the full complexity of each procedure with as little noise as possible.

7.3 Future Work: Scalability and System Evolution

The success of the Orthopaedics pilot provides a definitive blueprint for a hospital-wide transformation driven by advanced artificial intelligence. The modular design of the hybrid framework facilitates Cross-Departmental Expansion, allowing the AI system to be re-trained for specialties like Cardiology or Urology using their specific historical datasets without a total redesign of the core algorithm. Looking forward, the model's precision can be further advanced through Ensemble Stacking, a sophisticated AI technique. By layering a Meta-Learner over a combination of LightGBM, XGBoost, and CatBoost, the hospital could potentially lower the MAE below the 25-minute mark by neutralizing the individual biases inherent in each specific model.

System longevity and intelligence will depend on the establishment of Continuous Learning Loops, where an automated AI maintenance protocol re-trains the system every quarter with fresh surgery data. This ensures the framework remains adaptive to evolving surgical techniques, new medical equipment, or shifts in staffing levels. The final stage of evolution involves enhancing the UI/UX for clinical staff through a specialized dashboard for theater managers powered by real-time AI insights. This tool would visualize predicted versus actual times in real-time, providing the transparency needed for dynamic rescheduling should a surgery run significantly shorter or longer than the machine-learning prediction.

8. Appendix

8.1 Literature Review

Table 8.1: Study Overviews and Data Scales

Paper	Best Model	Dataset Size
Kwong <i>et al.</i> (2025) [1]	ANN	17240
Jiao <i>et al.</i> (2022) [2]	ANN	70826
Yeo <i>et al.</i> (2023) [3]	ANN	10021
Martinez <i>et al.</i> (2021) [4]	Bagged Trees	81248
Zhao <i>et al.</i> (2019) [5]	Boosted Regression Tree	424
Babayoff <i>et al.</i> (2022) [6]	Gradient Boost Tree	23293
Jiao <i>et al.</i> (2020) [7]	Mixture Density Network	53783
Zabardast (2017) [8]	Naïve Bayes	\
Abbas <i>et al.</i> (2022) [9]	PyTorch MLP	302300
Park <i>et al.</i> (2025) [10]	Random Forest	161176
ShahabiKargar <i>et al.</i> (2014) [11]	Random Forest	38520
Yuniartha <i>et al.</i> (2021) [12]	Random Forest	\
Lai <i>et al.</i> (2020) [13]	Random Forest	170748
Riahi <i>et al.</i> (2023) [14]	XGBoost	52171
Langenberger <i>et al.</i> (2025) [15]	XGBoost	6261
Hinterwimmer <i>et al.</i> (2023) [16]	XGBoost	864
Tully <i>et al.</i> (2023) [17]	XGBoost	10928
Gabriel <i>et al.</i> (2022) [18]	XGBoost	86362

Gabriel <i>et al.</i> (2023) [19]	XGBoost	3189
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As shown in Table 8.1, the reviewed literature utilizes a wide array of machine learning architectures and varying dataset scales to predict surgery durations. The data indicates a significant shift toward ensemble learning and gradient boosting techniques, with XGBoost emerging as the most frequently cited "best model" across six different studies. Other ensemble methods, such as Random Forest and ANN/MLP structures, also demonstrate strong performance across diverse surgical specialties. The scale of the data utilized in these studies varies dramatically, ranging from smaller, procedure-specific cohorts of several hundred cases to massive multi-year datasets, such as the 302,300 cases analyzed by Abbas *et al.* (2022). This diversity in data scale and model selection highlights the ongoing effort to balance computational complexity with high-accuracy predictions tailored to specific institutional or surgical needs.

Table 8.2 Algorithmic Comparison

This section highlights the diversity of machine learning architectures explored across the literature, allowing for a quick comparison of model selection.

Paper	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]
AdaBoost								✓	✓			✓	✓						
Artificial Neural Network (ANN)	✓	✓	✓															✓	
Balance d Bagging Classifier																	✓		
Balance d Random Forest																	✓		
Bayesian Approach		✓																	
Bayesian Network-based Classification Models							✓	✓											
Bagged Trees				✓															
Bagging Regressor																			✓
Baseline Method					✓													✓	✓
CatBoost										✓									
Decision Tree							✓					✓							
Feedforward Neural Network																	✓		
Gradient Boosting Machine (GBM)	✓						✓												
Gradient Boosted Trees					✓	✓													
Hospital Estimation											✓								
k-Nearest Neighbors (kNN)			✓					✓	✓			✓	✓						

LightGBM										✓									
Linear Regression	✓			✓					✓	✓	✓	✓	✓	✓				✓	✓
Logistic Regression								✓								✓			
Lasso Regression	✓				✓														
Multilayer Perceptron						✓													
Multivariate Adaptive Regression Splines											✓								
Multivariable Linear Regression					✓										✓				
Mixture Density Network							✓												
Modular Artificial Neural Network (MANN)																			
Naive Bayes								✓											
Nearest Neighbor Regression															✓				
Neural Network				✓							✓								
Polynomial Regression															✓				
PyTorch MLP									✓										
Random Forest (RF)	✓		✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓
Regression Trees				✓															
Ridge Regression	✓				✓														
Scheduled Duration	✓	✓																	
Scikit-learn MLP									✓										
Stochastic Gradient Descent												✓							
Support Vector Machine				✓				✓				✓							
Support Vector Regression																			
XGBoost	✓								✓	✓			✓	✓	✓	✓	✓	✓	✓

Table 8.3 Performance Analysis of Studies

Paper	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]
Best MAE (mins)	31.8 (Neutral Network)			26 (Bagged Trees)		14.9 (GBT)				16.32 (Random Forest)		A: 25.143 (Linear Regression), B: 23.038 (Random Forest)	31.04 (RF)	20 (XGBoost)	10.87 (XGBoost, HA)			29.1 (XGBoost)	44.31 (XGBoost)
Worst MAE (mins)				37 (Regression Trees)		15.9 (Random Forest)				50.22 (Linear Regression)		A: 26.610, B: 25.633 (Linear Regression with patient features)	33.6 (linear Regression)	43 (Linear Regression)	13.61 (Regression, KA)			41.3 (baseline method)	N/A (baseline method)
BEST MSE (mins)	70.81								0.893 (PyTorch MLP)										
WORST MSE (mins)									1.156 (KNN)										
Best MdAE (mins)												A: 25.364 (Random Forest), B: 24.347 (SVM)		11 (XGBoost)				22.8 (XGBoost)	
Worst MdAE (mins)												A: 28.023, B: 28.308 (Classification methods with midpoint/mode mismatch)		25 (Linear Regression)				33.4 (baseline method)	
Best RMSE (mins)	49.7 (ANN)			26 (Bagged Trees)	80.2 (Boosted Regression Tree)	20.3 (GBT)		16.6 (Naive Bayes)							12.13 (XGBoost, HA)			44.5 (XGBoost)	52.95 (XGBoost)
Worst RMSE (mins)				(Regression Trees)	100.4 (Baseline)	21.7 (Random Forest)		40 (Logistic Regression)							13.61 (Regression, KA)			60.4 (baseline method)	N/A (baseline)
Best MAPE (%)	26% (Neutral Network, Random Forest)					16.4% (GBT)						31% (Random Forest)		32% (XGBoost)	22.66% (XGBoost, HA)			29.3% (XGBoost)	
Worst MAPE (%)	34% (Scheduled Duration)					18.1% (Random Forest)								76% (Linear Regression)	23.89% (Regression, KA)			41.6% (baseline method)	
Best R ²	0.78 (ANN)													0.82 (XGBoost)				0.835 (XGBoost)	0.770 (XGBoost)
Worst R ²	0.43 (Scheduled Duration)													0.06 (Linear Regression)				0.446 (baseline method)	0.213 (baseline method)
Best AUC			82% (ANN)													91.6% (XGBoost)	77.9% (XGBoost)		
Best CRPS		13.8 ± 35.4 (ANN)						18.1 (Mixture Density Network)											

As shown in table 8.3, the evaluation of these predictive models relies on a variety of metrics, with Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) serving as the most frequent indicators of accuracy. Across the 19 studies, Best MAE values typically range from approximately 10.87 minutes to 44.31 minutes, illustrating that predictive precision is often highly dependent on the specific surgical specialty and dataset complexity. Ensemble methods consistently outperform simpler alternatives; for instance, XGBoost and Random Forest often yield the lowest error rates and the highest

R^2 values, with R^2 peaks reaching 0.835 in some ambulatory settings. Conversely, traditional linear regression and simple baseline methods (such as scheduled duration) frequently produce the "Worst" results, characterized by higher RMSE and lower explained variance. Furthermore, specialized metrics like AUC (reaching up to 91.6% in complication prediction) and CRPS are utilized in more recent studies to assess model robustness and probabilistic forecasting reliability.

Table 8.4 Feature Studying

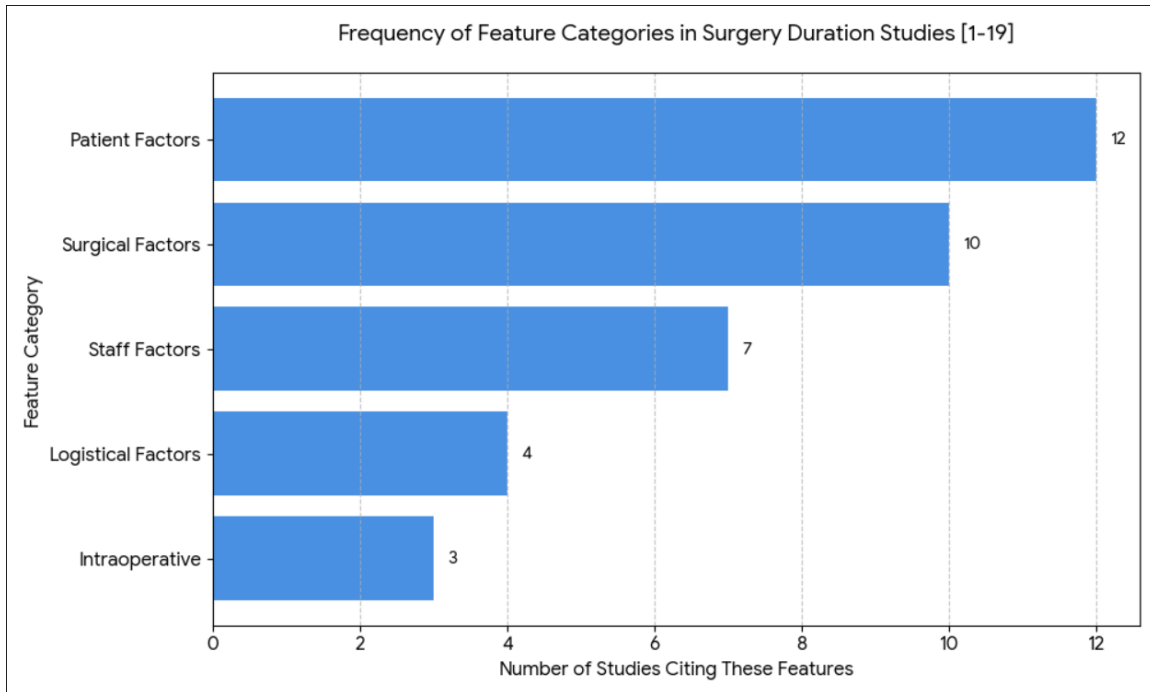


Table 8.4 illustrated that Patient Factors remain the primary foundational input for surgery duration prediction, with indicators such as Age, BMI, ASA Score, and comorbidities like Hypertension or OSA appearing in approximately 63% of the cited studies. Beyond baseline patient characteristics, Surgical Factors—including Procedure Codes, Specialty, and complexity metrics like the Number of Levels—are critical for narrowing down duration variance.

The "human element" is also a significant consideration, as roughly 37% of the research incorporates Staff Factors, specifically the Surgeon's ID and experience level, to account for individual technical impacts on operative time. While Logistical Factors (such as Scheduled Duration or Day of Week) and Intraoperative data (including Vital Signs or Real-time technical parameters) are utilized less frequently, they provide essential context for institutional efficiency and high-fidelity modeling.

The following table summarizes the key features identified across the five primary categories of predictive variables:

Category	Key Features
Patient Factors	Age, BMI, ASA Score, Gender, Comorbidities (Hypertension, OSA)
Surgical Factors	Surgeon ID, Surgeon Experience, Team Size, Anesthesia Type
Logistical Factors	Scheduled Duration, Day of Week, Time of Day, Year/Season
Intraoperative	Vital Signs, Medications used, Real-time technical parameters

8.2 Data Preprocessing Python Scripts

8.2.1 Missing Value Imputation

```
def reimpute_host_team(spec: str) -> None:
    mask_spec = df["host_spec"].astype(str) == spec
    base = mask_spec & (~in_window) & df["host_team"].notna()
    if not base.any():
        return
    dist = df.loc[base, "host_team"].value_counts(normalize=True)
    targets = df.index[mask_spec & in_window]
    if len(targets) == 0:
        return
    df.loc[targets, "host_team"] = rng.choice(
        dist.index.to_numpy(),
        size=len(targets),
        p=dist.to_numpy(),
    )

reimpute_host_team("O&T")
reimpute_host_team("SURG")
m = df["host_spec"].isin(["DENT", "ENT"]) & df["host_team"].isna()
df.loc[m, "host_team"] = df.loc[m, "host_spec"]
```

Fig 8.2.1: Handling the missing value of host_team data

```
# Structural quality filters
null_pct = df.isnull().mean() * 100
df = df.loc[:, null_pct <= 40].copy()
df = df.loc[:, df.nunique(dropna=False) > 1].copy()
df = df.loc[df.isnull().sum(axis=1) < 4].copy()
```

Fig.8.2.2: Filter out the features with missing rate excess 40% and the row with missing value

8.2.2 Tokenizing

```
def build_subword_encode(
    texts: pd.Series,
    embed: pd.DataFrame,
    surgery: pd.Series,
    out_path: Path,
    model_type: str,
    vocab_size: int = 1024,
    pctl: float = 1.0,
    maxlen_cap: int = 25,
    maxlen_min: int = 5,
) -> None:
    """
    Local subword tokenizer export (inlined from tokenizer_variants_common).
    Writes tok_* columns + embed columns + surgery_duration to CSV.
    """
    lines = texts.fillna("").astype(str).tolist()
    lengths = [len(x.split()) for x in lines]
    max_len = max(maxlen_min, min(maxlen_cap, _percentile_int(lengths, pctl)))

    if model_type == "bpe":
        tokenizer = Tokenizer(models.BPE(unk_token="<UNK>"))
        trainer = trainers.BpeTrainer(
            vocab_size=vocab_size,
            special_tokens=["<PAD>", "<UNK>"],
            min_frequency=2,
        )
    elif model_type == "unigram":
        tokenizer = Tokenizer(models.Unigram())
        trainer = trainers.UnigramTrainer(
            vocab_size=vocab_size,
            special_tokens=["<PAD>", "<UNK>"],
            unk_token="<UNK>",
        )
    else:
        raise ValueError(f"Unsupported model_type: {model_type}")

    tokenizer.normalizer = normalizers.Sequence([normalizers.NFKC(), normalizers.Lowercase()])
    tokenizer.pre_tokenizer = pre_tokenizers.Whitespace()
```

Fig. 8.2.3: The BPE tokenizer for the px_description

8.3 Manual Dictionary

short_term	full
lscs	lower segment cesarean section
bso	bilateral salpingo-oophorectomy
phaco	phacoemulsification
ppv	pars plana vitrectomy
turp	transurethral resection of the prostate
eb	endometrial biopsy
iol	intraocular lens
le	left eye
re	right eye
lap	laparoscopy / laparoscopic
tkr	total knee replacement
el	endolaser
ursl	ureteroscopic lithotripsy
tah	total abdominal hysterectomy
h&c	hysteroscopy and curettage
handc	hysteroscopy and curettage
mp	membrane peeling
tlh	total laparoscopic hysterectomy
pciol	posterior chamber intraocular lens
turbt	transurethral resection of bladder tumor
btl	bilateral tubal ligation
cryo	cryotherapy
pfr	pelvic floor repair
rfa	radiofrequency ablation
thr	total hip replacement
sfiol	scleral fixated intraocular lens
vh	vaginal hysterectomy
mmc	mitomycin c
usg	ultrasonography
pcnl	percutaneous nephrolithotomy
ukr	unicompartmental knee replacement
av	arteriovenous
la	local anesthesia
thbso	total hysterectomy and bilateral salpingo-oophorectomy
rirs	retrograde intrarenal surgery
ssf	sacrospinous fixation
rezum	rezum water vapor therapy
iucd	intrauterine contraceptive device

rp	retrograde pyelogram / radical prostatectomy
acl	anterior cruciate ligament
k-wire	kirschner wire
scp	sacrocolpopexy
tvf	tension-free vaginal tape
vhpf	vaginal hysterectomy and pelvic floor repair
pfna	proximal femoral nail antirotation
xr	x-ray
jj	double j (stent)

Table 8.3.1

8.4 Feature Distribution Tables

Table 8.4.1: Hosting Specialty Distribution by Surgery Time (Minute)

Hosting Specialty	Dataset	Count	Mean	Median	Min	Max
anae	fill	200	53.29	46	12	219
anae	filter	141	49.68	43	12	144
dent	fill	359	226.05	136	1	567
dent	filter	154	133.94	104.5	16	425
ent	fill	1237	80.60	28	2	567
ent	filter	886	51.48	23	2	425
eye	fill	1808	80.70	66	12	370
eye	filter	994	62.73	49	12	286
o&g	fill	2181	102.94	74	6	567
o&g	filter	1153	91.49	57	6	425
o&t	fill	3256	157.45	143	3	567
o&t	filter	1777	138.72	129	3	425
surg	fill	5591	115.78	82	1	567
surg	filter	3705	91.43	66	1	425

Table 8.4.2: Hosting Team Distribution by Surgery Time (Minute)

Hosting Team	Dataset	Count	Mean	Median	Min	Max
a	fill	1368	189.14	164.5	2	567
a	filter	708	147.86	132.5	2	425
b	fill	1910	140.35	111	3	567
b	filter	1126	130.60	103.5	3	425
c	fill	1756	142.76	135	3	567
c	filter	983	129.98	119	3	425
d	fill	1407	156.34	137	3	567
d	filter	813	132.20	123	3	425
ent	fill	1237	80.60	28	2	567
ent	filter	886	51.48	23	2	425
eye	fill	1279	73.40	55	12	370
eye	filter	708	53.50	43	12	244
u	fill	2406	67.55	37.5	1	567
u	filter	1852	53.06	27	1	425
other	fill	1616	157.99	123	1	567
other	filter	1734	90.89	65	6	425
vr	fill	529	98.34	92	17	300
day	fill	564	37.60	31	6	239
obs	fill	560	71.07	73	8	207

Table 8.4.3: Patient Type Distribution by Surgery Time (Minute)

Patient Type	Dataset	Count	Mean	Median	Min	Max
day patient	fill	6308	48.47	37	1	341
day patient	filter	4262	39.46	30	1	244
Inpatient	fill	4137	174.11	143	4	567
Inpatient	filter	2214	151.56	127	4	425
Outpatient	fill	2	219.50	219.5	187	252
post holiday case	fill	218	224.00	198	35	567
post holiday case	filter	91	196.89	172	35	425
same day admission	fill	3967	163.00	143	1	567
same day admission	filter	2243	135.84	126	3	425

Table 8.4.4: Session Type Distribution by Surgery Time (Minute)

Session Type	Dataset	Count	Mean	Median	Min	Max
a	fill	4778	89.15	68.5	2	567
a	filter	2885	70.04	49	2	425
p	fill	3294	55.05	36	1	449
p	filter	2025	31.44	24	1	234
w	fill	6560	169.91	140	1	567
w	filter	3900	143.74	124	3	425

Table 8.4.5: Theatre Code Distribution by Surgery Time (Minute)

Theatre Code	Dataset	Count	Mean	Median	Min	Max
dot2	fill	548	70.18	71	8	236
dot2	filter	256	66.42	67	13	180
ot1	fill	1687	51.58	41	1	429
ot1	filter	1125	40.17	35	1	244
ot10	fill	1099	193.79	148	5	567
ot10	filter	630	171.34	137	5	425
ot11	fill	1240	158.75	125.5	5	567
ot11	filter	744	126.35	106	6	425
ot2	fill	1773	77.88	63	2	488
ot2	filter	1050	58.22	46	2	295
ot3	fill	1015	169.61	156	4	567
ot3	filter	542	147.08	137	4	425
ot5	fill	1317	139.46	135	5	527
ot5	filter	840	128.81	125.5	5	425
ot6	fill	1340	133.79	101.5	2	567
ot6	filter	786	109.82	78	2	425
ot7	fill	1633	94.47	56	1	567
ot7	filter	1082	63.95	38.5	1	425
ot8	fill	1670	92.30	41.5	2	567
ot8	filter	1097	60.02	28	2	425
ot9	fill	1158	168.46	161	4	567
ot9	filter	572	155.78	148.5	4	425
other	fill	152	65.49	38	1	359
other	filter	86	31.05	15	1	313

8.5 Final Parameter of LightGBM

```
BASELINE_PARAMS = {  
    'n_estimators': 1649,  
    'learning_rate': 0.0188110732863311,  
    'num_leaves': 72,  
    'max_depth': 3,  
    'subsample': 0.6534148339754011,  
    'colsample_bytree': 0.5608161866429593,  
    'reg_alpha': 9.64479796634143,  
    'reg_lambda': 14.744169048139852,  
    'random_state': 42,  
    'n_jobs': -1,  
    'verbose': -1  
}
```

Fig 8.5.1 final Parameter of LightGBM

9. Reference

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10. Contribution

10.1 Nature of Contribution

Name	Description
WONG Man Chun	1) Project Planning and Background Research; 2) Designed End-to-end Project Architecture; 3) Data Quality Check and Analysis Programming; 4) Data Preprocessing Programming; 5) Single Model Developing and Analysis Programming; 6) Stacking Model Developing and Analysis Programming; 7) Report Writing for Session: 4, 5 and 6.
TIN Yiu Tsun	1)Preprocessing Programming; 2)Background Research; 3)Report Writing for Session 3, 4.1.2, 8; 4)Project Video; 5)Project Structure Organizing.
TANG Cheuk Him	1) Background Research; 2) Data Quality Check and Analysis Programming; 3) Report Writing for Session: 1, 2, 3.2.3, 7, 9; 4) Report Formatting and Structing; 5) Presentation Flow Design and Formatting.
WEN Velma	1) Background Research; 2) Data Quality Check and Analysis Programming; 3) Report Writing for Session: 1, 2, 3.2.3, 7, 9; 4) Report Formatting and Structing; 5) PowerPoint Design and Formatting.

10.2 Percentage of Project Contribution

Name	Percentage (%)
WONG Man Chun	40
TIN Yiu Tsun	30
TANG Cheuk Him	15
WEN Velma	15

10.3 Percentage of Write-up Contribution

Name	Percentage (%)
WONG Man Chun	40
TIN Yiu Tsun	30
TANG Cheuk Him	15
WEN Velma	15